Efficient Sieve Minimum Distance Estimation of Semiparametric Conditional Moment Models*

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Abstract

We propose an estimation method for models of conditional moment restrictions that contain finite dimensional unknown parameters (θ) as well as infinite dimensional unknown functions (h), where h could depend on endogenous variables. Our proposal is to approximate h with sieves and then estimate θ and the sieve parameters jointly by applying the method of minimum distance. We show that: (1) the sieve estimator of h is consistent with rate under certain metric; (2) the estimator of θ is $\sqrt{n}$ consistent and asymptotically normally distributed; (3) the estimator for the asymptotic variance of the θ estimator is consistent and easy to compute; and (4) the optimally weighted minimum distance estimator of θ attains the semiparametric efficiency bound. These results extend the work of Hansen (1982) and Newey (1990, 1993) to a semiparametric conditional moment restrictions setting.

1. Introduction

A general framework for analyzing data (Y, X) is to presume that the data satisfy some conditional moment restrictions:

$$E[\rho(Z, \alpha_o)|X] = 0 \tag{1.1}$$

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where $Z \equiv (Y', X_z')'$, X_z is a subset of X, α_o is a (column) vector of finite dimensional unknown parameters, $\rho(\cdot,\cdot)$ is a (column) vector of known functions, and $E[\rho(Z,\alpha_o)|X]$ is the conditional expectation of $\rho(Z,\alpha_o)$ given X, with the conditional distribution of Y given X unspecified. A common method for estimating the parameters of interest, α_o, is the generalized method of moments (GMM) estimation proposed by Hansen (1982). The GMM first converts the conditional moment restrictions (1.1) into the unconditional moment restrictions:

$$E[W\rho(Z,\alpha_o)] = 0, \tag{1.2}$$

where W are instruments (e.g. X and nonlinear measurable transformations of X), and then estimates α_o by solving the sample analog of (1.2). Under some sufficient conditions, Hansen (1982) shows that the GMM estimator of α_o is $\sqrt{n}$ consistent and asymptotically normally distributed, where n is the sample size. Moreover, he establishes that the optimally weighted GMM estimator attains the efficiency bound of the unconditional moment restrictions model (1.2). Chamberlain (1987) derives the efficiency bound of the conditional moment restrictions model (1.1). He shows that the optimally weighted GMM estimator attains his bound if W are the optimal instruments[1]. Newey (1990, 1993) propose to estimate the optimal instruments nonparametrically and shows that his estimators attain Chamberlain's (1987) bound for model (1.1). In this paper, we want to extend all these results to the semiparametric conditional moment restrictions setting

$$E[\rho(Z,\theta_o,h_o(\cdot))|X] = 0, \tag{1.3}$$

where the parameters of interest $\alpha_o = (\theta_o, h_o)$ now contain both a vector of finite dimensional unknown parameters θ_o and a vector of infinite dimensional unknown functions $h_o(\cdot) = (h_{o1}(\cdot),...,h_{oq}(\cdot))'$, [the dot "$\cdot$" in $h_{oj}(\cdot)$ denotes its generic argument which may vary with applications and the subscript j]. Our primary focus is the consistent estimation of α_o, the $\sqrt{n}$ asymptotic distribution and efficiency of the estimator of θ_o.

The inclusion of $h_o(\cdot)$ in the parameters enables (1.3) to encompass many important classes of semiparametric and nonparametric models. The popular partially linear regression $E[Y - X_1'\theta_o - h_o(X_2)|X_1, X_2] = 0$ studied by Robinson (1988), the semiparametric regression $E[\rho(Z,\theta_o,h_{o1}(\delta_1(X)),...,h_{oq}(\delta_q(X)))|X] = 0$ studied by Chamberlain (1992), and the widely used index models $E[Y - X_1'\theta_{o1} - h_o(X_2'\theta_{o2})|X_1, X_2] = 0$ studied by Powell, Stock and Stoker (1989), and Ichimura (1993) are all special cases of the following class:

$$E[\rho(Z,\theta_o,h_{o1}(\delta_1(X,\theta_o)),...,h_{oq}(\delta_q(X,\theta_o)))|X] = 0 \tag{1.4}$$

where $\delta_j(X,\theta_o), j = 1,...,q$ are known measurable functions of X up to unknown θ_o. Clearly this class (1.4) fits into our general framework (1.3).

[1] The optimal instruments are often unknown functions of X.

The shape-invariant Engel curve model with endogenous total expenditure (e.g., $E[Y_{1j} - h_{oj}(Y_2 - \delta(X_1, \theta_{o1})) - X_1'\theta_{o2j}|X_1, X_2] = 0$, $j = 1, ..., q$, Ai, Blundell and Chen, 2001), some transformation model with unknown link (e.g. $E[h_o(Y) - X_1 - X_2'\theta_o|X_1, X_2] = 0$ with h_o monotone[2]) and others, are examples where the unknown functions h_o depend on the endogenous variables Y. These examples are special cases of the following more general class:

$$E[\rho(Z, \theta_o, h_{o1}(\delta_1(Z, \theta_o)), ..., h_{oq}(\delta_q(Z, \theta_o)))|X] = 0 \tag{1.5}$$

where $\delta_j(Z, \theta_o), j = 1, ..., q$ are known measurable functions of Z up to unknown θ_o. Our general framework (1.3) encompasses (1.5) and much more. It permits dependence of $\delta_j(\cdot)$ on unknown functions, thereby allows h_{oj} to be a function of other unknown functions. Nonparametric censored regression models are examples where an unknown function depends on another unknown function. Our general framework also permits dependence of $\delta_j(\cdot)$ on unobserved variables. Random coefficients models are examples where the unknown density of the random coefficients is a function of unobserved random coefficients. Additional examples that fit into (1.3) can be found in Powell (1994) and Horowitz (1998).

Under the assumption that model (1.3) identifies the parameters of interest $\alpha_o = (\theta_o, h_o)$, we propose a sieve minimum distance (hereafter SMD) estimator of α_o. The proposed SMD estimator can be interpreted as a GMM estimator, thereby is very intuitive and easy to implement. Under a set of sufficient conditions, we show that the SMD estimator $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ for $\alpha_o = (\theta_o, h_o)$ is consistent with convergence rate under the Fisher-like metric (to be defined later), and the estimator for the parametric component, $\widehat{\theta}_n$, is $\sqrt{n}$ consistent and asymptotically normally distributed. These results extend the work of Shen (1997) and Chen and Shen (1998) to the semiparametric conditional moments framework (1.3). In addition, we provide a consistent and easy to compute estimator for the asymptotic covariance of $\widehat{\theta}_n$ and show that, when the weighting matrix $\Sigma(X)$ is set to $\Sigma_o(X) \equiv Var[\rho(Z, \alpha_o)|X]$ (assumed to be positive-definite), the corresponding estimator $\widehat{\theta}_n$ (hereafter the optimally weighted SMD estimator) attains the semiparametric efficiency bound of model (1.3). The latter result generalizes the efficiency bound result of Chamberlain (1992) to the much more general model (1.3).

The literature on general theory of semiparametric efficient estimation is large. The papers most closely related to ours are those by Severini and Wong (1992), Shen (1997), Murphy and Van der Vaart (2000), Hansen (1982) and Newey (1990, 1993). The first three sets of authors studied efficient estimation in the semiparametric maximum likelihood setting.[3] Hansen proposed efficient estimation for the

[2] We thank W. Newey for suggesting this example. Notice that the true residual function $\rho(Z, \theta_o, h_o) = h_o(Y) - X_1 - X_2'\theta_o$ is only assumed to have conditional mean zero given the regressor $X = (X_1, X_2')'$, not the more commonly seen assumption of independence between $\rho(Z, \theta_o, h_o)$ and X, say as in Horowitz (1996).

[3] It is known that the semiparametric efficiency bounds and efficient estimators with known

parametric unconditional moments restrictions (1.2), while Newey investigated efficient estimation of the parametric conditional moments restrictions (1.1). Our results extend all these studies to the semiparametric conditional moments restrictions (1.3).

After our initial submission, an anonymous referee drew our attention to the working paper of Newey and Powell (1989) on nonparametric instrumental variables model $E[\rho(Z, h_o(\delta(Z)))|X] = 0$. Our approach is very similar to theirs in that both papers use sieves to approximate the unknown functions h_o and estimate the unknown sieve parameters by applying the minimum distance procedure. The two papers differ, however, in their focus. While they focus on the identification and consistent estimation of the nonparametric component, we focus on the $\sqrt{n}$ asymptotic distribution and efficiency of the parametric component. Thus our work can be viewed as complement to theirs. There are also papers on semiparametric efficient estimation of particular conditional moment restrictions models, such as Robinson's (1988) partially linear regression, Ichimura's (1993) single index model, and Klein and Spady's (1993) binary choice model. These papers utilize the specific structures of their respective models, hence their estimation procedures might be simpler and their results might be derived under weaker sufficient conditions than ours. However, there is no published work prior to ours on semiparametric efficient estimation of conditional moment models (1.3) where the unknown function h_o depends on the endogenous variables Y.

The rest of the paper is organized as follows. Section 2 formally introduces the SMD estimator and discusses its GMM interpretation. Section 3 obtains consistency and convergence rate of $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ to $\alpha_o = (\theta_o, h_o)$ under the Fisher-like metric (to be defined later). Section 4 derives the $\sqrt{n}$ asymptotic normality of $\widehat{\theta}_n$. Section 5 provides a consistent estimator for the asymptotic variance of $\widehat{\theta}_n$. Section 6 studies the semiparametric efficiency property of $\widehat{\theta}_n$. Section 7 first computes convergence rates of the nonparametric component $\widehat{h}_n$ to h_o under the sup-norm metric and the mean squared error metric. It then specializes our general theory to the partially linear nonparametric IV regression $E[Y_1 - X_1'\theta_o - h_o(Y_2)|X_1, X_2] = 0$. Although served as a simple illustration, we note that there is no published work on semiparametric efficient estimation of θ_o for this example. Section 8 evaluates the finite sample properties of the SMD estimator via a small scale Monte Carlo study.[4] Section 9 concludes the paper with suggestions for future research. All technical proofs are relegated to the Appendices.

functional form of $F_{Y|X}$ (conditional distribution of Y given X) are in general different than those with unknown functional form of $F_{Y|X}$.

[4] See Ai, Blundell and Chen (2001) for a more careful application of the SMD estimator to a system of shape-invariant semi/nonparametric Engel curve model with endogenous total expenditure.

2. Estimator

Throughout the paper, we assume that the sample observations $\{(Y_i, X_i) : i = 1, 2, \cdots, n\}$ are drawn independently from the distribution of (Y, X) on $\mathcal{Y} \times \mathcal{X}$ with $\mathcal{X}$ a compact convex subset of $\mathcal{R}^s$, and that the unknown distribution of (Y, X) satisfies the conditional moment restriction (1.3), where $\rho : \mathcal{Z} \times \mathcal{A} \to \mathcal{R}^r$ is a $r \times 1$ known mapping, $Z \equiv (Y', X_z')' \in \mathcal{Z} \equiv \mathcal{Y} \times \mathcal{X}_z$, $\mathcal{X}_z \subseteq \mathcal{X}$, and $\alpha_o \equiv (\theta_o, h_o) \in \mathcal{A} \equiv \Theta \times \mathcal{H}$, with Θ a compact (with nonempty interior) subset of $\mathcal{R}^b$ and $\mathcal{H} \equiv \mathcal{H}^1 \times \cdots \times \mathcal{H}^q$ a functional space of continuous functions.

Under the assumption that model (1.3) identifies the parameters of interest α_o, we propose an estimator of α_o that has desirable large sample properties. Our approach is to apply the method of minimum distance. Heuristically, if the functional forms of $h_o(\cdot)$ and the conditional distribution of Y given X, $F_{Y|X}$, were known, then the functional form of the conditional expectation

$$m(x, \alpha) \equiv \int \rho(y, x_z, \theta, h(\cdot)) dF_{Y|X=x}(y)$$

would be known. The minimum distance estimation of θ_o would be appropriate since condition (1.3) implies that α_o solves

$$\inf_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}} E\left[m(X, \alpha)'[\Sigma(X)]^{-1} m(X, \alpha)\right] \tag{2.1}$$

where $\Sigma(X)$ is a positive definite matrix for any given X, (and " $'$ " denotes the matrix transpose in this paper). Since the functional forms of $h_o(\cdot)$ and $F_{Y|X}$ (hence $m(\cdot, \cdot)$) are in fact unknown, we estimate $m(X, \alpha)$ nonparametrically with $\widehat{m}(X, \alpha)$ using observations on $\rho(Z, \alpha)$ and X. The true values α_o would then be estimated by minimizing the sample analog of (2.1) with $m(X, \alpha)$ replaced by $\widehat{m}(X, \alpha)$. However, the resulting estimator could be inconsistent or converge arbitrarily slowly when the parameter space $\mathcal{H}$ is too large. To remedy this problem, we follow the sieve literature (Grenander, 1981) by approximating $\mathcal{H} \equiv \mathcal{H}^1 \times \cdots \times \mathcal{H}^q$ with a sequence of sieve spaces $\mathcal{H}_n \equiv \mathcal{H}_n^1 \times \cdots \times \mathcal{H}_n^q$. The sieve spaces are computable (often finite-dimensional compact) parameter spaces which may increase with sample size and become dense in the original functional space $\mathcal{H}$ (i.e., for any $h \in \mathcal{H}$, there exists a $\Pi_n h \in \mathcal{H}_n$ such that $d(h, \Pi_n h) \to 0$ as $n \to \infty$, where d is a pseudo distance). Example sieve spaces include power series, Fourier series, splines, wavelets, neural networks and many more. The sieve method is natural and easy to implement since once $h \in \mathcal{H}$ is replaced by $h_n \in \mathcal{H}_n$ the estimation problem becomes a parametric one. The sieve method is especially convenient when h_o enters $\rho(Z, \theta_o, h_o)$ nonlinearly or satisfies some shape restrictions (e.g. monotonicity, concavity, additivity, exclusion restrictions etc.). All one needs to do is to choose a shape preserving sieve $\mathcal{H}_n$ and then replace $h \in \mathcal{H}$ by $h_n \in \mathcal{H}_n$.[5]

[5] See Newey, Powell and Vella (1999) for using the spline sieve to estimate unknown functions

To summarize, the SMD estimator of α_o is obtained by minimizing the sample analog of a nonparametric version of (2.1) with h restricted to the sieve space $\mathcal{H}_n$:

$$\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n) : \min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \widehat{m}(X_i,\alpha)'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}(X_i,\alpha), \tag{2.2}$$

where $\widehat{\Sigma}(X)$ is a consistent estimator of some positive definite matrix $\Sigma(X)$ for every X, and the weighting matrix $\Sigma(X)$ is introduced to address potential heteroscedasticity in the model. Although $\widehat{m}(X,\alpha)$ could be any nonparametric estimator of $m(X,\alpha)$ such as kernel estimator in our first version of the paper, here we consider a linear sieve estimator of $m(X,\alpha) = (m_1(X,\alpha),...,m_r(X,\alpha))'$:

$$\widehat{m}_l(X,\alpha) = \sum_{j=1}^{n} \rho_l(Z_j,\alpha)p^{k_n}(X_j)'(P'P)^{-1}p^{k_n}(X), \quad l = 1,...,r \tag{2.3}$$

where $\{p_{0j}(X), j = 1,2,...\}$ denotes a sequence of known basis functions that can approximate any measurable function of X well (to be more precise later), $p^{k_n}(X) = (p_{01}(X),...,p_{0k_n}(X))'$ and $P = (p^{k_n}(X_1),...,p^{k_n}(X_n))'$ for some integer k_n, with $k_n \to \infty$ and $k_n/n \to 0$ as $n \to \infty$.

Note that with $\widehat{\Sigma}(X_i) = I$, our SMD estimator $\widehat{\alpha}_n$ is the solution to:

$$\min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \left(\sum_{j=1}^{n} \rho(Z_j,\alpha)\otimes p^{k_n}(X_j)\right)' \left(I\otimes(P'P)^{-1}\right) \left(\sum_{j=1}^{n} \rho(Z_j,\alpha)\otimes p^{k_n}(X_j)\right) \tag{2.4}$$

where $\otimes$ denotes the Kronecker product and I the $r\times r-$identity matrix. Thus, $\widehat{\alpha}_n$ is also the GMM estimator based on the following increasing number of unconditional moment restrictions:

$$E[p_{0j}(X)\rho(Z,\theta_o,h_o)] = 0, \ j = 1,2,...,k_n. \tag{2.5}$$

With non-identity weighting, the SMD estimator can still be interpreted as a GMM estimator. This GMM interpretation implies that the total number of moment restrictions must be larger than or equal to the total number of unknown coefficients:

$$rk_n \geq \dim(\theta) + \dim(\mathcal{H}_n) = \dim(\theta) + \sum_{j=1}^{q} \dim(\mathcal{H}_n^j)$$

where $\dim(\mathcal{H}_n^j)$ denotes the number of unknown sieve coefficients in $\mathcal{H}_n^j$. For instance if $r \equiv \dim(\rho) = 1$, then the number of linear sieve terms k_n in the

with additive structure; Chen, Hansen and Scheinkman (1997) and Chen and Conley (2000) for using the shape-preserving wavelet cardinal B-spline sieves to estimate a monotone eigenfunction of a diffusion process and an isotropic conditional covariance function respectively.

approximation of the conditional mean function $m(X, \alpha)$ is precisely the number of unconditional moment restrictions in (2.5), hence it must be larger than or equal to the total number of unknown coefficients to be estimated (i.e., the number of θ_o coefficients plus the number of sieve coefficients).

3. Consistency and Convergence Rates

In this section, we first introduce some definitions, then present a set of sufficient conditions for consistency and general convergence rate of $\widehat{\alpha}_n$ to α_o.

Throughout this paper, we adopt the following notation: $|\cdot|$ denotes the absolute value operator and, when applied to a vector-valued function $a(Z)$, $|a(Z)|$ is the element by element operation. $||\cdot||_E$ denotes the Euclidean norm: $||a(Z)||_E^2 = trace\{a(Z)'a(Z)\}$ for any matrix $a(Z)$. The inequality $a(Z) \leq c$ implies that each component of $a(Z)$ is no larger than c. For any pair of matrices A and B, $A > (\geq) B$ means that $A - B$ is a positive definite (semidefinite) matrix. We also denote $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ as the smallest and the largest eigenvalues of a symmetric matrix A.

3.1. Definitions

We define the (pathwise) directional derivative of $\rho(Z, \alpha)$ with respect to α evaluated at α_o at the direction $[\alpha - \alpha_o]$ as:

$$\rho_{\alpha_o}[Z, \alpha - \alpha_o] \equiv \lim_{\tau \to 0} \left\{ \frac{\rho(Z, \alpha(\tau)) - \rho(Z, \alpha_o)}{\tau} \right\} \tag{3.1}$$

where $\alpha(\tau)$ is an one-dimensional continuous path in $\mathcal{A}$ satisfying $\alpha(0) = \alpha_o$ and $\alpha(1) = \alpha \in \mathcal{A}$. For example, the directional derivative for the class (1.5) is

$$\rho_{\alpha_o}[Z, \alpha - \alpha_o] = \frac{d\rho(Z, \alpha_o)}{d\theta'}(\theta - \theta_o) + \sum_{j=1}^{q} \frac{\partial \rho(Z, \alpha_o)}{\partial h_j}(h_j(\delta_j(Z, \theta_o)) - h_{oj}(\delta_j(Z, \theta_o)))$$

where $\frac{d\rho(Z,\alpha_o)}{d\theta'} = \frac{\partial \rho(Z,\alpha_o)}{\partial \theta'} + \sum_{j=1}^{q} \frac{\partial \rho(Z,\alpha_o)}{\partial h_j} \frac{\partial h_j(\delta_j(Z,\theta_o))}{\partial \theta'}$ is the total derivative with respect to the transpose of θ and $\frac{\partial \rho(Z,\alpha_o)}{\partial \theta'}$ and $\frac{\partial \rho(Z,\alpha_o)}{\partial h_j}$ are usual partial derivatives.

Also define

$$m_{\alpha_o}[X, \alpha - \alpha_o] \equiv E\{\rho_{\alpha_o}[Z, \alpha - \alpha_o]|X\}.$$

For any α_1, $\alpha_2 \in \mathcal{A}$, we define a Fisher-like metric $\|\alpha_1 - \alpha_2\|$ as

$$\|\alpha_1 - \alpha_2\| \equiv \sqrt{E\{m_{\alpha_o}[X, \alpha_1 - \alpha_2]'\Sigma(X)^{-1} m_{\alpha_o}[X, \alpha_1 - \alpha_2]\}}. \tag{3.2}$$

Definition 3.1. *$\widehat{\alpha}_n$ is consistent (under the Fisher-like metric) for α_o if $||\widehat{\alpha}_n - \alpha_o|| \to 0$ in probability, denoted as $||\widehat{\alpha}_n - \alpha_o|| = o_p(1)$. Let $\{\eta_n\}$ denote a positive sequence that decreases to zero as $n \to \infty$. $\widehat{\alpha}_n$ is said to be consistent for α_o at a rate (strictly) faster than η_n if $||\widehat{\alpha}_n - \alpha_o||/\eta_n \to 0$ in probability, denoted as $||\widehat{\alpha}_n - \alpha_o|| = o_p(\eta_n)$.*

We adopt the Fisher-like metric $\|\alpha_1 - \alpha_2\|$ here over more popular mean-squared-error metric and sup-norm metric for the following reasons. First, the convergence rate under Fisher-like metric is what we need to derive the $\sqrt{n}$ asymptotic normality and efficiency of $\widehat{\theta}_n$, (our main focus of the paper). Second, Newey and Powell (1989) already established consistency (without rates) under the mean-squared-error metric and the sup-norm metric for the model $E[\rho(Z, h_o(\delta(Z))|X] = 0$, and their consistency result applies to our general model (1.3) after minor modification. However, as noted by Newey and Powell (1989), it is difficult to directly obtain convergence rates under the mean-squared-error metric and the sup-norm metric for their model $E[\rho(Z, h_o(\delta(Z))|X] = 0$. Convergence rates under the Fisher-like metric, on the other hand, is easier to obtain. Moreover, in many applications, some crude convergence rates under the mean-squared-error metric and the sup-norm metric can be obtained from the convergence rates under the Fisher-like metric, see Section 7 for detail. Finally, when the nonparametric part is not uniquely (pointwisely) identified, the Fisher-like metric still permits the convergence of the SMD estimator to a set of values satisfying the conditional moment restrictions (1.3) and the asymptotic normality of $\sqrt{n}(\widehat{\theta}_n - \theta_o)$ is still valid as long as θ_o is uniquely identified.

To aid presentation, we also introduce the following definitions:

Definition 3.2. *(**Envelope**) A functional $g(Z, \alpha)$ satisfies an envelope condition with coefficient c_{1n} over $\mathcal{A}_n$ if there exists a measurable function $c_1(Z) : \mathcal{Z} \to [0, \infty)$ satisfying: $E[c_1(Z)^p] < \infty$ for some $p \geq 4$; $E[c_1(Z)^2|X]$ exists; and $|g(Z, \alpha)| \leq c_{1n} c_1(Z)$ for all $\alpha \in \mathcal{A}_n$ and $Z \in \mathcal{Z}$.*

Definition 3.3. *(**Lipschitz**) A functional $g(Z, \alpha)$ is Lipschitz continuous with coefficient c_{2n} over $\mathcal{A}_n$ if there exists a measurable function $c_2(Z) : \mathcal{Z} \to [0, \infty)$ satisfying: $E[c_2(Z)^2] < \infty$ such that $|g(Z, \alpha_1) - g(Z, \alpha_2)| \leq c_{2n} c_2(Z) \|\alpha_1 - \alpha_2\|$ holds for all $Z \in \mathcal{Z}$ and $\alpha_1, \alpha_2 \in \mathcal{A}_n$.*

Definition 3.4. *(**Conditional smoothness**) A functional $g(Z, \alpha)$ satisfies conditional smoothness over $\mathcal{A}_n$ if there is a $a_0 > 0$ satisfying: for every $\alpha \in \mathcal{A}_n$ there exists a $\pi(\alpha)$ such that $E[g(Z, \alpha)|X] = p^{k_n}(X)'\pi(\alpha) + O(c_{1n} k_n^{-a_0})$ holds uniformly over $X \in \mathcal{X}$, $\alpha \in \mathcal{A}_n$.*

It is worth noting that our envelope and Lipschitz conditions differ from the familiar envelope and Lipschitz conditions in the parametric micro-econometrics literature in that (possibly increasing) coefficients c_{1n} and c_{2n} are introduced.

These coefficients are equal to one in the usual envelope and Lipschitz conditions, but they could be increasing with the sieve space here. In many nonparametric applications, the envelope coefficient c_{1n} could still be equal to the constant one.[6] The Lipschitz coefficient c_{2n}, on the other hand, usually grows with sample size n. This is because in an infinite dimensional parameter space, $g(Z,\alpha_1) - g(Z,\alpha_2)$ is often at the same order as $\alpha_1 - \alpha_2$, but $\alpha_1 - \alpha_2$ is generally not at the same order as $\|\alpha_1 - \alpha_2\|$ (the Fisher-like metric). Hence the usual Lipschitz condition (with $c_{2n} = 1$) does not hold in an infinite dimensional parameter space. The inclusion of possibly growing coefficient c_{2n} is one way to extend the usual Lipschitz condition. Another closely related generalization is to assume: for any $Z \in \mathcal{Z}$ and $\alpha_1, \alpha_2 \in \mathcal{A}_n$, there exists a constant $\epsilon \in (0,1]$ such that $|g(Z,\alpha_1) - g(Z,\alpha_2)| \leq c_2(Z)\,\|\alpha_1 - \alpha_2\|^{\epsilon}$, see e.g. Newey and Powell (1989) and Chen and Shen (1998) for Lipschitz condition of this sort.

The conditional smoothness condition assumes that the uniform approximation error of the linear sieve $p^{k_n}(X) = (p_{01}(X), ..., p_{0k_n}(X))'$ to $E[g(Z,\alpha)|X]$ is at the order $O(c_{1n}k_n^{-a_0})$ (uniformly over $X \in \mathcal{X}$ and $\alpha \in \mathcal{A}_n$). Here the value c_{1n} is simply the envelope coefficient of $|g(Z,\alpha)|$ over $\alpha \in \mathcal{A}_n$, which is at most the order of $\ln(1 + \dim(\mathcal{A}_n))$. And $O(k_n^{-a_0})$ is the uniform approximation error (uniformly over $X \in \mathcal{X}$) for arbitrarily fixed $\alpha \in \mathcal{A}_n$, which is determined by the smoothness (γ_0) of $E[g(Z,\alpha)|X]$, the dimension (s) of X, and the type of basis functions used to approximate $E[g(Z,\alpha)|X]$. For example, if for any fixed $\alpha \in \mathcal{A}_n$, $E[g(Z,\alpha)|X]$ belongs to the Hölder space $\Lambda^{\gamma_0}(\mathcal{X})$ with smoothness $\gamma_0 > s/2$, and X has density bounded above and below by positive constants over its support $\mathcal{X}$, and $p^{k_n}(X)$ is some tensor-product sieve with $k_n \approx (J_n)^s$, and J_n denotes the number of univariate B-splines (or power series or Fourier series) basis functions for the individual components of X, then the approximation error is simply $E[g(Z,\alpha)|X] - p^{k_n}(X)'\pi(\alpha) = O([J_n]^{-\gamma_0}) = O(k_n^{-\frac{\gamma_0}{s}})$ uniformly over $X \in \mathcal{X}$, see e.g. Newey (1997).

3.2. Convergence rate under the Fisher-like metric

In the following we define $\xi_{0n} \equiv \sup_{X\in\mathcal{X}} \left\|p^{k_n}(X)\right\|_E$ and $\xi_{1n} \equiv \sup_{x\in\mathcal{X}} \left\|\frac{\partial p^{k_n}(x)}{\partial x'}\right\|_E$. The constants ξ_{0n} and ξ_{1n} are obviously related to the linear sieve basis functions $p^{k_n}(X)$. We also denote ξ_{2n}, ξ_{3n} and ξ_{4n} as some constants that may increase with sample size. These constants are related to the smoothness of $\rho(Z,\alpha)$ over the sieve $\mathcal{A}_n$ and the size of the sieve $\mathcal{A}_n$. All these values ξ_{0n}, ξ_{1n}, ξ_{2n}, ξ_{3n} and ξ_{4n} can be computed in applications when the sieves $p^{k_n}(X)$ and $\mathcal{A}_n$ are specified. Let $\delta_n = n^{-1/4}\xi_{4n}^{-1/2}$ denote a sequence of constants that decrease to zero as sample size n goes to infinity, ($\xi_{4n} = \xi_{2n}$ in most applications). We also denote Π_n as a projection from the original parameter space $\mathcal{A} \equiv \Theta \times \mathcal{H}$ to the sieve space $\mathcal{A}_n \equiv \Theta \times \mathcal{H}_n$, $N(\mathcal{A}_n, \varepsilon)$ as the minimal number of radius ε covering

[6] The envelope coefficient c_{1n}, if allowed to increase with the sample size, usually grows very slowly say at the order $\ln(1 + \dim(\mathcal{A}_n))$.

balls of $\mathcal{A}_n$ under the Fisher-like metric, and denote $\dim(\mathcal{A}_n) = b + \dim(\mathcal{H}_n)$, with $\dim(\mathcal{H}_n)$ the number of unknown sieve coefficients in $h_n \in \mathcal{H}_n$.

To establish the consistency of the SMD estimator $\widehat{\alpha}_n$ to α_o under the Fisher-like metric at a rate faster than δ_n, we need conditions to ensure that, respectively at a rate faster than δ_n, $\mathcal{H}_n$ approximates $\mathcal{H}$, $\widehat{\Sigma}(X)$ converges to $\Sigma(X)$ uniformly over X, and $\widehat{m}(X,\alpha)$ converges to $m(X,\alpha)$ uniformly over $X \in \mathcal{X}, \alpha \in \mathcal{A}_n$. We now state such a set of sufficient conditions.

Assumption 3.1. *(i) The data $\{(Y_i, X_i)_{i=1}^n\}$ are i.i.d.; and (ii) $\mathcal{X}$ is compact with non-empty interior.*

Assumption 3.2. *The smallest and the largest eigenvalues of $E\{p^{k_n}(X)p^{k_n}(X)'\}$ are bounded and bounded away from zero for all k_n.*

Assumption 3.3. *$E[\rho(Z,\alpha)|X] = 0$ holds for any $\alpha \in \mathcal{A}$ iff $\alpha = \alpha_o$.*

Assumption 3.4. *(i) $\widehat{\Sigma}(X) = \Sigma(X) + o_p(\delta_n)$ uniformly over $X \in \mathcal{X}$; (ii) there exist some positive constants c_1 and c_2 such that $c_1 \leq \lambda_{\min}(\Sigma(X)) \leq \lambda_{\max}(\Sigma(X)) \leq c_2$ for all $X \in \mathcal{X}$.*

Assumption 3.5. *There exist some positive constants c_1 and c_2 such that*

$$c_1 E\{m(X,\alpha)'\Sigma(X)^{-1}m(X,\alpha)\} \leq \|\alpha - \alpha_o\|^2 \leq c_2 E\{m(X,\alpha)'\Sigma(X)^{-1}m(X,\alpha)\}$$

hold for all $\alpha \in \mathcal{A}_n$ and $\|\alpha - \alpha_o\| \leq o(1)$ and $X \in \mathcal{X}$.

Assumption 3.6. *$\rho(Z,\alpha)$ satisfies the Lipschitz (with coefficient ξ_{2n}), envelope (with coefficient ξ_{3n}), and conditional smoothness conditions over $\mathcal{A}_n$.*

Assumption 3.7. *(i) $\frac{n\delta_n^2}{\ln[(\frac{\xi_{1n}\xi_{3n}}{\delta_n})^s N(\mathcal{A}_n, \frac{\delta_n}{\xi_{0n}\xi_{2n}})] \times \max\{\xi_{0n}^2\xi_{3n}^2, \xi_{0n}^{2+2/p}\delta_n^{1-2/p}\xi_{3n}^{1+2/p}\}} \to +\infty$; (ii) $\xi_{3n}k_n^{-a_0} = o(\delta_n)$; (iii) $\delta_n\xi_{4n}^2 = O(1)$; and (iv) $rk_n \geq \dim(\mathcal{A}_n)$.*

Assumption 3.8. *For any $\alpha \in \mathcal{A}$, there exists $\Pi_n\alpha \in \mathcal{A}_n$ such that $\|\alpha - \Pi_n\alpha\| = o(\delta_n)$ as $n \to \infty$.*

Denote $\mathcal{F}_n = \{m(X,\alpha)'\Sigma(X)^{-1}m(X,\alpha) : \alpha \in \mathcal{A}_n, \|\alpha - \alpha_o\| \leq \sqrt{\delta_n}\}$ and $\kappa_n \in (0,1]$ as a measure of the size of the sieve space:

$$\kappa_n \equiv \inf_{\kappa}\left\{\kappa \in (0,1] : \frac{1}{\kappa^2}\int_{\kappa^2}^{\kappa}\sqrt{\log N(\mathcal{F}_n,\omega)}d\omega \leq O(\sqrt{n})\right\}.$$

Assumption 3.9. $\kappa_n = o(\delta_n)$.

Assumptions 3.1 - 3.9 are sufficient for the SMD estimator to be consistent at a rate faster than $\sqrt{\delta_n}$. To obtain a sharper rate, we need an additional local condition on $m(X,\alpha)$:

Assumption 3.10. *$\frac{m(X,\alpha)}{\eta_n}$ satisfies the envelope (with coefficient ξ_{4n}) condition over $\alpha \in \mathcal{A}_n$ and $\|\alpha - \alpha_o\| \leq o(\eta_n)$ for any constant $\delta_n \leq \eta_n \leq \delta_n^{3/4}$.*

We now state consistency and convergence rate under the Fisher-like metric.

Theorem 3.1. *Under Assumptions 3.1-3.10, we obtain:* $\|\widehat{\alpha}_n - \alpha_o\| = o_p(\delta_n)$ *with* $\delta_n = n^{-1/4}\xi_{4n}^{-1/2}$.

Remark 3.1. **(i)** Theorem 3.1 only gives a crude convergence rate of the proposed estimator. This rate is what we need to establish the $\sqrt{n}-$asymptotic normality and efficiency of $\widehat{\theta}_n$. The rate is established under mild conditions since all the coefficients $\xi_{0n}, \xi_{1n}, \xi_{2n}, \xi_{3n}, \xi_{4n}$ are allowed to grow with sample size, and Assumptions 3.7-3.9 allow for a wide range of growth orders of the smoothing parameters k_n and $\dim(\mathcal{A}_n)$. However, this rate is not sharp enough if one mainly cares about estimation and inference of the nonparametric component h_o, and one would prefer the exact convergence rate of $\widehat{h}_n$ to h_o instead. Notice that for the general nonlinear model (1.3) especially when h_o depends on the endogenous variables Y, computing the exact convergence rate is beyond the scope of this paper.

(ii) Theorem 3.1 still holds true even if α_o is not uniquely (pointwisely) identified. Suppose that Assumption 3.3 is replaced by a weaker identification condition: $m(X, \alpha) = 0$ holds for any $\alpha \in \mathcal{A}$ if and only if $\|\alpha - \alpha_o\| = 0$. Also, suppose that the sieve space $\mathcal{A}_n$ is chosen so that $E\{m(X, \alpha)'m(X, \alpha)\}$ is uniquely minimized over $\mathcal{A}_n$. Let α_n denote the minimizer of $E\{m(X, \alpha)'m(X, \alpha)\}$ over $\mathcal{A}_n$. Suppose that α_n under some sup-norm converges to some α^* satisfying $\|\alpha^* - \alpha_o\| = 0$. The above theorem holds with α_o replaced by α^*.

(iii) In some applications such as Robinson's partially linear regression and Ichimura's single index model, the directional derivative $\rho_{\alpha_o}[Z, \alpha - \alpha_o]$ does not depend on the endogenous variables Y, (hence $m_{\alpha_o}[X, \alpha - \alpha_o] = \rho_{\alpha_o}[Z, \alpha - \alpha_o]$). For this class of models, the initial nonparametric estimation of the conditional mean function $m(X, \alpha)$ can be omitted, and the unknown α_o is now estimated by a sieve generalized least squares (SGLS) regression:

$$\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n) : \min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \rho(Z_i, \alpha)'[\widehat{\Sigma}(X_i)]^{-1}\rho(Z_i, \alpha). \tag{3.3}$$

With slight modification, Theorem 3.1 still can be proved for the SGLS estimator where the Fisher-like metric now becomes $\|\alpha - \alpha_o\|^2 = E\{\rho_{\alpha_o}[Z, \alpha - \alpha_o]'\Sigma(X)^{-1}\rho_{\alpha_o}[Z, \alpha - \alpha_o]\}$. For this simpler class of models, Chen and Shen (1998) obtain the convergence rate $\|\widehat{\alpha}_n - \alpha_o\| = O_p\left(\max\{\|\alpha_o - \Pi_n\alpha_o\|, \kappa_n\}\right)$.

Discussion of Assumptions.

Assumption 3.1(i) rules out dependent data. This condition, though restrictive, is not critical for our results on consistency of $\widehat{\alpha}$ and later $\sqrt{n}$ asymptotic normality of $\widehat{\theta}$. These results still can be proved for weakly dependent data using the approach developed here and those developed in Chen and Shen (1998). The independence assumption is, however, critical for the semiparametric efficiency

result below. Assumption 3.1(ii) restricts the conditioning regressors X to be bounded. This is not critical for Theorem 3.1 since one can always trim large values of conditioning variables X. Notice that the endogenous variables Y are allowed to have unbounded supports, see Section 7 for examples.

Assumption 3.2 imposes no near multicollinearity among the basis functions that are used to estimate the conditional mean function as sample increases. This condition is satisfied if the basis functions are orthonormal. If the basis functions are not orthonormal, there always exists a transformation such that the transformed basis functions are orthonormal. Since we are only concerned with the fitted values and the fitted values are invariant under orthonormal transformation and the constants ξ_{0n} and ξ_{0n} are invariant under a nonsingular transformation, this condition is not as restrictive as it appears. Indeed, similar conditions are often imposed in the econometric literature, see Andrews (1994) and Newey (1997).

Assumption 3.3 is a global identification condition. Verification of this condition is often nontrivial and requires the knowledge of the structures of the specific models, see Newey and Powell (1989) for examples. As we pointed out before, if one only cares about unique identification and estimation of the parametric component θ_o, this condition can be weakened to permit a set of true values of the nonparametric component h_o.

Assumption 3.4(i) requires that the estimator of the weighting matrix converges to the weighting matrix uniformly over the regressors at a rate faster than δ_n. This assumption is not restrictive and can be satisfied by many estimators. For example, the identity weighting matrix satisfies this condition. The estimator proposed in Section 5 also satisfies this condition. Assumption 3.4(ii) requires the weighting matrix to be bounded from above and below. Conditions of this sort are common in the weighted LS regression literature.

Assumption 3.5 requires that the population criterion function can be approximated locally by a quadratic function. This condition is satisfied if $\rho(\cdot)$ is linear in $h(\cdot)$. If $\rho(\cdot)$ is nonlinear in $h(\cdot)$, this condition restricts the class of nonlinear models to which our results apply. This condition is not needed for the consistency of $\widehat{\alpha}_n$ but is very useful in deriving a fast convergence rate of $||\widehat{\alpha}_n - \alpha_o||$ and later the $\sqrt{n}-$normality of $\widehat{\theta}_n - \theta_o$.

Assumptions 3.6 and 3.7(i)(ii) are conditions related to the linear sieve estimation of the conditional mean function $m(X, \alpha) = E[\rho(Z, \alpha)|X]$. Assumption 3.6's conditional smoothness and Assumption 3.7(ii) together require that the bias part of the estimator $\widehat{m}(X, \alpha)$ shrinks to zero at the order $O(\xi_{3n} k_n^{-a_0}) = o_p(\delta_n)$ uniformly over $X \in \mathcal{X}, \alpha \in \mathcal{A}_n$; while Assumption 3.6's Lipschitz and envelope conditions and Assumption 3.7(i) together determine the rate $o_p(\delta_n)$ at which the residual part $\widehat{m}(X, \alpha) - E\{\widehat{m}(X, \alpha)\}$ goes to zero uniformly over $X \in \mathcal{X}, \alpha \in \mathcal{A}_n$.

Assumptions 3.8 and 3.9 are conditions related to the sieve approximation of $\alpha_o \in \mathcal{A}$ by $\Pi_n \alpha_o \in \mathcal{A}_n$. Assumption 3.8 requires that the sieve approximation error must shrink to zero at a rate faster than δ_n while Assumption 3.9 requires that the size of the sieve space $\mathcal{A}_n$ should not be too large. To ver-

ify these conditions, we must compute the sieve approximation error $\|\alpha - \Pi_n \alpha\|$ and the covering number $N(\mathcal{A}_n, \omega)$. Both calculations can be carried out using techniques developed in many mathematics books including Van der Vaart and Wellner (1996). Such calculations can also be found in econometrics applications such as Fenton and Gallant (1996) for Hermite polynomials, Newey (1997) for spline and power series, Chen, Hansen and Scheinkman (1997) for shape-preserving wavelet B-spline, and Chen and White (1998) for multivariate neural networks. In many interesting applications, the approximation error $\|\alpha - \Pi_n \alpha\|$ is of the order $\max\left\{(\dim(\mathcal{H}_n^1))^{-a_1}, ..., (\dim(\mathcal{H}_n^q))^{-a_q}\right\}$ where the constant $a_j > 0$ depends on the smoothness of $\mathcal{H}^j$ and the dimension of the augment of $h_{oj}(\cdot)$ for $j = 1, ..., q$. For linear sieves, the covering number $N(\mathcal{A}_n, \omega)$ is a power function of ω and Assumption 3.9 is automatically satisfied with $\kappa_n = O\left(\sqrt{n^{-1}\dim(\mathcal{A}_n)}\right)$. For nonlinear sieves such as neural networks, we usually have $\kappa_n = O\left(\sqrt{n^{-1}\dim(\mathcal{A}_n)\log(\dim(\mathcal{A}_n))}\right)$, see Chen and Shen (1998) for calculations of κ_n for many sieves.

Remark 3.2. There are two sets of smoothing parameters: the number of linear sieve terms in the approximation of $m(X, \alpha)$ (i.e. k_n) and the number of sieve terms in the approximation of $\mathcal{H}$ (i.e. $\dim(\mathcal{H}_n)$), that need to be determined. Assumption 3.7 alone can not determine k_n since it involves the constants ξ_{2n}, ξ_{3n}, and ξ_{4n} which may depend on $\dim(\mathcal{H}_n)$. On the other hand, Assumptions 3.8 and 3.9 do not involve constants that depend on k_n. Thus the smoothing parameter $\dim(\mathcal{H}_n)$ can be determined by Assumptions 3.8 and 3.9. For instance, one may wish to use $\|\alpha - \Pi_n \alpha\| = O(\kappa_n)$ or $o(\kappa_n)$ to determine the order of $\dim(\mathcal{H}_n)$, and then use Assumption 3.7 to determine the order of k_n. This simple way of choosing the smoothing parameters guarantees that Assumptions 3.7-3.9 are satisfied and hence Theorem 3.1 holds. But this ad-hoc rule might not produce the "best" smoothing parameters for the proposed SMD estimator, nor does it produce a unique set of smoothing parameters. Applied researchers still face the question of choosing the smoothing parameters. Like the current semiparametric and nonparametric literature, our theory is not very helpful in this regard. In practice, applied researchers may wish to use some generalized cross-validation methods to decide the smoothing parameters. An alternative approach is to extend the higher order expansion theory such as those of Donald and Newey (2000) and Linton (2001) to our estimator and then choose the smoothing parameters by optimizing the higher order performance of the SMD estimator. But it is beyond the scope of this paper to develop theoretical justifications of some cross-validation methods and/or the higher order expansion theory for the SMD estimator of the general model (1.3).

4. Asymptotic Normality

We now turn to the asymptotic distribution of $\widehat{\theta}_n$. The standard approach for deriving the asymptotic distribution is to assume that $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ is an interior solution and then apply Taylor expansion. This approach may not work well here for two reasons. First, while it is reasonable to expect $\widehat{\theta}_n$ to be an interior solution, $\widehat{h}_n$ often is not. Hence the first order condition may not hold. Second, even if the first order condition were satisfied, we need to compute the projection of the derivatives with respect to θ onto the closed linear span of the derivatives with respect to h. Because h is infinite dimensional, computing such projection is often complicated and infeasible. An alternative approach is to apply the Riesz representation theorem, which directly links a plug-in estimator of any smooth functional (such as $\widehat{\theta}_n$) to the derivatives of the sample criterion function without relying on the first order condition. This approach was used in papers such as Shen (1997) and Chen and Shen (1998) and is adopted here.

Let $\overline{\mathcal{A}} \equiv \Theta \times \overline{\mathcal{H}}$ denote the linear completion of $\Theta \times \mathcal{H}$ under the Fisher-like metric $\|\cdot\|$, and let $\langle \cdot, \cdot \rangle$ denote the inner product induced by the norm $\|\cdot\|$ on $\overline{\mathcal{A}}$. A linear functional $f : \overline{\mathcal{A}} \to \mathcal{R}$ is bounded (i.e. continuous) if and only if

$$\sup_{\|\alpha-\alpha_o\|>0, \alpha\in\overline{\mathcal{A}}} \frac{|f(\alpha) - f(\alpha_o)|}{\|\alpha - \alpha_o\|} < \infty.$$

The Riesz representation theorem states that for any bounded linear functional $f : \overline{\mathcal{A}} \to \mathcal{R}$, there exists a representor $v^* \in \overline{\mathcal{A}}$ satisfying:

$$\|v^*\| \equiv \sup_{\|\alpha-\alpha_o\|>0, \alpha\in\overline{\mathcal{A}}} \frac{|f(\alpha) - f(\alpha_o)|}{\|\alpha - \alpha_o\|} < \infty$$

and

$$f(\alpha) - f(\alpha_o) = \langle \alpha - \alpha_o, v^* \rangle \text{ for any } \alpha \in \overline{\mathcal{A}}.$$

Under some conditions and for any bounded linear functional $f(\alpha)$, one could establish the following link of $f(\widehat{\alpha}_n) - f(\alpha_o)$ to the pathwise directional derivatives of the sample criterion function:

$$\sqrt{n}(f(\widehat{\alpha}_n) - f(\alpha_o)) = -\frac{1}{\sqrt{n}} \sum_{i=1}^{n} m_{\alpha_o}[X_i, v^*]'[\Sigma(X_i)]^{-1} \rho(Z_i, \alpha_o) + o_p(1). \quad (4.1)$$

Then the Lindeberg-Levy central limit theorem implies that $\sqrt{n}\,(f(\widehat{\alpha}_n) - f(\alpha_o))$ is asymptotically normally distributed with mean zero and variance $Var(f)$:

$$Var(f) = E\{m_{\alpha_o}[X_i, v^*]'\Sigma(X_i)^{-1}\Sigma_o(X)\Sigma(X_i)^{-1} m_{\alpha_o}[X_i, v^*]\} \quad (4.2)$$

with $\Sigma_o(X) \equiv Var\left[\rho(Z, \alpha_o)|X\right]$.

We now apply this approach to derive the asymptotic distribution of $\sqrt{n}(\widehat{\theta}_n - \theta_o)$. Notice that for any fixed non-zero $\lambda \in \mathcal{R}^b$, $f(\alpha) = \lambda'\theta$ clearly is a linear functional on $\overline{\mathcal{A}}$. We just need to provide sufficient conditions to ensure: (a) $f(\alpha) = \lambda'\theta$ is bounded so we can compute its corresponding representor $v^* = (v^*_\theta, v^*_h) \in \overline{\mathcal{A}} = \Theta \times \overline{\mathcal{H}}$ and $m_{\alpha_o}[X_i, v^*]$ in the variance formulae (4.2); and (b) the linkage equation (4.1) holds.

We first consider the boundedness of the linear functional $f(\alpha) = \lambda'\theta$. Write the directional derivative as:

$$\rho_{\alpha_o}[Z, \alpha - \alpha_o] = \frac{d\rho(Z, \alpha_o)}{d\theta'}(\theta - \theta_o) + \frac{d\rho(Z, \alpha_o)}{dh'}[h - h_o], \tag{4.3}$$

where $\frac{d\rho(Z,\alpha_o)}{d\theta'}$ is the ordinary (total) derivative with respect to $\theta' = (\theta_1, ..., \theta_b)$ evaluated at α_o, and $\frac{d\rho(Z,\alpha_o)}{dh'}[h-h_o]$ denotes the directional derivative with respect to $h' = (h_1, ..., h_q)$ evaluated at α_o at the direction $h - h_o$. To simplify notation we let $\mathcal{W} = \mathcal{H} - h_o$ and $\overline{\mathcal{W}} = \overline{\mathcal{H}} - h_o$. Note that $\overline{\mathcal{W}}$ can be expressed as a linear combination of itself. To see this, consider any $q \times 1-$vector valued function $w^j \in \overline{\mathcal{W}}$, $j = 1, ..., b$. We form the $q \times b-$matrix valued function $w \equiv (w^1, ..., w^b)$. Because $\overline{\mathcal{W}}$ is linear, we have $-w(\theta - \theta_o) \in \overline{\mathcal{W}}$ for any $b \times 1-$Euclidean vector $\theta - \theta_o \in \Theta - \theta_o$. Thus there must exist $h - h_o \in \overline{\mathcal{W}}$ such that $h - h_o = -w(\theta - \theta_o)$. On the other hand, for any $h - h_o \in \overline{\mathcal{W}}$, we can always write $h - h_o = \frac{h-h_o}{\theta_j - \theta_{oj}}(\theta_j - \theta_{oj})$ if the j-th element $\theta_j - \theta_{oj}$ of $\theta - \theta_o$ is non-zero. Then with $w^j = -\frac{h-h_o}{\theta_j - \theta_{oj}}$ and $w^l = 0$ for $l \neq j$, we have $h - h_o = -(w^1, ..., w^b)(\theta - \theta_o)$. Hence, the space $\overline{\mathcal{W}}$ is the same as the linear combination of itself, and we can always replace $h - h_o$ by $-w(\theta - \theta_o)$ in equation (4.3) to obtain:

$$\rho_{\alpha_o}[Z, \alpha - \alpha_o] = (\frac{d\rho(Z, \alpha_o)}{d\theta'} - \frac{d\rho(Z, \alpha_o)}{dh'}[w])(\theta - \theta_o),$$

where $\frac{d\rho(Z,\alpha_o)}{dh'}[w] = (\frac{d\rho(Z,\alpha_o)}{dh'}[w^1], ..., \frac{d\rho(Z,\alpha_o)}{dh'}[w^b])$.

Let $D_w(X)$ denote the $r \times b-$matrix valued function:

$$D_w(X) \equiv E\left[\frac{d\rho(Z, \alpha_o)}{d\theta'} - \frac{d\rho(Z, \alpha_o)}{dh'}[w]|X\right]. \tag{4.4}$$

Then we have

$$m_{\alpha_o}[X, \alpha - \alpha_o] \equiv E\{\rho_{\alpha_o}[Z, \alpha - \alpha_o]|X\} = D_w(X)(\theta - \theta_o)$$

and the Fisher-like norm:

$$\|\alpha - \alpha_o\|^2 = (\theta - \theta_o)'E\left[D_w(X)'\Sigma(X)^{-1}D_w(X)\right](\theta - \theta_o). \tag{4.5}$$

Note that $E[D_w(X)'\Sigma(X)^{-1}D_w(X)]$ is quadratic in w. There exists a $w^* \equiv (w^{*1}, ..., w^{*b}) \in \prod_{j=1}^b \overline{\mathcal{W}}$ such that

$$E[D_w(X)'\Sigma(X)^{-1}D_w(X)] - E[D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)] \tag{4.6}$$

is positive semi-definite for any $w \equiv (w^1, ..., w^b) \in \prod_{j=1}^{b} \overline{\mathcal{W}}$. Then, for $f(\alpha) = \lambda'\theta$, we have

$$\sup_{\|\alpha-\alpha_o\|>0, \alpha\in\overline{\mathcal{A}}} \frac{|f(\alpha) - f(\alpha_o)|}{\|\alpha - \alpha_o\|} = \lambda' \left(E[D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)]\right)^{-1} \lambda. \tag{4.7}$$

Thus, $f(\alpha) = \lambda'\theta$ is bounded if and only if $E[D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)]$ is finite positive-definite. Below we formally state this condition.

Assumption 4.1. *$E[D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)]$ is finite and positive-definite.*

This condition ensures not only the boundedness of the linear functional $f(\alpha) = \lambda'\theta$, thereby the existence of its Riesz representor, but also the uniqueness of the true value θ_o. The identification part can be understood in much the same way as the identification of a nonlinear least squares estimator. $\widehat{\theta}_n - \theta_o$ can be thought of obtained from the weighted LS regression of $\rho(Z, \alpha_o)$ on $D_{w^*}(X)$ with $\Sigma(X)^{-1}$ as the weighting matrix. Thus, Assumption 4.1 is considered as a regularity condition.

Given Assumption 4.1, equations (4.5) and (4.7), we find the Riesz representor $v^* = (v^*_\theta, v^*_h) \in \Theta \times \overline{\mathcal{H}}$ for the bounded linear functional $f(\alpha) = \lambda'\theta$ as:

$$\begin{aligned} v^*_\theta &= \left(E\left[D_{w^*}(X)'[\Sigma(X)]^{-1}D_{w^*}(X)\right]\right)^{-1} \lambda \\ v^*_h &= -w^*(\cdot) \left(E\left[D_{w^*}(X)'[\Sigma(X)]^{-1}D_{w^*}(X)\right]\right)^{-1} \lambda. \end{aligned}$$

Hence

$$m_{\alpha_o}[X, v^*] = D_{w^*}(X) \left(E\left[D_{w^*}(X)'[\Sigma(X)]^{-1}D_{w^*}(X)\right]\right)^{-1} \lambda. \tag{4.8}$$

Assuming the linkage equation (4.1) holds, and substituting (4.8) into the general variance formulae (4.2), we obtain that $\sqrt{n}(\lambda'\widehat{\theta}_n - \lambda'\theta_o)$ is asymptotically normally distributed with mean zero and variance $\lambda' V^{-1}\lambda$, where

$$V^{-1} \equiv \begin{bmatrix} \left(E\{D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)\}\right)^{-1} \times \\ \left(E\{D_{w^*}(X)'\Sigma(X)^{-1}\Sigma_o(X)\Sigma(X)^{-1}D_{w^*}(X)\}\right) \times \\ \left(E\{D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)\}\right)^{-1} \end{bmatrix} \tag{4.9}$$

Since λ is arbitrary, $\widehat{\theta}_n$ is then $\sqrt{n}$ consistent and asymptotically normally distributed with covariance V^{-1}.

To ensure that the linkage equation (4.1) holds, we require that the SMD estimator converges under the Fisher-like metric sufficiently fast, and the linear approximation error term $r_\rho[Z, \alpha, \alpha_o]$ has sufficient curvature around α_o, where

$$r_\rho[Z, \alpha, \alpha_o] \equiv \rho(Z, \alpha) - \rho(Z, \alpha_o) - \rho_{\alpha_o}[Z, \alpha - \alpha_o].$$

In the following, we define $r_m[X, \alpha, \alpha_o] \equiv E\{r_\rho[Z, \alpha, \alpha_o]|X\}$. Let $\varepsilon_n = n^{-1/2}$, $u^* = \pm v^*$, and $\alpha^* = \alpha + \varepsilon_n u^*$. Define $\mathcal{N}_n = \{\alpha \in \mathcal{A}_n : \|\alpha - \alpha_o\| \leq o(\delta_n)\}$. We make the following sufficient conditions for (4.1):

Assumption 4.2. *There exist some positive constants c_1 and c_2 such that $c_1 \leq \lambda_{\min}(\Sigma_o(X)) \leq \lambda_{\max}(\Sigma_o(X)) \leq c_2$ for all $X \in \mathcal{X}$.*

Assumption 4.3. *(i) For some positive value c, $\lambda_{\max}(Var\{\rho_{\alpha_o}[Z, \Pi_n v^*]|X\}) \leq c$ for all X and n; (ii) there exists a π_n such that $m_{\alpha_o}[X, \Pi_n v^*] = p^{k_n}(X)'\pi_n + o(\delta_n)$ holds for all X and n.*

Assumption 4.4. *(i) $\rho_{\alpha_o}[Z, \alpha - \alpha_o]$ satisfies the Lipschitz (with coefficient ξ_{2n}), envelope (with coefficient ξ_{3n}) and conditional smoothness conditions over $\alpha \in \mathcal{A}_n$; (ii) $\frac{\rho_{\alpha_o}[Z, \alpha - \alpha_o]}{\delta_n}$ satisfies the envelope (with coefficient ξ_{4n}) condition over $\alpha \in \mathcal{N}_n$.*

Assumption 4.5. *(i) $\frac{r_\rho[Z, \alpha, \alpha_o]}{\delta_n^2}$ satisfies the envelope (with coefficient $\xi_{5n} = o\left(\delta_n^{-1}\right)$) condition over $\alpha \in \mathcal{N}_n$; (ii) there exist a positive sequence $\xi_{6n} = O(\xi_{4n})$ and a measurable function $C(Z)$ with $E\{[C(Z)]^2|X\} < +\infty$ such that for all $\alpha \in \mathcal{N}_n$, $|r_\rho[Z, \alpha, \alpha_o] - r_\rho[Z, \Pi_n \alpha^*, \alpha_o]| \leq \varepsilon_n \delta_n \xi_{6n} C(Z)$.*

Assumption 4.2 is familiar in the microeconometric literature. It requires that the conditional variance-covariance matrix is bounded from above and below in a positive semidefinite matrix sense. Assumptions 4.4-4.5 are further restrictions on the local curvature of $\rho(\cdot)$. Assumption 4.4(i) is very similar to Assumption 3.6. Assumptions 4.4(ii) and 4.5(i) are respectively similar to the Taylor expansion up to the first and second order terms, except that the coefficients ξ_{4n} and ξ_{5n} may grow with sample size. The conditions $\xi_{4n} = O(\delta_n^{-1/2})$ (Assumption 3.7(iii)) and $\xi_{5n} = o\left(\delta_n^{-1}\right)$ (Assumption 4.5(i)) imply that in an infinite-dimensional parameter space, the first order term in a Taylor expansion is at the order δ_n^ϵ for some $0 < \epsilon \leq 1$, while the second order term is at the order $\delta_n^{1+\epsilon}$ for some $0 < \epsilon \leq 1$. Assumptions 4.5(ii) is very mild given the definitions of $r_\rho[Z, \alpha, \alpha_o]$ and $\Pi_n \alpha^* = \Pi_n(\alpha + \varepsilon_n u^*)$ for any $\alpha \in \mathcal{N}_n$.

Theorem 4.1. *Under Assumptions 3.1-3.10 and 4.1-4.5, $\sqrt{n}(\widehat{\theta}_n - \theta_o) \Longrightarrow N(0, V^{-1})$ with V^{-1} given in (4.9).*

Remark 4.1. (i) Theorem 4.1 shows that the proposed SMD estimator for the parametric component (θ_o) attains the parametric rates. It provides an asymptotic distribution for the parametric component, which is needed for statistical inference. The asymptotic covariance has three terms. If we can interpret $D_{w^*}(X)$ as regressors in a system of equations with $\rho(Z, \alpha_o)$ as dependent variables, then V^{-1} has an interpretation of the covariance of the weighted LS estimator. The three terms in V^{-1} correspond to the three terms in the covariance of the (incorrectly) weighted LS estimator in the presence of heteroskedasticity.

(ii) In some applications such as Robinson's partial linear regression and Ichimura's single index model, the nonparametric component (h_o) can be concentrated out analytically. Although the proposed SMD procedure and Theorems 3.1 and 4.1 are still applicable to these applications, it may be worth taking advantage of the concentrated out nonparametric component. Let $h^*(\cdot,\theta)$ denote the concentrated out nonparametric component. $h^*(\cdot,\theta)$ often depends on some unknown functions. Suppose that, for each fixed θ, $h^*(\cdot,\theta)$ is estimated by some nonparametric estimator $\widehat{h^*}(\cdot,\theta)$. The estimator for the parametric component (θ_o) is then obtained as a solution to

$$\widehat{\theta}_n : \min_{\theta\in\Theta} \sum_{i=1}^{n} \widehat{m}(X_i,\theta,\widehat{h^*}(\cdot,\theta))'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}(X_i,\theta,\widehat{h^*}(\cdot,\theta)). \tag{4.10}$$

Suppose that $\widehat{h^*}(\cdot,\theta)$ is differentiable with respect to θ and that $\widehat{h^*}(\cdot,\theta)$ and its derivatives with respect to θ converge to the true values in probability uniformly at rates faster than δ_n. Theorem 4.1 still can be proved using the techniques developed in Newey (1994), Andrews (1994) and Pakes and Olley (1995).

(iii) In Remark 3.1(iii), we noted that the SGLS estimator (3.3) can consistently estimate α_o in models where the directional derivatives $\rho_{\alpha_o}[Z,\alpha-\alpha_o]$ do not depend on the endogenous variables Y. Theorem 4.1 can be proved for such an estimator, see e.g. Chen and Shen (1998).

5. Covariance Estimator

To conduct statistical inference on the parametric component, a consistent covariance estimator is needed. In this section, we provide such an estimator by consistently estimating each component of V^{-1} given in (4.9). Consistent estimator of $\Sigma(X)$ already exists. The true conditional variance $\Sigma_o(X) = Var\left(\rho(Z,\alpha_o)|X\right)$ can be estimated by any nonparametric procedure such as the kernel method in our first version of the paper. Here we use the sieve regression of $\rho(Z_i,\widehat{\alpha}_n)\rho(Z_i,\widehat{\alpha}_n)'$ on $p^{k_n}(X_i)$. For any $X\in\mathcal{X}, \alpha\in\mathcal{A}_n$, we define

$$\Sigma_o(X,\alpha) \equiv E\left[\rho(Z,\alpha)\rho(Z,\alpha)'|X\right] \equiv \{\sigma_{ojl}(X,\alpha)\}_{j,l=1,\ldots,r}$$

and

$$\begin{aligned}\widehat{\Sigma}_o(X,\alpha) &\equiv \{\widehat{\sigma}_{ojl}(X,\alpha)\}_{j,l=1,\ldots,r}, \\ \widehat{\sigma}_{ojl}(X,\alpha) &\equiv \sum_{i=1}^{n}\rho_j(Z_i,\alpha)\rho_l(Z_i,\alpha)p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X). \end{aligned} \tag{5.1}$$

Then $\widehat{\Sigma}_o(X) = \widehat{\Sigma}_o(X,\widehat{\alpha}_n)$ is a consistent estimator of $\Sigma_o(X)$ under the following condition:

Assumption 5.1. *(i) $\rho(Z,\alpha)\rho(Z,\alpha)'-\rho(Z,\alpha_o)\rho(Z,\alpha_o)'$ satisfies the conditional smoothness condition (with coefficient $c_{1n}=\xi_{3n}/\xi_{0n}$) over $\alpha\in\mathcal{N}_n$; and (ii) $O(c_{1n}k_n^{-a_0}) = o(\delta_n)$.*

Lemma 5.1. *Under Assumptions 3.1-3.2, 3.7, 4.4-4.5 and 5.1, we have:* $\widehat{\Sigma}_o(X,\alpha) = \Sigma_o(X,\alpha) + o_p(\delta_n)$ *uniformly over* $X \in \mathcal{X}$, $\alpha \in \mathcal{N}_n$.

To estimate the $r \times b-$matrix valued function $D_{w^*}(X)$, we need to estimate the $q \times b-$matrix valued function $w^*(\cdot)$ first. Recall that $D_{w^j}(X)$ and $w^j(\cdot)$ denote the j-th columns of $D_w(X)$ and $w(\cdot)$ respectively. Then, $w^{*j}(\cdot)$, $j = 1, ..., b$ is the solution to:

$$\inf_{w^j(\cdot)\in\overline{\mathcal{W}}} E\left[D_{w^j}(X)'[\Sigma(X)]^{-1}D_{w^j}(X)\right].$$

The $w^{*j}(\cdot)$ can be estimated by applying a nonparametric least squares regression. Specifically, let θ_j denote the j-th element of θ and let

$$\widehat{D}_{w^j}(X) \equiv \sum_{i=1}^{n}\left(\frac{d\rho(Z_i,\widehat{\alpha}_n)}{d\theta_j} - \frac{d\rho(Z_i,\widehat{\alpha}_n)}{dh'}[w^j(\cdot)]\right)p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X)$$

denote a sieve estimator of $D_{w^j}(X)$. We estimate $w^{*j}(\cdot)$ by $\widehat{w^{*j}}(\cdot)$, the solution to

$$\min_{w^j(\cdot)\in\mathcal{H}_n}\sum_{i=1}^{n}\widehat{D}_{w^j}(X_i)'[\widehat{\Sigma}(X_i)]^{-1}\widehat{D}_{w^j}(X_i).$$

Let $\widehat{w^*}$ denote the $q \times b-$matrix whose j-th column is $\widehat{w^{*j}}(\cdot)$. Then, $D_{w^*}(X)$ is estimated by $\widehat{D}_{\widehat{w^*}}(X)$. Finally we denote $\widehat{V^{-1}}$ as the estimator of V^{-1} given in (4.9):

$$\widehat{V^{-1}} \equiv \begin{bmatrix} \left(\frac{1}{n}\sum_{i=1}^{n}\widehat{D}_{\widehat{w^*}}(X_i)'\widehat{\Sigma}(X_i)^{-1}\widehat{D}_{\widehat{w^*}}(X_i)\right)^{-1} \times \\ \left(\frac{1}{n}\sum_{i=1}^{n}\widehat{D}_{\widehat{w^*}}(X_i)'\widehat{\Sigma}(X_i)^{-1}\widehat{\Sigma}_o(X_i)\widehat{\Sigma}(X_i)^{-1}\widehat{D}_{\widehat{w^*}}(X_i)\right) \times \\ \left(\frac{1}{n}\sum_{i=1}^{n}\widehat{D}_{\widehat{w^*}}(X_i)'\widehat{\Sigma}(X_i)^{-1}\widehat{D}_{\widehat{w^*}}(X_i)\right)^{-1}. \end{bmatrix}$$

In the next assumption we denote $\|w^j(\cdot)\| = \|(0,0) - (0, w^j(\cdot))\|$.

Assumption 5.2. *There is a measurable function* $C(Z)$ *with* $E\{[C(Z)]^2|X\} < \infty$ *such that:* $|\frac{d\rho(Z,\alpha)}{d\theta_j} - \frac{d\rho(Z,\alpha_o)}{d\theta_j}| \leq C(Z)\xi_{4n}\delta_n$ *and* $|\frac{d\rho(Z,\alpha)}{dh'}[w^j(\cdot)] - \frac{d\rho(Z,\alpha_o)}{dh'}[w^j(\cdot)]| \leq C(Z)\xi_{4n}\delta_n$ *hold for all* $Z \in \mathcal{Z}$, $\alpha \in \mathcal{N}_n$, $w^j \in \overline{\mathcal{W}}$ *with* $\|w^j(\cdot)\| \leq c < \infty$, *and* $j = 1, ..., b$.

Theorem 5.1. *Under Assumptions 3.1-3.10, 4.1-4.2, and 5.1-5.2, we have:* $\widehat{V^{-1}} = V^{-1} + o_p(1)$.

Theorem 5.1 shows that the covariance estimator is consistent. It is worth noting that, for linear sieves, computing $\widehat{w^{*j}}(\cdot)$ does not require nonlinear optimization. A simple pooled and weighted LS regression of the derivative of ρ with respect to θ_j on the derivatives of ρ with respect to h is all we need. Thus, our covariance estimator is easy to compute.

6. Efficiency

Having established the asymptotic distribution of the parametric component, we now investigate the efficiency of $\widehat{\theta}_n$. Clearly, the efficiency depends on the choice of the weighting matrix $\Sigma(X)$. It is straightforward to show that the optimal weighting matrix in the sense of minimizing the asymptotic variance of the estimator $\widehat{\theta}_n$ is $\Sigma(X) = \Sigma_o(X) \equiv Var(\rho(Z, \alpha_o)|X)$. Thus, when $\Sigma(X)$ is set to $\Sigma_o(X)$, the corresponding estimator is the best in the class of all SMD estimators, and hence called the *optimally weighted SMD estimator*. The asymptotic covariance of the optimally weighted SMD estimator is V_o^{-1}, where

$$V_o = E\left[D_{w_o}(X)'[\Sigma_o(X)]^{-1}D_{w_o}(X)\right] = \inf_{w(\cdot)} E\left[D_w(X)'[\Sigma_o(X)]^{-1}D_w(X)\right]. \quad (6.1)$$

The issue now is whether the optimally weighted SMD estimator is also the best in the class of all $\sqrt{n}$ consistent and regular estimators for model (1.3). To address this issue, one would normally have to compute the semiparametric efficiency bound for model (1.3) and then compare the semiparametric efficiency bound with the asymptotic covariance matrix of the optimally weighted SMD estimator. Unfortunately, in our general setup (1.3), there is no analytical expression of the semiparametric efficiency bound, see Chamberlain (1992) for examples. Fortunately, we do not need the explicit expression of the bound and are still able to present the following efficiency result:

Theorem 6.1. *Suppose that Assumptions 3.1-3.10 and 4.1-4.5 hold with* $\Sigma(X) = \Sigma_o(X)$. *Then the optimally weighted SMD estimator for the parametric component in model* (1.3) *attains the semiparametric efficiency bound* V_o *given in (6.1).*

Theorem 6.1 presents two results. First, the optimally weighted SMD estimator is efficient. Second, the V_o defined in (6.1) is the semiparametric efficiency bound for model (1.3). The first result provides an efficient estimator without actually computing the efficiency bound. This result is very useful since computing efficiency bound is often difficult. The second result extends Chamberlain's (1992) efficiency bound result to the much more general model (1.3).

It is known that the generalized LS estimator is efficient for a parametric regression model when the variance of the error term is used as the weighting matrix. In a more general parametric conditional moment restrictions model (1.1), the GMM estimator is efficient when the moment functions are weighted by the covariance matrix of the moment functions (Newey 1990, 93). In a semiparametric conditional moment restrictions setting (1.3), one would expect the same efficiency result to hold when the covariance of the moment functions is used as the weighting matrix. Theorem 6.1 simply confirms this intuition.

To obtain a feasible efficient SMD estimator, one can implement the following simple three-step procedure:

Step 1. Obtain an initial consistent SMD estimator $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ by

$$\min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \widehat{m}(X_i,\alpha)'\widehat{m}(X_i,\alpha).$$

Step 2. Compute the optimal weighting matrix $\widehat{\Sigma}_o(X) = \widehat{\Sigma}_o(X, \widehat{\alpha}_n)$ by (5.1).

Step 3. Obtain the optimally weighted SMD estimator $\widetilde{\alpha}_n = (\widetilde{\theta}_n, \widetilde{h}_n)$ by

$$\min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \widehat{m}(X_i,\alpha)' \left(\widehat{\Sigma}_o(X_i)\right)^{-1} \widehat{m}(X_i,\alpha).$$

By Theorem 3.1, the initial estimator $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ in Step 1 converges to $\alpha_o = (\theta_o, h_o)$ at a rate faster than δ_n. By Lemma 5.1, $\widehat{\Sigma}_o(X)$ in Step 2 converges to $\Sigma_o(X)$ uniformly over $X \in \mathcal{X}$ at a rate faster than δ_n. The following theorem now follows directly from Theorem 4.1.

Theorem 6.2. *Under Assumptions 3.1-3.3, 3.5-3.10, 4.1-4.5 with $\Sigma(X) = \Sigma_o(X)$ and 5.1, the three-step SMD estimator satisfies $\sqrt{n}(\widetilde{\theta}_n - \theta_o) \Longrightarrow N(0, V_o^{-1})$.*

Remark 6.1. **(i)** In the three-step SMD estimation procedure, one can repeat Step 2 using $\widetilde{\alpha}_n$ to re-estimate the weighting matrix, and then Step 3 to obtain a new estimator. The new estimator for θ_o is asymptotically as efficient as $\widetilde{\theta}_n$. Thus, there is no first-order asymptotic efficiency gain by repeating Step 2 and Step 3. This remains true even if Step 2 and Step 3 are repeated many times. It is possible, however, that the new estimator has better finite sample properties.

(ii) Hansen, Heaton and Yaron (1996) and Newey and Smith (2000) have shown that for the parametric unconditional moment models (1.2), continuously updated GMM procedures have better finite sample properties. One might expect the same result holds for the following continuously updated SMD procedure:

$$\min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \widehat{m}(X_i,\alpha)' \left(\widehat{\Sigma}_o(X_i,\alpha)\right)^{-1} \widehat{m}(X_i,\alpha), \tag{6.2}$$

where $\widehat{m}(X_i,\alpha)$ and $\widehat{\Sigma}_o(X_i,\alpha)$ are given by (2.3) and (5.1) respectively. We shall investigate this in another paper.

(iii) When the directional derivative $\rho_{\alpha_o}[Z, \alpha - \alpha_o]$ does not depend on the endogenous variables Y, the following 3-step SGLS procedure can be used:

(1). Obtain an initial consistent SGLS estimator $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ by

$$\min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \rho(Z_i,\alpha)'\rho(Z_i,\alpha).$$

(2). Compute the optimal weighting matrix $\widehat{\Sigma}_o(X) = \widehat{\Sigma}_o(X, \widehat{\alpha}_n)$ by (5.1).

(3). Obtain the optimally weighted SGLS estimator $\widetilde{\alpha}_n = (\widetilde{\theta}_n, \widetilde{h}_n)$ by

$$\min_{\alpha=(\theta,h)\in\Theta\times\mathcal{H}_n} \frac{1}{n}\sum_{i=1}^{n} \rho(Z_i,\alpha)' \left(\widehat{\Sigma}_o(X_i)\right)^{-1} \rho(Z_i,\alpha).$$

To illustrate our efficiency result, consider Klein and Spady's (1993) binary response model: $Y = 1\{\delta(X,\theta_o) \geq U\}$, where the conditional distribution of error term U given regressor X is a unknown function of $\delta(X,\theta_o)$, denoted as $h_o(\delta(X,\theta_o))$. Clearly this model fits into our class (1.4) with $\rho(Z,\theta,h) = Y - h(\delta(X,\theta))$. The directional derivative $\rho_{\alpha_o}[Z,\alpha-\alpha_o] = -\nabla h(\delta(X,\theta_o))\frac{\partial\delta(X,\theta_o)}{\partial\theta'}(\theta-\theta_o) - [h(\delta(X,\theta_o)) - h_o(\delta(X,\theta_o))]$ does not depend on Y. Hence we apply the simple 3-step SGLS procedure to obtain an $\sqrt{n}$ efficient estimator $\widetilde{\theta}_n$ of θ_o, with asymptotic variance V_o^{-1}:

$$\begin{aligned} V_o &= \inf_w E\left\{D_w(X)'\Sigma_o(X)^{-1}D_w(X)\right\}, \qquad (6.3)\\ D_w(X) &= \nabla h_o(\delta(X,\theta_o))\frac{\partial\delta(X,\theta_o)}{\partial\theta'} - w(\delta(X,\theta_o)), \end{aligned}$$

where $\Sigma_o(X) = h_o(\delta(X,\theta_o))[1 - h_o(\delta(X,\theta_o))]$. We can solve the minimization problem in (6.3) analytically with $w^* = \nabla h(\delta(X,\theta_o))E[\frac{\partial\delta(X,\theta_o)}{\partial\theta'}|\delta(X,\theta_o)]$. Hence

$$V_o = E\left[\frac{\{\nabla h_o(\delta(X,\theta_o))\}^2 Var\left(\frac{\partial\delta(X,\theta_o)}{\partial\theta}|\delta(X,\theta_o)\right)}{\Sigma_o(X)}\right],$$

which coincides with the asymptotic covariance matrix of Klein and Spady's (1993) efficient estimator of θ_o.

As another example, consider the partially additive IV regression of Chamberlain (1992, model (3.5), page 579)): $E[Y_1 - Y_2'\theta_o - \sum_{j=1}^q h_{oj}(X_j)|X_0, X_1, ..., X_q] = 0$. The directional derivative $\rho_{\alpha_o}[Z,\alpha-\alpha_o] = -Y_2'(\theta-\theta_o) - \sum_{j=1}^q [h_j(X_j) - h_{oj}(X_j)]$ depends on Y. Hence we apply our general 3-step SMD procedure to obtain an $\sqrt{n}$ efficient estimator $\widetilde{\theta}_n$ of θ_o, with asymptotic variance V_o^{-1}:

$$V_o = \inf_{w_j\in\overline{\mathcal{H}}^j - h_{oj}} E\left\{\{E[Y_2'|X] - \sum_{j=1}^q w_j(X_j)\}'\Sigma_o(X)^{-1}\{E[Y_2'|X] - \sum_{j=1}^q w_j(X_j)\}\right\}. \qquad (6.4)$$

Notice that when $Y_2 = X_0$, this model reduces to the popular partially additive regression $E[Y - X_0'\theta_o - \sum_{j=1}^q h_{oj}(X_j)|X] = 0$. The directional derivative $\rho_{\alpha_o}[Z,\alpha-\alpha_o]$ is now independent of Y. Hence we apply the simple 3-step SGLS procedure to obtain an $\sqrt{n}$ efficient estimator $\widetilde{\theta}_n$ of θ_o, with asymptotic variance V_o^{-1} given in (6.4) except now $E[Y_2|X] = X_0$. We note that even for these simple partially additive models, the minimization problem in (6.4) can not be solved analytically. Nevertheless, our method leads to an efficient estimator $\widetilde{\theta}_n$ (of θ_o) and a consistent estimator of its asymptotic variance V_o^{-1}.

7. Applications

7.1. Convergence rates of nonparametric components

In Section 3, we establish that $\widehat{\alpha}_n$ converges to α_o at a rate $o_p(\delta_n)$ under the Fisher-like metric. However in some applications, applied researchers may be interested in the convergence rates of the nonparametric part $\widehat{h}_n$ to h_o under the usual mean-squared-error and the sup-norm metric. In the following, we provide a formula to compute the convergence rates under these metrics.

For any $\alpha_1, \alpha_2 \in \mathcal{A}$, we define the mean-squared-error metric as:

$$\begin{aligned} \|\alpha_1 - \alpha_2\|_2^2 &\equiv ||\theta_1 - \theta_2||_E^2 + \|h_1 - h_2\|_2^2 \\ \|h_1 - h_2\|_2^2 &= \sum_{j=1}^{q} E\{[h_{1j}(\bullet) - h_{2j}(\bullet)]^2\}, \end{aligned}$$

and define the (weighted) sup-norm metric as:

$$\begin{aligned} \|\alpha_1 - \alpha_2\|_{\infty,\omega} &\equiv ||\theta_1 - \theta_2||_E + \|h_1 - h_2\|_{\infty,\omega} \\ \|h_1 - h_2\|_{\infty,\omega} &= \sum_{j=1}^{q} \|h_{1j} - h_{2j}\|_{\infty,\omega_j} = \sum_{j=1}^{q} \sup_{\bullet}\{|[h_{1j}(\bullet) - h_{2j}(\bullet)]\omega_j(\bullet)|\} \end{aligned}$$

where $\omega_j(\cdot), j = 1, ..., q$ denote some weighting functions. The weighting function $\omega_j(\bullet)$ is needed when the density of the argument $\bullet$ of h_j is not bounded below from zero[7] and/or not bounded above from infinity[8].

For any $\alpha_1 = (\theta_1', h_1')'$, $\alpha_2 = (\theta_2', h_2')' \in \mathcal{A}$, we define:

$$\|\theta_1 - \theta_2\|^2 \equiv (\theta_1 - \theta_2)' E\left(E\{\frac{d\rho(Z,\alpha_o)}{d\theta'}|X\}' E\{\frac{d\rho(Z,\alpha_o)}{d\theta'}|X\}\right)(\theta_1 - \theta_2),$$

$$||h_1 - h_2||^2 \equiv E\left(\left\|E\{\frac{d\rho(Z,\alpha_o)}{dh'}[h_1(\cdot) - h_2(\cdot)]|X\}\right\|_E^2\right).$$

Before we compute the convergence rates, we make the following assumptions:

Assumption 7.1. *For any $h \in \mathcal{H}$, there exists $\Pi_n h \in \mathcal{H}_n$ such that $\|h - \Pi_n h\|_{\infty,\omega} = o(\delta_n)$.*

Assumption 7.2. *(i) $E\left(E\{\frac{d\rho(Z,\alpha_o)}{d\theta'}|X\}' E\{\frac{d\rho(Z,\alpha_o)}{d\theta'}|X\}\right)$ is finite positive-definite; (ii) $E\left(\left\|E\{\frac{d\rho(Z,\alpha_o)}{dh'}[u(\cdot)]|X\}\right\|_E^2\right) \leq c < \infty$ for any direction $u(\cdot) \in \mathcal{H}$ with $\|u\|_{\infty,\omega} = 1$.*

Assumption 7.3. *(i) For all $j = 1, ..., q$, the support of the argument $\cdot$ of $h_{oj}(\cdot)$ does not depend on $\theta_o \in \Theta$; (ii) $E[\{\omega_j(\bullet)\}^{-2}] < \infty$ for $j = 1, ..., q$.*

[7] This clearly happens when the argument $\bullet$ of h_j has unbounded support.

[8] This could happen when the models involve solving partial differential equations such as some diffusion models in finance.

It is easy to see that under Assumption 7.2(ii), we have $\|h_1 - h_2\| \leq c \|h_1 - h_2\|_{\infty,\omega}$ over $\mathcal{H}$; while under Assumption 7.3, we have $\|h_1 - h_2\|_2 \leq c \|h_1 - h_2\|_{\infty,\omega}$ over $\mathcal{H}$. Notice that the Fisher-like metric is generally weaker than the root mean-squared-error metric except that in few cases the two become equivalent. Under the above assumptions, we can define:

$$\tau_{\infty n} \equiv \sup_{h \in \mathcal{H}_n : h \neq \Pi_n h_o} \frac{\|h - \Pi_n h_o\|_{\infty,\omega}}{\|h - \Pi_n h_o\|} < \infty \text{ for each } n \tag{7.1}$$

and

$$\tau_{2n} \equiv \sup_{h \in \mathcal{H}_n : h \neq \Pi_n h_o} \frac{\|h - \Pi_n h_o\|_2}{\|h - \Pi_n h_o\|} \leq \tau_{\infty n} \text{ for each } n. \tag{7.2}$$

These and Theorem 3.1 immediately lead to the following result.

Theorem 7.1. *Suppose that Assumptions 3.1-3.7, 3.9-3.10, 4.1, 7.1 and 7.2 hold. Then: (i)* $\max\{||\widehat{\theta}_n - \theta_o||, ||\widehat{h}_n - h_o||\} = o_p(\delta_n)$*, and (ii)* $||\widehat{h}_n - h_o||_{\infty,\omega} = o_p(\tau_{\infty n}\delta_n)$*. (iii) If further, Assumption 7.3 holds, then* $||\widehat{h}_n - h_o||_2 = o_p(\tau_{2n}\delta_n)$*.*

To compute the convergence rates under the sup-norm metric and the mean-squared-error metric, one must compute the constants $\tau_{\infty n}$ and τ_{2n}. We now demonstrate how to compute these constants when linear sieves are used. Given Assumption 7.1, we can without loss of generality assume that the parameter space is given by $\mathcal{A} = \Theta \times \mathcal{H} = \{\alpha = (\theta', h')' : ||\alpha||_{com} < \infty\}$, where $||\alpha||_{com} = ||\theta||_E + \sum_{j=1}^q ||h_j||_{com}$ is some norm which is relatively compact with respect to the $||\alpha||_{\infty,\omega} = ||\theta||_E + \sum_{j=1}^q ||h_j||_{\infty,\omega_j}$-norm, (i.e., for any $\alpha \in \mathcal{A}$ with $||\alpha||_{com} \leq const.$, there exists a sequence $\{\alpha_n\}$ such that $||\alpha_n||_{com} \leq const.$ and $||\alpha - \alpha_n||_{\infty,\omega} \to 0$ as $n \to \infty$). Often $||h_j||_{com}$ is some norm involves higher order derivatives such as Hölder norm or Sobolev norm. Define the sieve space $\mathcal{A}_n = \Theta \times \mathcal{H}_n$ with $\mathcal{H}_n \equiv \mathcal{H}_n^1 \times ... \times \mathcal{H}_n^q$

$$\mathcal{H}_n^j = \left\{ h_{jn}(\cdot) = p^{k_{jn}}(\cdot)'\beta^{k_{jn}} : ||h_{jn}||_{com} \leq const. \right\}, \ j = 1, ..., q$$

where $p^{k_{jn}}(\cdot) = (p_{j1}(\cdot), ..., p_{jk_{jn}}(\cdot))'$ denote some base functions, $\beta^{k_{jn}} = (\beta_{j1}, ..., \beta_{jk_{jn}})'$ are sieve coefficients to be estimated. Also we denote $\Pi_n h_{oj} = p^{k_{jn}}(\cdot)'\beta^{ok_{jn}} \in \mathcal{H}_n^j$. We require the sieve base $p^{k_{jn}}(\cdot)$ to satisfy: (i) for any $h \in \mathcal{H}$, there exists $\Pi_n h \in \mathcal{H}_n$ such that $||h - \Pi_n h||_{\infty,\omega} = o(1)$; (ii) the frame property, i.e., norm of the function $h_j \in \mathcal{H}^j$ can be expressed in terms of the corresponding norm of the coefficients $\{\beta_{jk} : k = 1, 2, ...\}$. Examples of frame bases include spline, wavelets etc. Then we have for any $h \in \mathcal{H}_n$:

$$\|h - \Pi_n h_o\|_{\infty,\omega} \leq \sum_{j=1}^q \{\sup \left\| p^{k_{jn}}(\cdot)\omega_j(\cdot) \right\|_E\} ||\beta^{k_{jn}} - \beta^{ok_{jn}}||_E$$

$$\|h - \Pi_n h_o\|_2^2 = \sum_{j=1}^q (\beta^{k_{jn}} - \beta^{ok_{jn}})' E[p^{k_{jn}}(\cdot)p^{k_{jn}}(\cdot)'](\beta^{k_{jn}} - \beta^{ok_{jn}})$$

$$\|h-\Pi_n h_o\|^2 = E\left(\left\|\sum_{j=1}^{q} E\{\frac{d\rho(Z,\alpha_o)}{dh_j}[h_j(\cdot)-\Pi_n h_{oj}(\cdot)]|X\}\right\|_E^2\right)$$

For $j=1,2,...,q$, we denote

$$\frac{d\rho(Z,\alpha_o)}{dh_j}[p^{k_{jn}}(\cdot)'] = \left(\frac{d\rho(Z,\alpha_o)}{dh_j}[p_{j1}(\cdot)],...,\frac{d\rho(Z,\alpha_o)}{dh_j}[p_{jk_{jn}}(\cdot)]\right),$$

$$B_n(X) = E\left\{\left(\frac{d\rho(Z,\alpha_o)}{dh_1}[p^{k_{1n}}(\cdot)'],...,\frac{d\rho(Z,\alpha_o)}{dh_q}[p^{k_{qn}}(\cdot)']\right) \mid X\right\};$$

and $B_n = E\{B_n(X)'B_n(X)\}$. Then:

$$\|h-\Pi_n h_o\|^2 = (\beta^{k_{1n}\prime}-\beta^{ok_{1n}\prime},...,\beta^{k_{qn}\prime}-\beta^{ok_{qn}\prime})B_n(\beta^{k_{1n}\prime}-\beta^{ok_{1n}\prime},...,\beta^{k_{qn}\prime}-\beta^{ok_{qn}\prime})'.$$

Hence we have the following crude upper bounds on $\tau_{\infty n}$ and τ_{2n}:

$$\tau_{\infty n} \leq \sqrt{\frac{\max_{j=1}^{q}\{\sup.\,|p^{k_{jn}}(\cdot)'p^{k_{jn}}(\cdot)[\omega_j(\cdot)]^2|\}}{\lambda_{\min}(E\{B_n(X)'B_n(X)\})}} \tag{7.3}$$

and

$$\tau_{2n} \leq \sqrt{\frac{\max_{j=1,...q}\{\lambda_{\max}\left(E[p^{k_{jn}}(\cdot)p^{k_{jn}}(\cdot)']\right)\}}{\lambda_{\min}(E\{B_n(X)'B_n(X)\})}}. \tag{7.4}$$

7.2. Example

In this section we illustrate the SMD procedure with a detailed application to a partially linear nonparametric IV regression with heteroskedastic conditional variance of unknown form:

$$\begin{aligned} Y_{1i} &= X_{1i}'\theta_o + h_o(Y_{2i}) + \varepsilon_i, \\ E[\varepsilon_i|X_{1i},X_{2i}] &= 0, \quad E[\varepsilon_i^2|X_{1i},X_{2i}] = \Sigma_o(X_{1i},X_{2i}) > 0 \end{aligned}$$

where $\alpha_o = (\theta_o, h_o)$, $Z = (Y, X_z)$, $Y = (Y_1, Y_2')'$, $X_z = X_1$ and $X = (X_1', X_2')'$. Without loss of generality, we assume that $\dim(X_1) = b = \dim(\theta)$, $\dim(X_2) = d$ and $\dim(X) = s = b + d$, also $\dim(Y_1) = 1$, $\dim(Y_2) = d$. It is easy to see that $E[\rho(Z,\alpha_o)|X] = 0$ with $\rho(Z,\alpha) = Y_1 - X_1'\theta - h(Y_2)$. Notice that this model is an instrumental variable alternative to the empirical example studied by Newey, Powell and Vella (1999, page 584, equation (7.1)). To our knowledge there is no published results on $\sqrt{n}$ consistent and efficient estimation of this model.

We use the tensor-product sieves to approximate multivariate unknown functions $E[\rho(Z,\alpha)|X]$ and $h(Y_2)$. A tensor-product sieve is simply the product of univariate sieves. For example let $\{\phi_i\}_{i=0}^{\infty}$ denote spline (Fourier series or power series) basis functions for $L_2([0,1])$ (the space of square-integrable univariate functions), then the tensor product $\{\phi_{\overrightarrow{i}}(x) = \prod_{l=1}^{s}\phi_{i_l}(x_l) : \overrightarrow{i} = (i_1,...,i_l,...,i_s) \in$

$\{0,1,...,\}^s\}$ is a spline (or Fourier series or power series) basis function for $L_2([0,1]^s)$ (the space of square-integrable s-variate functions). In this example, we use the tensor-product B-spline sieve $\{p_{0k}(x) = \prod_{l=1}^{s} \phi_{k_l}(x_l) : k = 1,...,k_n\}$ to approximate the unknown conditional mean function $E[\rho(Z,\alpha)|X]$, where $\{\phi_{k_l}(x_l) : k_l = 1,...,J_n+\gamma_0+1\}$ are univariate B-spline basis functions of order (γ_0+1) over the support $\mathcal{X}_l$, and x_l denotes the l-th component of x, $l = 1,...,s$. The smoothing parameter k_n is given by $k_n = (J_n+\gamma_0+1)^s$, see Schumaker (1981) for more details about B-splines.

We assume the unknown functions belong to some Hölder spaces. Let $\mathcal{S} \subseteq \mathcal{R}^s$. For any $\overrightarrow{i} = (i_1,.....,i_s) \in \{0,1,...,\}^s$, we write $|\overrightarrow{i}| = \sum_{j=1}^{s} i_j$, and for any $x = (x_1,...,x_s)' \in \mathcal{S}$, we denote the $|\overrightarrow{i}|$ -th derivative of $g()$ as:

$$\nabla^{\overrightarrow{i}} g(x) = \frac{\partial^{|\overrightarrow{i}|}}{\partial x_1^{i_1} \cdot ... \cdot \partial x_s^{i_s}} g(x).$$

Let $\Lambda^\gamma(\mathcal{S})$ denote a Hölder space with (total) smoothness $\gamma \equiv [\gamma-1]+\varsigma$, where $[\gamma-1]$ denotes the integer part of $\gamma-1 \geq 0$ and $\varsigma \in (0,1]$ is the Hölder exponent. $\Lambda^\gamma(\mathcal{S})$ is a space of functions with up to $[\gamma-1]$-th order continuous derivatives, and the highest derivatives are Hölder-Lipschitz continuous with the Hölder exponent $\varsigma \in (0,1]$:

$$\Lambda^\gamma(\mathcal{S}) = \left\{ g \in C^{[\gamma-1]}(\mathcal{S}) : \max_{|\overrightarrow{i}|=[\gamma-1]} \sup_{x \neq \widetilde{x}} \frac{\left\| \nabla^{\overrightarrow{i}} g(x) - \nabla^{\overrightarrow{i}} g(\widetilde{x}) \right\|_E}{(\|x-\widetilde{x}\|_E)^\varsigma} < \infty \right\}.$$

The Hölder space becomes a Banach space when endowed with the Hölder norm:

$$||g||_{\Lambda^\gamma} = \sup_x |g(x)| + \max_{|\overrightarrow{i}|=[\gamma-1]} \sup_{x \neq \widetilde{x}} \frac{\left\| \nabla^{\overrightarrow{i}} g(x) - \nabla^{\overrightarrow{i}} g(\widetilde{x}) \right\|_E}{(\|x-\widetilde{x}\|_E)^\varsigma} < \infty.$$

We call $\Lambda_c^\gamma(\mathcal{S}) \equiv \{g \in \Lambda^\gamma(\mathcal{S}) : ||g||_{\Lambda^\gamma} \leq c < \infty\}$ a Hölder ball (with radius c).

For convenience, we restate Assumptions 3.1 and 3.2 as follows:

Assumption 3.1': *(i) The data $\{(Y_i, X_i)_{i=1}^n\}$ are i.i.d.; (ii) X has density bounded above and below over its compact support $\mathcal{X} = \mathcal{X}_1 \times ... \times \mathcal{X}_s$, and $\mathcal{X}$ has a piecewise-smooth boundary.*

Assumption 3.2': *Assumption 3.2 holds with $p^{k_n}(x) = (p_{01}(x),...,p_{0k_n}(x))'$ as the tensor product B-splines $p_{0k}(x) = \prod_{l=1}^{s} \phi_{k_l}(x_l)$, where $\{\phi_{k_l}(x_l) : k_l = 1,...,J_n+\gamma_0+1\}$ are the univariate B-spline basis functions of order (γ_0+1) over the support $\mathcal{X}_l$, $l = 1,...,s$, and $k_n = (J_n+\gamma_0+1)^s$.*

Assumptions 3.1'(ii) and 3.2' imply that $\xi_{0n} = \xi_{1n} = \sqrt{k_n}$, see e.g. Newey (1997).

The parameter space for the nonparametric component h_o is assumed to be the Hölder space $\mathcal{H} = \Lambda^\gamma(\mathcal{R}^d)$ with $\gamma > d/2$. Since the support of h_o can be

unbounded, we use the following univariate B-spline with order $\varpi > \gamma$:

$$B_{\varpi}(u) = \frac{1}{(\varpi - 1)!} \sum_{i=0}^{\varpi} (-1)^i \begin{pmatrix} \varpi \\ i \end{pmatrix} [\max(0, u - i)]^{\varpi - 1}. \tag{7.5}$$

Then $\{2^{k/2} B_{\varpi}(2^k Y_{2,l} - i) : i \in \mathbf{Z}, k = 0, ..., K_n\}$ forms a spline-wavelet sieve basis for functions belong to $\Lambda^{\gamma}(\mathcal{R})$. The tensor-product then forms a sieve basis for functions belong to $\mathcal{H} = \Lambda^{\gamma}(\mathcal{R}^d)$. We denote the tensor-product sieve basis as $p^{k_{1n}}(Y_2)$, a $k_{1n} \times 1$ -vector with $k_{1n} = (2^{K_n})^d$. Thus, the sieve space for the nonparametric component is:

$$\mathcal{H}_n = \{h \in \Lambda^{\gamma}(\mathcal{R}^d) : h(Y_2) = p^{k_{1n}}(Y_2)' \beta^{k_{1n}} \text{ with } ||h||_{\Lambda^{\gamma}} \leq \ln(1 + k_{1n})\}. \tag{7.6}$$

We estimate $\alpha_o = (\theta_o, h_o)$ by applying the SMD procedure (2.2) with $\mathcal{H}_n$ being the sieve (7.6). In the following we denote $f_{X,Y_2}(x, y_2)$ as the joint density of $(X', Y_2')'$, $f_X(x)$ and $f_{Y_2}(y_2)$ as the marginal densities of X and Y_2 respectively.

Proposition 7.1. *Suppose that Assumptions 3.1', 3.2' and the followings are satisfied:*

A7.1.1. (i) the conditional distribution of Y_2 given X is complete; (ii) $h_o \in \mathcal{H} = \Lambda^{\gamma}(\mathcal{R}^d)$, $\gamma > d/2$, and $E[h_o(Y_2)|X] \notin$ linear span (X_1); (iii) $E[X_1 X_1']$ is finite positive-definite.

A7.1.2. for each fixed y_2, $\frac{f_{X,Y_2}(\cdot, y_2)}{f_X(\cdot) f_{Y_2}(y_2)}$ belongs to $\Lambda^{\gamma_0}(\mathcal{X})$, $\gamma_0 > s/2$, and

$$\int_{\mathcal{R}^d} \left[\left\| \frac{f_{X,Y_2}(\cdot, y_2)}{f_X(\cdot) f_{Y_2}(y_2)} \right\|_{\Lambda^{\gamma_0}(\mathcal{X})} \right]^2 f_{Y_2}(y_2) dy_2 < \infty.$$

A7.1.3. $E[|Y_1|^p] < \infty$ for some $p \geq 4$; $E[|Y_1|^2|X] < \infty$

A7.1.4. $E[\{\omega(Y_2)\}^{-p}] < \infty$ with $\omega(y_2) = (1 + ||y_2||_E^2)^{-a/2}$ for some $a > \gamma > 0$ and $p \geq 4$; $E[\{\omega(Y_2)\}^{-2}|X] < \infty$

A7.1.5. $\max\left\{k_{1n}^{-\gamma/d}, \sqrt{\frac{k_{1n}}{n}}\right\} = o(n^{-1/4} \xi_{4n}^{-1/2})$.

A7.1.6. (i) $\frac{\sqrt{n}}{\xi_{4n} k_n k_{1n} [\ln(1+k_{1n})]^2 \ln n} \to +\infty$; (ii) $k_n^{-(\frac{\gamma_0}{s} + \frac{1}{2})} \ln(1+k_{1n}) = o(n^{-1/4} \xi_{4n}^{-1/2})$; (iii) $k_n \geq b + k_{1n}$.

A7.1.7. $\inf_w E\left[\{X_1' - E[w(Y_2)|X]\}'[\Sigma_o(X)]^{-1}\{X_1' - E[w(Y_2)|X]\}\right]$ *is finite positive-definite.*

Let $\widehat{\alpha}_n = (\widehat{\theta}_n, \widehat{h}_n)$ be the SMD estimator with identity weighting. Then: (1) $\|\widehat{\alpha}_n - \alpha_o\| = o_p(n^{-1/4} \xi_{4n}^{-1/2})$ with $\xi_{4n} = \sqrt{k_{1n}} \times \tau_{2n}$, $\tau_{2n} = k_{1n}^{\frac{\gamma_0}{s} + \frac{1}{2}} \ln(1 + k_{1n})$; $||\widehat{h} - h_o||_2 = o_p\left(n^{-1/4} \xi_{4n}^{-1/2} \tau_{2n}\right)$; $||\widehat{h} - h_o||_{\infty,\omega} = o_p\left(n^{-1/4} \xi_{4n}^{1/2}\right)$.

In addition, suppose the following holds:

A7.1.8. $E[\{Y_1 - X_1'\theta_o - h_o(Y_2)\}^2|X]$ is bounded from above and below from zero for all $X \in \mathcal{X}$.

Then: (2) the optimally weighted SMD estimator $\widetilde{\theta}_n$ satisfies $\sqrt{n}(\widetilde{\theta}_n - \theta_o) \Longrightarrow N(0, V_o^{-1})$, with

$$V_o = \inf_w E\left[\{X_1' - E[w(Y_2)|X]\}'[\Sigma_o(X)]^{-1}\{X_1' - E[w(Y_2)|X]\}\right] \tag{7.7}$$

where $w = (w^1, ..., w^b)$, w^j is a square measurable function of Y_2 and $w^j \in$ the completion of $\mathcal{H} - h_o$ for $j = 1, ..., b$.

Remark 7.1. **(i)** Condition A7.1.2 implies that there are orthonormal bases $\{\phi_{0k} : k = 1, ..., \infty\}$ (for $\Lambda^{\gamma_0}(\mathcal{X})$ and $L_2(\mathcal{X}, f_X)$), $\{\phi_{1k} : k = 1, ..., \infty\}$ (for $L_2(\mathcal{R}^d, f_{Y_2})$), and corresponding singular numbers $\{\mu_k\}$ with $\mu_1 = 1 \geq \mu_2 \geq \mu_3 \geq ...$ such that

$$\begin{aligned} \frac{f_{X,Y_2}(x, y_2)}{f_X(x) f_{Y_2}(y_2)} &= \sum_{k=1}^{\infty} \mu_k \phi_{0k}(x)\phi_{1k}(y_2), \\ c_1[k^{\frac{\gamma_0}{s}+\frac{1}{2}} \ln(1+k)]^{-1} &\leq \mu_k \leq c_2 k^{-(\frac{\gamma_0}{s}+\frac{1}{2})}, \end{aligned}$$

see e.g. Birman and Solomjak (1980, theorem 3.2 and corollaries 3.5 and 3.7).[9] Hence this condition implies that τ_{2n} defined in (7.2) satisfies $\tau_{2n} = [\mu_{k_{1n}}]^{-1} \leq k_{1n}^{\frac{\gamma_0}{s}+\frac{1}{2}} \ln(1 + k_{1n})$. There are many other sufficient conditions besides condition A7.1.2 that lead to the decay rate of the singular numbers $\{\mu_k\}$. In this example we impose A7.1.2 since it is logically consistent with our SMD estimation procedure. In particular, A7.1.2 implies that $\rho(Z, \alpha) = Y_1 - X_1'\theta - h(Y_2)$ satisfies Assumption 3.6's conditional smoothness with $k_n^{-a_0} = (k_n + 1)^{-(\frac{\gamma_0}{s}+\frac{1}{2})}$, see the proof in the Appendix E.

(ii) Notice that the parameter γ measures the smoothness of the true unknown function $h_o \in \Lambda^{\gamma}(\mathcal{R}^d)$, while the parameter γ_0 measures the smoothness of the $\frac{f_{X,Y_2}(\cdot,y_2)}{f_X(\cdot)f_{Y_2}(y_2)} \in \Lambda^{\gamma_0}(\mathcal{X})$ for each fixed y_2. The higher the $\gamma > 0$ is, the faster the nonparametric rates under Fisher-like metric, root mean squared metric and sup-norm metric are. Also, the higher the $\gamma_0 > 0$ is, the faster the nonparametric rate under the Fisher-like metric is, but the slower the nonparametric rates under the root-mean squared metric and sup-norm metric are.

(iii) Notice that by setting $k_{1n}^{-\gamma/d} = \sqrt{\frac{k_{1n}}{n}}$ in A7.1.5, we obtain $k_{1n} = O(n^{d/(2\gamma+d)})$, and hence Stone's (1982) optimal convergence rate $O_p(n^{-\gamma/(2\gamma+d)})$ under the root mean-squared-error metric if Y_2 were not endogenous (i.e. if $E[Y_1 - \{X_1'\theta_o + h_o(Y_2)\}|X_1, Y_2] = 0$) and $\gamma > d/2$. However since Y_2 is endogenous, if we still set $k_{1n} = O(n^{d/(2\gamma+d)})$, then in order to satisfy Conditions A7.1.5 and A7.1.6(i)(iii), we need $\frac{\gamma}{d} > \frac{5}{2} + \frac{\gamma_0}{s} > 3$, and Condition A7.1.6 suggests $k_n = (1 + k_{1n})^{1+\epsilon}$ for some $\epsilon \geq 0$. To satisfy A7.1.6(ii) we then need $\frac{\gamma}{d} - (\frac{5}{2} + \frac{\gamma_0}{s}) > \epsilon \geq \frac{2s}{2\gamma_0+s}$, which

[9] Although theorem 3.2 and corollaries 3.5 and 3.7 in Birman and Solomjak (1980) only stated $\mu_k \leq c_2 k^{-(\frac{\gamma_0}{s}+\frac{1}{2})}$, but given the fact that μ_k equals to the k-th approximation number, we also have $\mu_k \geq c_1[k^{\frac{\gamma_0}{s}+\frac{1}{2}} \ln(1+k)]^{-1}$.

will be satisfied when $\frac{\gamma}{d} > \frac{5}{2} + \frac{\gamma_0}{s} + \frac{2s}{2\gamma_0+s} > 4$. This means that h_o must be sufficiently smooth if one use the order $k_{1n} = O(n^{d/(2\gamma+d)})$. If h_o is not very smooth (e.g. $d/2 < \gamma \leq 4d$), it is not a good idea to use $k_{1n}^{-\gamma/d} = \sqrt{\frac{k_{1n}}{n}}$ to determine the order of k_{1n}, instead it is better to use $k_{1n}^{-\gamma/d} = o(\sqrt{\frac{k_{1n}}{n}})$ in A7.1.5 to determine the order of k_{1n}, and then A7.1.6 to determine the order of k_n. Although the sufficient conditions for Theorems 3.1 and 7.1 are far from the optimal ones, this example illustrates the difficulty due to the endogenous Y_2 as the argument of the unknown function $h_o()$.

8. Simulation

To illustrate the implementation of our proposed SMD procedure, and to evaluate the finite sample performance of the SMD estimator $\widehat{\alpha}_n$ (especially the parametric part $\widehat{\theta}_n$), we conduct a small scale simulation study.

The simulation is based on a partially linear nonparametric IV regression:

$$Y_1 = X_1\theta_o + h_o(Y_2) + U$$

with true values $\theta_o = 1$ and $h_o(Y_2) = \exp(Y_2)/(1+\exp(Y_2))$. We assume that Y_2 is generated according to:

$$Y_2 = X_1 + X_2 + R \times U + \varepsilon$$

with $R = 0.9$ or 0.1. The regressors X_1 and X_2 are independent and uniformly distributed over $[0,1]$. ε is independent of $X = (X_1, X_2)'$ and has a standard normal distribution. Conditional on X, U has a normal distribution with mean zero and variance $X_1^2 + X_2^2$. Thus, by design, the regressor Y_2 is endogenous and the error term U is heteroscedastic. Conditional on X, the covariance between Y_2 and U is $R(X_1^2 + X_2^2)$. Thus, $R = 0.9$ indicates high level of correlation while $R = 0.1$ indicates low level of correlation.

The proposed procedure requires approximating the nonparametric part $h_o(Y_2)$ and the conditional mean function $E[Y_1 - X_1\theta - h(Y_2)|X]$ by sieves. For the nonparametric part, since it is bounded between zero and one, we approximate it by a power series multiplied by a standard normal cumulative distribution function. The number of terms in the approximation, however, is not uniquely determined by the sufficient conditions in Sections 3 and 7. Those conditions only suggest that the number of terms should be in the order of $n^{1/8}$. For a sample size of $n = 1000$, $n^{1/8}$ is roughly 3. We decide to use 4. Hence, we approximate $h_o(Y_2)$ by

$$h_o(Y_2) \approx \Phi(Y_2)(\pi_o + \pi_1 Y_2 + \pi_2 Y_2^2 + \pi_3 Y_2^3)$$

where $\Phi(Y_2)$ denotes the standard normal cumulative distribution function. Note that the true value of π_o is 1. Substituting for $h_o(Y_2)$ and ignoring the approximation error term, we have

$$Y_1 = X_1\theta_o + \pi_0\Phi(Y_2) + \pi_1\Phi(Y_2)Y_2 + \pi_2\Phi(Y_2)Y_2^2 + \pi_3\Phi(Y_2)Y_2^3 + U.$$

Now treating the above equation as if it were the true model, a natural estimation method is the instrumental variable estimation method. This is exactly what our procedure proposes to do. Since the error term U is orthogonal to the regressors X, any nonlinear transformation of X can be used as valid instruments. We will use power series as instruments. It is known from the instrumental variables literature that the number of instruments must be at least 5, the number of unknown coefficients, but the exact number of instruments is undetermined. We decide to use an ad-hoc rule that the number of instruments is twice the number of unknown coefficients. This leads to the following set of instruments: 1, X_1, X_2, X_1^2, X_1X_2, X_2^2, X_1^3, $X_1^2X_2$, $X_1X_2^2$, X_2^3. This set of instruments is the sieve used to approximate the conditional mean function. This same set of instruments is also used to estimate the optimal weighting matrix.

A sample of 1000 data points were generated according to the above design. The proposed procedure was applied to the data twice: first with identity weighting matrix $\Sigma(X) = I$ (corresponding to inefficient estimator), then with optimal weighting matrix (corresponding to efficient estimator). The estimated coefficients were recorded. Then, a new sample of data were drawn and the estimated coefficients were computed again. This procedure was repeated 1000 times. The mean (MEAN), standard deviation (STD), mean absolute deviation (MAD), mean squared error (MSE), and 25, 50, and 75 percentile of the θ_o estimator are reported in the Table 1 under the heading: power series for each of the two correlation cases. To evaluate the performance of the proposed SMD estimator of the nonparametric component, we also compute and report the integrated mean squared error and the integrated mean absolute deviation. The integrated mean squared error (IMSE) is computed according to the following discrete expression:

$$0.01 \sum_{j=0}^{399} mean\{[(h_o(-2 + 0.01 \times j) - \widehat{h}(-2 + 0.01 \times j)]^2\}$$

where mean{.} denotes the average over all 1000 SMD estimators $\widehat{h}$. The integrated mean absolute deviation (IMAD) is computed according to:

$$0.01 \sum_{j=0}^{399} mean\{|h_o(-2 + 0.01 \times j) - \widehat{h}(-2 + 0.01 \times j)|\}.$$

The power series may not be a good sieve to approximate an unknown function with unbounded support since the power series has erratic tail behavior. To investigate whether the power series has produced spurious results, we replaced the polynomial with a spline. The spline we used is some constant over $(-\infty, -2)$, a third degree polynomial over $[-2, 2]$, and some other constant over $(2, +\infty)$. The spline was chosen so that it is continuous over the entire real line. Specifically, we used

$$h_o(Y_2) \approx \Phi(Y_2)(\pi_0 + [1\{-2 \leq Y_2 \leq 2\}(Y_2 - 4)(Y_2 + 2)^2 - 32 \times 1\{Y_2 > 2\}]\pi_1)$$

where $1\{\cdot\}$ is the indicator function. The set of instruments is the same as before. The results are reported under the heading: Spline.

Glancing at the Table 1, we have the following findings. First, both the inefficient and efficient SMD estimators of the parametric component θ_o are mean and median biased. But the mean biases are within the standard deviation of the true value. The biases, however, depend on the correlation parameter (R) between the endogenous regressor and the error term. When there is small correlation between Y_2 and U (i.e., when $R = 0.1$), both estimators are essentially unbiased. Second, the efficient SMD estimator of θ_o performs slightly better than the inefficient SMD estimator. The bias of the efficient SMD estimator is generally smaller than the bias of the inefficient SMD estimator, and the standard deviation of the efficient one is generally about 30 percent lower than the standard deviation of the inefficient one. Third, the choice of sieves (spline or power series) seems to have little impact on the performance of the SMD estimators of the parametric component. Fourth, the SMD estimator of the nonparametric component appears fitting the true function well. Here the choice of estimation method and choice of sieves seem to have big impact on the performance of the SMD estimator of h_o. For instance, the efficient estimation method reduces the integrated mean squared error by about 60 to 90 percent, whereas the spline sieve reduces it by about 80 percent.

To summarize, the proposed SMD procedure seems to produce good estimate for the parametric component θ_o. Here the choice of sieves does not seem to matter while the efficient method seems to perform slightly better (for the θ_o estimate). The procedure also produces a reasonably good estimate for the nonparametric component h_o. But the choice of sieves and the choice of estimation methods (inefficient or efficient) do matter (for the h_o estimate). If this small Monte Carlo study is indicative of what might be true generally, then splines and efficient estimation method should be used if the nonparametric component is the primary focus. Interestingly, similar findings are reported in Ai, Blundell and Chen (2001) for an empirical study of the shape-invariant semi-nonparametric Engel curve models with endogenous total expenditure.

9. Conclusion

In this paper, we propose a SMD estimator for the general semiparametric conditional moments restrictions model (1.3). We present consistency and convergence rates under various metrics for the SMD estimator of the nonparametric component. We also derive the $\sqrt{n}$ asymptotic normality of the SMD estimator of the parametric component and provide a consistent estimator of its asymptotic covariance matrix. We show that the optimally weighted SMD estimator of the parametric component can attain the semiparametric efficiency bound of model (1.3), and provide a three-step procedure which leads to a feasible efficient estimator. Finally we provide an example and a small scale Monte Carlo to show

that our SMD procedure has both theoretical and practical values.

Our theoretical results can be extended in several directions. First, the model (1.3) assumes all moment restrictions hold conditional on the same set of regressors. This assumption may not hold in some applications (see, Newey, Powell, and Vella (1999) for an example). Typically, in those applications different set of moment restrictions hold conditional on different set of regressors. Our procedure and results need to be extended to models of this sort. Second, although our results are sufficient for statistical inference on the parametric component, they are not sufficient for statistical inference on the nonparametric component. In some applications, we may be interested in testing the functional forms of $h_o(.)$ (e.g. completely unspecified $h_o(.)$ against the semiparametric index form, or the semiparametric index form against a parametric specification of $h_o(.)$). We may also be interested in testing the joint restrictions of both the parametric component and the nonparametric component. We are currently undertaking a project to develop more efficient tests for all sorts of restrictions of the model (1.3). Third, as we noted before, although our results on convergence rates and $\sqrt{n}$ asymptotic normality can be easily extended to weakly dependent time series data, the problem of semiparametric efficiency bound with time series data is non-trivial. We plan to address this issue by extending the work of Hansen and Richard (1987) in a separate paper.

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Mathematical Appendices

This appendices consist of several parts. In Appendix A, we present some uniform convergence results for the sieve estimators. In Appendix B, we prove the consistency of the proposed estimator $\widehat{\alpha}_n$ and compute the convergence rates. In Appendix C, we prove several lemmas which are then used to establish the asymptotic distribution of the estimator $\widehat{\theta}_n$. In Appendix D, we prove the consistency of the covariance estimator $\widehat{V}^{-1}$ and verify the efficiency property of the optimally weighted SMD estimator. In Appendix E, we apply our general theory to the example.

Appendix A: Useful Uniform Convergence Results

Recall that $\|\cdot\|$ denotes the Fisher-like metric on $\mathcal{A}$, $\overline{\mathcal{A}}$ denotes the linear completion of $\mathcal{A}$ under the metric $\|\cdot\|$, $\mathcal{A}_n$ denotes a sieve approximation of $\mathcal{A}$, $N(\mathcal{A}_n, \varepsilon)$ denotes the minimal number of ε-radius covering balls of $\mathcal{A}_n$ under $\|\cdot\|$, and $\|\cdot\|_E$ denotes the Euclidean norm. Also, recall the definitions of envelope and Lipschitz conditions, and constants ξ_{0n} and ξ_{1n} in Section 3. In the following lemma, $g : \mathcal{Z} \times \overline{\mathcal{A}} \to \mathcal{R}$ denotes a generic measurable function of the data $Z \in \mathcal{Z}$ and the parameter $\alpha \in \overline{\mathcal{A}}$. Define $\varepsilon(Z, \alpha) = g(Z, \alpha) - E[g(Z, \alpha)|X]$ and $\varepsilon(\alpha) = (\varepsilon(Z_1, \alpha), ..., \varepsilon(Z_n, \alpha))'$. Lemma A.1 extends Ai's (1997) Lemma B.1 on kernel uniform convergence result to that based on a sieve.

Lemma A.1: (A) *Suppose that Assumptions 3.1-3.2 and the following conditions are satisfied:*

(i) $g(Z, \alpha)$ satisfies the envelope condition with coefficient c_{1n} over $\mathcal{A}_n$;
(ii) $g(Z, \alpha)$ satisfies the Lipschitz condition with coefficient c_{2n} over $\mathcal{A}_n$;
(iii) there exists a sequence of constants δ_{1n} satisfying $\delta_{1n} = o(1)$ and

$$\frac{n\delta_{1n}^2}{\ln[(\frac{\xi_{1n}c_{1n}}{\delta_{1n}})^s N(\mathcal{A}_n, \frac{\delta_{1n}}{\xi_{0n}c_{2n}})] \times \max\{\xi_{0n}^2 c_{1n}^2, \xi_{0n}^{2+2/p}\delta_{1n}^{1-2/p}c_{1n}^{1+2/p}\}} \to +\infty.$$

Then uniformly over $(X, \alpha) \in \mathcal{X} \times \mathcal{A}_n$:

$$p^{k_n}(X)'(P'P)^{-1}P'\varepsilon(\alpha) = o_p(\delta_{1n}).$$

(B) *Suppose that Assumptions 3.1-3.2 and the following condition are satisfied:*

(iv) there exists a sequence of constants δ_{2n} with $\delta_{2n} = o(1)$ such that for all $X \in \mathcal{X}$ and $\alpha \in \mathcal{A}_n$, there exists $\pi(\alpha)$ satisfying

$$E[g(Z, \alpha)|X] = p^{k_n}(X)'\pi(\alpha) + o(\delta_{2n}).$$

Then uniformly over $\alpha \in \mathcal{A}_n$:

$$\frac{1}{n}\sum_{i=1}^{n}\left(p^{k_n}(X_i)'(P'P)^{-1}\sum_{j=1}^{n}p^{k_n}(X_j)E[g(Z, \alpha)|X_j] - E[g(Z, \alpha)|X_i]\right)^2 = o_p(\delta_{2n}^2).$$

(C) *Suppose that Assumptions 3.1-3.2 and the following condition are satisfied:*

(v) for a fixed $\overline{\alpha} \in \overline{\mathcal{A}}$, $E[\varepsilon(Z,\overline{\alpha})^2|X]$ *is bounded for all* $X \in \mathcal{X}$. *Then*:

$$\frac{1}{n}\sum_{i=1}^{n}\left(p^{k_n}(X_i)'(P'P)^{-1}\sum_{j=1}^{n}p^{k_n}(X_j)\varepsilon(Z_j,\overline{\alpha})\right)^2 = O_p(\frac{k_n}{n}).$$

Proof. **(A)**: Let c denote a generic constant which may have different values in different expressions. Let $\mathcal{W}_n = \mathcal{X} \times \mathcal{A}_n$. For any pair $(X^1, \alpha^1) \in \mathcal{W}_n$ and $(X^2, \alpha^2) \in \mathcal{W}_n$, we write

$$\left|p^{k_n}(X^1)'(P'P)^{-1}P'\varepsilon(\alpha^1) - p^{k_n}(X^2)'(P'P)^{-1}P'\varepsilon(\alpha^2)\right| \leq$$
$$\left|[p^{k_n}(X^1) - p^{k_n}(X^2)]'(P'P)^{-1}P'\varepsilon(\alpha^1)\right| + \left|p^{k_n}(X^2)'(P'P)^{-1}P'[\varepsilon(\alpha^1) - \varepsilon(\alpha^2)]\right|.$$

Note that

$$\left\|p^{k_n}(X^1) - p^{k_n}(X^2)\right\|_E^2 \leq \xi_{1n}^2 \left\|X^1 - X^2\right\|_E^2.$$

It follows that

$$\left|[p^{k_n}(X^1) - p^{k_n}(X^2)]'(P'P)^{-1}P'\varepsilon(\alpha^1)\right| \leq \xi_{1n}\left\|X^1 - X^2\right\|_E \sqrt{\frac{1}{n\lambda_n}\sum_{i=1}^{n}\varepsilon(Z_i,\alpha^1)^2}$$

where λ_n denotes the smallest eigenvalues of $P'P/n$. Condition (i) implies

$$\frac{1}{n}\sum_{i=1}^{n}[\varepsilon(Z_i,\alpha^1)]^2 \leq \frac{c_{1n}^2}{n}\sum_{i=1}^{n}(c_1(Z_i) + E[c_1(Z_i)|X_i])^2.$$

Assumption 3.2 implies $\lambda_n = O_p(1)$ (see pp. 162 of Newey (1997)). By the weak law of large numbers and noting that $E\{(E[c_1(Z)|X])^2\} \leq E\{c_1(Z)^2\}$, we obtain $\frac{1}{n}\sum_{i=1}^{n}(c_1(Z_i) + E[c_1(Z)|X_i])^2 = O_p(1)$. Thus, there exists a constant c such that

$$P\left(\sqrt{\frac{1}{n\lambda_n}\sum_{i=1}^{n}(c_1(Z_i) + E[c_1(Z)|X_i])^2} > c\right) < \eta$$

holds for any small $\eta > 0$ and sufficiently large n. Condition (ii) implies

$$|p^{k_n}(X^2)'(P'P)^{-1}P'[\varepsilon(\alpha^1) - \varepsilon(\alpha^2)]| \leq \xi_{0n}c_{2n}\left\|\alpha^1 - \alpha^2\right\|\sqrt{\frac{1}{n\lambda_n}\sum_{i=1}^{n}c_2(Z_i)^2}$$

with $\frac{1}{n\lambda_n}\sum_{i=1}^{n}c_2(Z_i)^2 = O_p(1)$. It follows that for some constant c:

$$P\left(\sup_{X\in\mathcal{X},\alpha_1,\alpha_2\in\mathcal{A}_n}\left\{|p^{k_n}(X^2)'(P'P)^{-1}P'[\varepsilon(\alpha^1) - \varepsilon(\alpha^2)]|/\left\|\alpha^1 - \alpha^2\right\|\right\} > c\xi_{0n}c_{2n}\right) < \eta$$

for sufficiently large n.

For any small ϵ, partition $\mathcal{W}_n$ into b_n mutually exclusive subsets $\mathcal{W}_{nm}, m = 1, 2, \cdots, b_n$, where $(X^1, \alpha^1) \in \mathcal{W}_{nm}$ and $(X^2, \alpha^2) \in \mathcal{W}_{nm}$ imply $\|X^1 - X^2\|_E \leq \epsilon\delta_{1n}/(c_{1n}\xi_{1n}c)$ and $\|\alpha^2 - \alpha^1\| \leq \epsilon\delta_{1n}/(c_{2n}\xi_{0n}c)$. Then, with probability approaching one, we have:

$$\left|p^{k_n}(X^1)'(P'P)^{-1}P'\varepsilon(\alpha^1) - p^{k_n}(X^2)'(P'P)^{-1}P'\varepsilon(\alpha^2)\right| \leq 2\epsilon\delta_{1n}.$$

Let (X^m, α^m) denote a fixed point in $\mathcal{W}_{nm}$. For any (X, α), there exists a m such that $\|X - X^m\|_E \leq \epsilon\delta_{1n}/(c_{1n}\xi_{1n}c)$ and $\|\alpha - \alpha^m\| \leq \epsilon\delta_{1n}/(c_{2n}\xi_{0n}c)$. Then, with probability approaching one,

$$\sup_{(X,\alpha)\in\mathcal{X}\times\mathcal{A}_n}\left|p^{k_n}(X)'(P'P)^{-1}P'\varepsilon(\alpha)\right| \leq 2\epsilon\delta_{1n} + \max_m\left|p^{k_n}(X^m)'(P'P)^{-1}P'\varepsilon(\alpha^m)\right|.$$

Hence

$$\begin{aligned} & P\left(\sup_{(X,\alpha)\in\mathcal{X}\times\mathcal{A}_n}\left|p^{k_n}(X)'(P'P)^{-1}P'\varepsilon(\alpha)\right| > 4\epsilon\delta_{1n}\right) \\ < \quad & 2\eta + P\left(\max_m |p^{k_n}(X^m)'(P'P)^{-1}P'\varepsilon(\alpha^m)| > 2\epsilon\delta_{1n}\right). \end{aligned}$$

For some constant c, let $M_n = \left(\frac{c\xi_{0n}c_{1n}}{\delta_{1n}\epsilon\eta}\right)^{2/p}$. Define $d_{in} = 1\{c_1(Z_i) \leq M_n\}$. Define $g_1(Z_i, \alpha) = d_{in}g(Z_i, \alpha)$ and $g_2(Z_i, \alpha) = (1 - d_{in})g(Z_i, \alpha)$. Define $\varepsilon_1(Z_i, \alpha)$ and $\varepsilon_2(Z_i, \alpha)$ accordingly. It then follows that

$$\begin{aligned} & P\left(\max_m |p^{k_n}(X^m)'(P'P)^{-1}P'\varepsilon(\alpha^m)| > 2\epsilon\delta_{1n}\right) \\ \leq \quad & P\left(\max_m\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_1(Z_i, \alpha^m)\right| > \epsilon\delta_{1n}\right) \\ & +P\left(\max_m\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_2(Z_i, \alpha^m)\right| > \epsilon\delta_{1n}\right) \\ \equiv \quad & P_1 + P_2. \end{aligned}$$

Applying the Markov inequality yields

$$P_2 \leq \frac{E\left[\max_m\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_2(Z_i, \alpha^m)\right|\right]}{\epsilon\delta_{1n}}.$$

Note that

$$\begin{aligned} \frac{|\varepsilon_2(Z_i, \alpha^m)|}{c_{1n}} \quad & \leq \quad (1 - d_{in})c_1(Z_i) + E[(1 - d_{in})c_1(Z_i)|X_i] \\ & \leq \quad (1 - d_{in})c_1(Z_i) + \sqrt{E[(1 - d_{in})|X_i]}\sqrt{E[c_1(Z_i)^2|X_i]}. \end{aligned}$$

Then

$$\begin{aligned}
&\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_2(Z_i,\alpha^m)\right| \\
\leq\ & \frac{c_{1n}\xi_{0n}}{\lambda_n}\sqrt{\frac{1}{n}\sum_{i=1}^n\frac{|\varepsilon_2(Z_i,\alpha^m)|^2}{c_{1n}^2}} \\
\leq\ & \frac{2c_{1n}\xi_{0n}}{\lambda_n}\sqrt{\frac{1}{n}\sum_{i=1}^n(1-d_{in})c_1(Z_i)^2}+\frac{2c_{1n}\xi_{0n}}{\lambda_n}\sqrt{\frac{1}{n}\sum_{i=1}^n E[(1-d_{in})|X_i]E[c_1(Z_i)^2|X_i]}.
\end{aligned}$$

Applying the Markov inequality yields

$$\frac{1}{n}\sum_{i=1}^n(1-d_{in})c_1(Z_i)^2 \leq \sqrt{\frac{1}{n}\sum_{i=1}^n(1-d_{in})}\sqrt{\frac{1}{n}\sum_{i=1}^n c_1(Z_i)^4} = O_p(\frac{1}{M_n^p}).$$

Using $E\{E[u|X_i]^2\} \leq E\{u^2\}$ for any random variable u, we also prove

$$\frac{1}{n}\sum_{i=1}^n E[(1-d_{in})|X_i]E[c_1(Z_i)^2|X_i] \leq O_p(\frac{1}{M_n^p}).$$

Combining the above results proves $P_2 \leq \frac{c\xi_{0n}c_{1n}}{M_n^{p/2}\epsilon\delta_{1n}} = \eta$.

Some calculations yield

$$\sigma_m^2 \equiv nE\{[p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_1(Z_i,\alpha^m)]^2\} = O(c_{1n}^2\xi_{0n}^2)$$

and

$$|p^{k_n}(X^m)'(P'P/n)^{-1}p^{k_n}(X_i)\varepsilon_1(Z_i,\alpha^m)| \leq \frac{M_n\xi_{0n}^2c_{1n}}{\lambda_n}.$$

Note that

$$\begin{aligned}
&P\left(\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_1(Z_i,\alpha^m)\right| > \epsilon\delta_{1n}\right) = \\
&E\left[P\left(\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_1(Z_i,\alpha^m)\right| > \epsilon\delta_{1n}\ \mid\ X_1,...,X_n\right)\right].
\end{aligned}$$

Applying the Bernstein inequality for independent processes, see e.g. Ichimura and Lee (1991, p.26), we obtain:

$$\begin{aligned}
&P\left(\left|p^{k_n}(X^m)'(P'P)^{-1}\sum_{i=1}^n p^{k_n}(X_i)\varepsilon_1(Z_i,\alpha^m)\right| > \epsilon\delta_{1n}\right) \\
\leq\ & 2E[\exp\left(-n\varepsilon^2\delta_n^2/(c\sigma_m^2+M_n\xi_{0n}^2c_{1n}\lambda_n\varepsilon\delta_n)\right)]
\end{aligned}$$

where $E[.]$ is taken with respect to the joint distribution of $(X_1, ..., X_n)$. Hence,

$$P_1 < 2b_n E[\exp\left(-n\varepsilon^2\delta_{1n}^2/(c\sigma_m^2 + M_n\xi_{0n}^2 c_{1n}\lambda_n\varepsilon\delta_{1n})\right)]$$

which is arbitrarily small if $\frac{n\delta_{1n}^2}{\max\{\xi_{0n}^2 c_{1n}^2, M_n\xi_{0n}^2 c_{1n}\delta_{1n}\}} - \ln(b_n) \to +\infty$. Since $\mathcal{X}$ is a compact subset in $\mathcal{R}^s$, we have

$$b_n = O\left((\frac{\delta_{1n}}{c_{1n}\xi_{1n}})^{-s} N(\mathcal{A}_n, \frac{\delta_{1n}}{\xi_{0n}c_{2n}})\right).$$

Substituting for M_n and b_n, we obtain

$$\begin{aligned} & \frac{n\delta_{1n}^2}{\ln(b_n)\max\{\xi_{0n}^2 c_{1n}^2, M_n\xi_{0n}^2 c_{1n}\delta_{1n}\}} \\ = \; & O\left(\frac{n\delta_{1n}^2}{\ln[(\frac{\delta_{1n}}{\xi_{1n}c_{1n}})^{-s} N(\mathcal{A}_n, \frac{\delta_{1n}}{\xi_{0n}c_{2n}})] \times \max\{\xi_{0n}^2 c_{1n}^2, \xi_{0n}^{2+2/p}\delta_{1n}^{1-2/p} c_{1n}^{1+2/p}\}}\right) \end{aligned}$$

Thus, $P_1 < \eta$ for sufficiently large n by Condition (iii).

(B): It follows from Condition (iv) and

$$\begin{aligned} & \frac{1}{n}\sum_{i=1}^n \left(p^{k_n}(X_i)'(P'P)^{-1}\sum_{j=1}^n p^{k_n}(X_j)E[g(Z,\alpha)|X_j] - E[g(Z,\alpha)|X_i]\right)^2 \\ = \; & \frac{1}{n}\sum_{i=1}^n \left(E[g(Z,\alpha)|X_i] - p^{k_n}(X_i)'\pi(\alpha)\right)^2. \end{aligned}$$

(C): Note that

$$\begin{aligned} & E\left[\frac{1}{n}\sum_{i=1}^n \left(p^{k_n}(X_i)'(P'P)^{-1}\sum_{j=1}^n p^{k_n}(X_j)\varepsilon(Z_j,\overline{\alpha})\right)^2\right] \\ = \; & E\left[\frac{1}{n}\sum_{i=1}^n p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X_i) \times E[\varepsilon(Z,\overline{\alpha})^2|X_i]\right] \\ \leq \; & E\left[\frac{c}{n}\sum_{i=1}^n p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X_i)\right] = O(\frac{k_n}{n}). \end{aligned}$$

Part (C) now follows from applying the Markov inequality. ■

Parts A and B of Lemma A.1 imply that uniformly over $\alpha \in \mathcal{A}_n$,

$$\frac{1}{n}\sum_{i=1}^n \left(p^{k_n}(X_i)'(P'P)^{-1}\sum_{j=1}^n p^{k_n}(X_j)g(Z_j,\alpha) - E[g(Z,\alpha)|X_i]\right)^2 = o_p(\max\{\delta_{1n}^2, \delta_{2n}^2\}).$$

Corollary A.1: *(i) Under Assumptions 3.1-3.2, 3.6, 3.7(i)(ii), we have uniformly over* $\alpha \in \mathcal{A}_n$, $\frac{1}{n}\sum_{i=1}^n \|\widehat{m}(X_i,\alpha) - m(X_i,\alpha)\|_E^2 = o_p(\delta_n^2)$;
(ii) under Assumptions 3.1-3.3 and 4.2, $\frac{1}{n}\sum_{i=1}^n \|\widehat{m}(X_i,\alpha_o)\|_E^2 = O_p(\frac{k_n}{n})$.

Proof. (i) follows from applying Parts A and B of Lemma A.1 to each element of $\rho(Z,\alpha)$ with $c_{1n} = \xi_{3n}$, $c_{2n} = \xi_{2n}$, $\delta_{1n} = \delta_{2n} = \delta_n$, and noting that Conditions (i)-(iv) are satisfied by Assumptions 3.6 and 3.7. (ii) follows from applying Part C of Lemma A.1 to each element of $\rho(Z,\alpha_o)$. ■

Corollary A.2: *(i) Under Assumption 3.1-3.2, 3.6's Lipschitz, 3.7(i) and 3.10, we obtain uniformly over* $\alpha \in \mathcal{A}_n$, $\|\alpha - \alpha_o\| \leq o(\eta_n)$ *and* $\delta_n \leq \eta_n \leq \delta_n^{3/4}$:

$$\frac{1}{n}\sum_{i=1}^n \|m(X_i,\alpha)\|_E^2 - E\{\|m(X,\alpha)\|_E^2\} = o_p(\xi_{4n}^2\eta_n^2\delta_n).$$

(ii) Under Assumption 3.1-3.2, 3.4(ii), 3.5-3.7 and 3.10, we obtain uniformly over $\alpha \in \mathcal{A}_n$, $\|\alpha - \alpha_o\| \leq o(\eta_n)$ *and* $\delta_n \leq \eta_n \leq \delta_n^{3/4}$:

$$\frac{1}{n}\sum_{i=1}^n \|\widehat{m}(X_i,\alpha)\|_E^2 = o_p(\eta_n^2) \textit{and } \frac{1}{n}\sum_{i=1}^n \|m(X_i,\alpha)\|_E^2 = o_p(\eta_n^2).$$

Proof. (i) We obtain the result by applying Part A of Lemma A.1 to $\|m(X,\alpha)\|_E^2$ with $\delta_{1n} = \xi_{4n}^2\eta_n^2\delta_n$ and $p^{k_n}(x) = 1$, and notice that $c_{1n} = \xi_{4n}^2\eta_n^2$ by Assumption 3.10, and $c_{2n} = \xi_{2n}\xi_{4n}\eta_n$ by Assumption 3.10 and 3.6's Lipschitz condition, since for any $\alpha_1, \alpha_2 \in \mathcal{A}_n$,

$$\left|\|m(X,\alpha_1)\|_E^2 - \|m(X,\alpha_2)\|_E^2\right| \leq E[|\rho(Z,\alpha_1) - \rho(Z,\alpha_2)|\,|X] \times [|m(X,\alpha_1)| + |m(X,\alpha_2)|].$$

Part (ii) follows from Corollaries A.1(i) and A.2(i), $E\{\|m(X,\alpha)\|_E^2\} = o(\eta_n^2)$ by Assumptions 3.4(ii) and 3.5, and $\xi_{4n}^2\delta_n = O(1)$ by Assumption 3.7(iii). ■

Appendix B: Convergence Rates for Sieve Estimators

We now prove consistency and compute the convergence rates of the proposed estimator. We begin by introducing more notation. Define

$$\widehat{L}_n(\alpha) \equiv -\frac{1}{2n}\sum_{i=1}^n \widehat{m}(X_i,\alpha)'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}(X_i,\alpha);$$

$$L_n(\alpha) \equiv -\frac{1}{2n}\sum_{i=1}^n m(X_i,\alpha)'[\Sigma(X_i)]^{-1}m(X_i,\alpha).$$

We need the following result to prove Theorem 3.1.

Corollary A.3: *(i) Under Assumptions 3.1-3.2, 3.4, 3.6-3.7,* $\widehat{L}_n(\alpha) - L_n(\alpha) = o_p(\delta_n)$ *uniformly over* $\alpha \in \mathcal{A}_n$; *(ii) Under Assumptions 3.1-3.4, 3.6-3.7 and 3.10,*

$$\widehat{L}_n(\alpha) - \widehat{L}_n(\alpha_o) - \{L_n(\alpha) - L_n(\alpha_o)\} = o_p(\eta_n\delta_n)$$

uniformly over $\alpha \in \mathcal{A}_n$ *with* $\|\alpha - \alpha_o\| \leq o(\eta_n)$ *and* $\delta_n \leq \eta_n \leq \delta_n^{3/4}$.

Proof. (i) follows from Corollary A.1(i) and Assumption 3.4. For (ii), note

$$\begin{aligned}
&\widehat{L}_n(\alpha) - \widehat{L}_n(\alpha_o) - \{L_n(\alpha) - L_n(\alpha_o)\} = -\frac{1}{2n}\sum_{i=1}^{n} \\
&(\widehat{m}(X_i,\alpha) - \widehat{m}(X_i,\alpha_o) - m(X_i,\alpha))' \left[\widehat{\Sigma}(X_i)\right]^{-1} (\widehat{m}(X_i,\alpha) - \widehat{m}(X_i,\alpha_o) - m(X_i,\alpha)) \\
&-\frac{1}{2n}\sum_{i=1}^{n} m(X_i,\alpha)' \left(\left[\widehat{\Sigma}(X_i)\right]^{-1} - \left[\Sigma(X_i)\right]^{-1}\right) m(X_i,\alpha) \\
&-\frac{1}{2n}\sum_{i=1}^{n} m(X_i,\alpha)' \left[\widehat{\Sigma}(X_i)\right]^{-1} (\widehat{m}(X_i,\alpha) - \widehat{m}(X_i,\alpha_o) - m(X_i,\alpha)).
\end{aligned}$$

Corollary A.1, Assumptions 3.4 and 3.7(i) imply the first term on the right hand side is $o_p(\eta_n\delta_n)$. Corollary A.2(ii) and Assumption 3.4 imply the second term is $o_p(\eta_n\delta_n)$. Corollaries A.1 and A.2(ii), Assumptions 3.4, 3.7(i) and 3.10 imply the third term is $o_p(\delta_n\eta_n)$. This completes the proof. ■

We now are ready to prove Theorem 3.1

Proof. **(Theorem 3.1):** Let $\eta_{0n} = \delta_n$ and $\delta_{0n} = 2\sqrt{\eta_{0n}} = O(\delta_n^{1/2})$. For any fixed $k > 1$,

$$\begin{aligned}
& P\left(\|\widehat{\alpha}_n - \alpha_o\| \geq k\delta_{0n}\right) \leq P\left(\sup_{\|\alpha-\alpha_o\|\geq k\delta_{0n},\alpha\in\mathcal{A}_n} \widehat{L}_n(\alpha) \geq \widehat{L}_n(\Pi_n\alpha_o)\right) \leq \\
& P\left(\sup_{\alpha\in\mathcal{A}_n} \left|\widehat{L}_n(\alpha) - L_n(\alpha)\right| > \eta_{0n}\right) + \\
& P\left(\left\{\sup_{\alpha\in\mathcal{A}_n} \left|\widehat{L}_n(\alpha) - L_n(\alpha)\right| \leq \eta_{0n}\right\} \bigcap \left\{\sup_{\|\alpha-\alpha_o\|\geq k\delta_{0n},\alpha\in\mathcal{A}_n} \widehat{L}_n(\alpha) \geq \widehat{L}_n(\Pi_n\alpha_o)\right\}\right) \\
\leq\ & P\left(\sup_{\alpha\in\mathcal{A}_n} \left|\widehat{L}_n(\alpha) - L_n(\alpha)\right| > \eta_{0n}\right) \\
& + P\left(\sup_{\|\alpha-\alpha_o\|\geq k\delta_{0n},\alpha\in\mathcal{A}_n} L_n(\alpha) - L_n(\alpha_o) \geq L_n(\Pi_n\alpha_o) - L_n(\alpha_o) - 2\eta_{0n}\right) \\
\equiv\ & P_1 + P_2.
\end{aligned}$$

Corollary A.3(i) implies $P_1 \to 0$ as $n \to \infty$. To show $P_2 \to 0$, we notice that all conditions A.1-A.4 in Chen and Shen (1998) are trivially satisfied given Assumptions 3.1, 3.3, 3.4(ii), 3.5, 3.6's Lipschitz and envelope conditions, 3.8 and 3.9. Hence $P_2 \to 0$ by the theorem 1 in Chen and Shen (1998). This proves: $\|\widehat{\alpha}_n - \alpha_o\| = o_p(\delta_{0n}) = o_p(\delta_n^{1/2})$.

Next, we refine the convergence rate by exploiting the local curvature of $\widehat{L}_n(\alpha)$ around α_o. Let $\eta_{1n} = \delta_n\delta_{0n} = \delta_n^{(1+1/2)}$ and $\delta_{1n} = 2\sqrt{\eta_{1n}} = O(\delta_n^{(1/2+1/4)})$. For any fixed $k > 1$, we have:

$$
\begin{aligned}
& P\left(\|\widehat{\alpha}_n-\alpha_o\|\geq k\delta_{1n}\right) \\
\leq\ & P\left(\sup_{\delta_{0n}\geq\|\alpha-\alpha_o\|\geq k\delta_{1n}}\widehat{L}_n(\alpha)-\widehat{L}_n(\alpha_o)\geq\widehat{L}_n(\Pi_n\alpha_o)-\widehat{L}_n(\alpha_o)\right) \\
\leq\ & P\left(\sup_{\delta_{0n}\geq\|\alpha-\alpha_o\|\geq k\delta_{1n}}\left|\widehat{L}_n(\alpha)-\widehat{L}_n(\alpha_o)-\{L_n(\alpha)-L_n(\alpha_o)\}\right|>\eta_{1n}\right) \\
& +P\left(\begin{array}{c}\left\{\sup_{\delta_{0n}\geq\|\alpha-\alpha_o\|\geq k\delta_{1n}}\left|\widehat{L}_n(\alpha)-\widehat{L}_n(\alpha_o)-\{L_n(\alpha)-L_n(\alpha_o)\}\right|\leq\eta_{1n}\right\} \\ \bigcap\left\{\sup_{\delta_{0n}\geq\|\alpha-\alpha_o\|\geq k\delta_{1n}}\widehat{L}_n(\alpha)-\widehat{L}_n(\alpha_o)\geq\widehat{L}_n(\Pi_n\alpha_o)-\widehat{L}_n(\alpha_o)\right\}\end{array}\right) \\
\leq\ & P\left(\sup_{\delta_{0n}\geq\|\alpha-\alpha_o\|\geq k\delta_{1n}}\left|\widehat{L}_n(\alpha)-\widehat{L}_n(\alpha_o)-\{L_n(\alpha)-L_n(\alpha_o)\}\right|>\eta_{1n}\right) \\
& +P\left(\sup_{\delta_{0n}\geq\|\alpha-\alpha_o\|\geq k\delta_{1n}}L_n(\alpha)-L_n(\alpha_o)\geq L_n(\Pi_n\alpha_o)-L_n(\alpha_o)-2\eta_{1n}\right) \\
\equiv\ & P_3+P_4
\end{aligned}
$$

Corollary A.3(ii) implies $\widehat{L}_n(\alpha)-\widehat{L}_n(\alpha_o)-\{L_n(\alpha)-L_n(\alpha_o)\}=o_p(\delta_n\delta_{0n})=o_p(\eta_{1n})$, hence $P_3\to 0$ as $n\to\infty$. Applying Theorem 1 of Chen and Shen again yields $P_4\to 0$ as $n\to\infty$. This proves $\|\widehat{\alpha}_n-\alpha_o\|=o_p(\delta_{1n})=o_p(\delta_n^{(1/2+1/4)})$.

Repeat the above proof infinite times, we obtain $\|\widehat{\alpha}_n-\alpha_o\|=O_p(\delta_{\infty n})$ with $\delta_{\infty n}=o(\delta_n^{(1/2+1/2^2+\ldots)})=o(\delta_n)$. This completes the proof. ■

Appendix C: Asymptotic Normality

In this section, we derive the asymptotic distribution of $\widehat{\theta}_n$. Recall that $f(\alpha)=\lambda'\theta$ is a smooth linear functional, v^* is its Riesz representor, and $f(\widehat{\alpha}_n)-f(\alpha_o)=\langle\widehat{\alpha}_n-\alpha_o,v^*\rangle$. Define

$$
\begin{aligned}
\widehat{m}_{\alpha_o}[X,\alpha-\alpha_o] &\equiv \sum_{i=1}^n\rho_{\alpha_o}[Z_i,\alpha-\alpha_o]p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X) \\
\widehat{r}_m[X,\alpha,\alpha_o] &\equiv \sum_{i=1}^n r_\rho[Z_i,\alpha,\alpha_o]p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X).
\end{aligned}
$$

Define $\widetilde{m}_{\alpha_o}[X,\alpha-\alpha_o]$ by replacing $\rho_{\alpha_o}[Z_i,\alpha-\alpha_o]$ in $\widehat{m}_{\alpha_o}[X,\alpha-\alpha_o]$ with $m_{\alpha_o}[X_i,\alpha-\alpha_o]$, and define $\widetilde{r}_m[X,\alpha,\alpha_o]$ by replacing $r_\rho[Z_i,\alpha,\alpha_o]$ in $\widehat{r}_m[X,\alpha,\alpha_o]$ with $r_m[X_i,\alpha,\alpha_o]$. We have:

$$m(X,\alpha)=m_{\alpha_o}[X,\alpha-\alpha_o]+r_m[X,\alpha,\alpha_o].$$

and

$$\widehat{m}(X,\alpha)-\widehat{m}(X,\alpha_o)=\widehat{m}_{\alpha_o}[X,\alpha-\alpha_o]+\widehat{r}_m[X,\alpha,\alpha_o]$$

Denote

$$L_{n\alpha_o}[\alpha-\alpha_o]\equiv-\frac{1}{n}\sum_{i=1}^n m_{\alpha_o}[X_i,\alpha-\alpha_o]'[\Sigma(X_i)]^{-1}m(X_i,\alpha_o)=0.$$

and

$$R[\alpha,\alpha_o] \equiv L_n(\alpha) - L_n(\alpha_o) - L_{n\alpha_o}[\alpha-\alpha_o] = L_n(\alpha)$$

Likewise we denote

$$\widehat{L}_{n\alpha_o}[\alpha-\alpha_o] \equiv -\frac{1}{n}\sum_{i=1}^{n}\widehat{m}_{\alpha_o}[X_i,\alpha-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}(X_i,\alpha_o).$$

and

$$\widehat{R}[\alpha,\alpha_o] \equiv \widehat{L}_n(\alpha) - \widehat{L}_n(\alpha_o) - \widehat{L}_{n\alpha_o}[\alpha-\alpha_o].$$

We can write

$$\begin{aligned}
-2R[\alpha,\alpha_o] &= \frac{1}{n}\sum_{i=1}^{n} m_{\alpha_o}[X_i,\alpha-\alpha_o]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\alpha-\alpha_o] + \\
&\quad \frac{2}{n}\sum_{i=1}^{n} m_{\alpha_o}[X_i,\alpha-\alpha_o]'[\Sigma(X_i)]^{-1}r_m[X_i,\alpha,\alpha_o] + \\
&\quad \frac{1}{n}\sum_{i=1}^{n} r_m[X_i,\alpha,\alpha_o]'[\Sigma(X_i)]^{-1}r_m[X_i,\alpha,\alpha_o] \\
&\equiv R1 + 2\times R3 + R4;
\end{aligned}$$

$$\begin{aligned}
-2\widehat{R}[\alpha,\alpha_o] &= \frac{1}{n}\sum_{i=1}^{n} \widehat{m}_{\alpha_o}[X_i,\alpha-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}_{\alpha_o}[X_i,\alpha-\alpha_o] + \\
&\quad \frac{2}{n}\sum_{i=1}^{n} \widehat{m}(X_i,\alpha_o)'[\widehat{\Sigma}(X_i)]^{-1}\widehat{r}_m[X_i,\alpha,\alpha_o] + \\
&\quad \frac{2}{n}\sum_{i=1}^{n} \widehat{m}_{\alpha_o}[X_i,\alpha-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{r}_m[X_i,\alpha,\alpha_o] + \\
&\quad \frac{1}{n}\sum_{i=1}^{n} \widehat{r}_m[X_i,\alpha,\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{r}_m[X_i,\alpha,\alpha_o] \\
&\equiv \widehat{R}1 + 2\times\widehat{R}2 + 2\times\widehat{R}3 + \widehat{R}4.
\end{aligned}$$

Corollary A.4: *Under Assumptions 3.1-3.4, 3.6-3.7, 3.10, 4.1-4.2, we have:*

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n} \widehat{m}_{\alpha_o}[X_i,\Pi_n v^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}(X_i,\alpha_o) \\
=\ &\frac{1}{n}\sum_{i=1}^{n} m_{\alpha_o}[X_i,v^*]'[\Sigma(X_i)]^{-1}\rho(Z_i,\alpha_o) + o_p(\frac{1}{\sqrt{n}}).
\end{aligned}$$

Proof. Applying Corollaries A.1(ii) and A.5(i), Assumptions 3.4 and 3.7(i), we have

$$\frac{1}{n}\sum_{i=1}^{n}(\widehat{m}_{\alpha_o}[X_i,\Pi_n v^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}(X_i,\alpha_o)-\widetilde{m}_{\alpha_o}[X_i,\Pi_n v^*]'[\Sigma(X_i)]^{-1}\widehat{m}(X_i,\alpha_o))$$
$$=\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{m}_{\alpha_o}[X_i,\Pi_n v^*]-\widetilde{m}_{\alpha_o}[X_i,\Pi_n v^*]\right)'\left[\widehat{\Sigma}(X_i)\right]^{-1}\widehat{m}(X_i,\alpha_o)$$
$$+\frac{1}{n}\sum_{i=1}^{n}\widetilde{m}_{\alpha_o}[X_i,\Pi_n v^*]'\left\{[\widehat{\Sigma}(X_i)]^{-1}-[\Sigma(X_i)]^{-1}\right\}\widehat{m}(X_i,\alpha_o)=o_p(\frac{1}{\sqrt{n}}).$$

Denote $g(X,\Pi_n v^*)=m_{\alpha_o}[X,\Pi_n v^*]'[\Sigma(X)]^{-1}$ and $\widehat{g}(X,\Pi_n v^*)$ as the linear sieve estimator of $\widetilde{m}_{\alpha_o}[X,\Pi_n v^*]'[\Sigma(X)]^{-1}$:

$$\widehat{g}(X,\Pi_n v^*)=p^{k_n}(X)'(P'P)^{-1}\sum_{i=1}^{n}p^{k_n}(X_i)\widetilde{m}_{\alpha_o}[X_i,\Pi_n v^*]'[\Sigma(X_i)]^{-1}.$$

By exchanging summation, we obtain

$$\frac{1}{n}\sum_{i=1}^{n}\widetilde{m}_{\alpha_o}[X_i,\Pi_n v^*]'[\Sigma(X_i)]^{-1}\widehat{m}(X_i,\alpha_o)=\frac{1}{n}\sum_{j=1}^{n}\widehat{g}(X_j,\Pi_n v^*)\rho(Z_j,\alpha_o)$$
$$=\frac{1}{n}\sum_{i=1}^{n}\{\widehat{g}(X_i,\Pi_n v^*)-g(X_i,\Pi_n v^*)\}\rho(Z_i,\alpha_o)+$$
$$\frac{1}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,\Pi_n v^*-v^*]'[\Sigma(X_i)]^{-1}\rho(Z_i,\alpha_o)$$
$$+\frac{1}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,v^*]'[\Sigma(X_i)]^{-1}\rho(Z_i,\alpha_o).$$

Notice that $\widehat{g}(X_i,\Pi_n v^*)$ converges to $g(X_i,\Pi_n v^*)$ and $E\left[\rho(Z_i,\alpha_o)|X_1,X_2,...,X_n\right]=0$. Applying the Markov inequality we obtain the first two terms on the right-hand side are $o_p(\frac{1}{\sqrt{n}})$. This completes the proof. ■

In the remainder of the appendix, we denote $\varepsilon_n=n^{-1/2}$, $\delta_n=\frac{1}{n^{1/4}\sqrt{\xi_{4n}}}$, $u^*=\pm v^*$, and $\alpha^*=\alpha+\varepsilon_n u^*$. Then, $\alpha-\Pi_n\alpha^*=-\varepsilon_n\Pi_n u^*$.

Corollary A.5 *(i) Under Assumptions 3.1-3.2, 3.7(i) and 4.3, we have:*

$$\frac{1}{n}\sum_{i=1}^{n}\|\widehat{m}_{\alpha_o}[X_i,\Pi_n v^*]-m_{\alpha_o}[X_i,\Pi_n v^*]\|_E^2=o_p(\delta_n^2);$$

(ii) Under Assumptions 3.1-3.2, 3.7 and 4.4(i), we have: uniformly over $\alpha\in\mathcal{A}_n$

$$\frac{1}{n}\sum_{i=1}^{n}\|\widehat{m}_{\alpha_o}[X_i,\alpha-\alpha_o]-m_{\alpha_o}[X_i,\alpha-\alpha_o]\|_E^2=o_p(\delta_n^2);$$

(iii) Under Assumptions 3.1-3.2, 3.5, 3.7 and 4.4, we have: uniformly over $\alpha \in \mathcal{N}_n$

$$\frac{1}{n}\sum_{i=1}^{n} \|m_{\alpha_o}[X_i, \alpha - \alpha_o]\|_E^2 = o_p(\delta_n^2);$$

(iv) Under Assumptions 3.1-3.2, 3.7 and 4.4, we have: uniformly over $\alpha \in \mathcal{N}_n$ and X

$$\widehat{m}_{\alpha_o}[X, \alpha - \alpha_o] - \widetilde{m}_{\alpha_o}[X, \alpha - \alpha_o] = o_p(\varepsilon_n).$$

Proof. Part (i) follows from applying Markov inequality and parts B and C of lemma A.1 to $m_{\alpha_o}[X_i, \Pi_n v^*]$ and noting that $\frac{k_n}{n} = o(\delta_n^2)$ by Assumption 3.7(i). Part (ii) follows from applying parts A and B of Lemma A.1 to $\rho_{\alpha_o}[Z, \alpha - \alpha_o]$ with $c_{1n} = \xi_{3n}$, $c_{2n} = \xi_{2n}$, and $\delta_{1n} = \delta_{2n} = \delta_n$.

Part (iii) follows from applying part A of Lemma A.1 to $\|m_{\alpha_o}[X, \alpha - \alpha_o]\|_E^2$ with $p^{k_n}(X) = 1$ and $\delta_{1n} = \xi_{4n}^2 \delta_n^3$, and notice that $c_{1n} = \xi_{4n}^2 \delta_n^2$, $c_{2n} = \xi_{2n}\xi_{4n}\delta_n$ by Assumption 4.4 since for any $\alpha_1, \alpha_2 \in \mathcal{A}_n$, $\left| \|m_{\alpha_o}[X, \alpha_1 - \alpha_o]\|_E^2 - \|m_{\alpha_o}[X, \alpha_2 - \alpha_o]\|_E^2 \right| \leq E[|\rho_{\alpha_o}[Z, \alpha_1 - \alpha_2]| \, |X] \times [|m_{\alpha_o}[X, \alpha_1 - \alpha_o]| + |m_{\alpha_o}[X, \alpha_2 - \alpha_o]|]$, and $|m_{\alpha_o}[X, \alpha - \alpha_o]| \leq E[|\rho_{\alpha_o}[Z, \alpha - \alpha_o]| \, |X]$. The result follows by noting $E\{\|m_{\alpha_o}[X, \alpha - \alpha_o]\|_E^2\} = O\left(\|\alpha - \alpha_o\|^2\right) = o(\delta_n^2)$ by Assumption 3.5 and $\xi_{4n}^2 \delta_n = O(1)$ by Assumption 3.7(iii).

Part (iv) follows from applying part A of Lemma A.1 to $\rho_{\alpha_o}[Z_i, \alpha - \alpha_o]$ with $c_{1n} = \xi_{4n}\delta_n$, $c_{2n} = \xi_{2n}$ and $\delta_{1n} = \xi_{4n}\delta_n^2 = n^{-1/2}$. ■

Corollary A.6. *Under Assumptions 3.1-3.2, 3.6-3.7, 3.10, 4.4-4.5, we have uniformly over $\alpha \in \mathcal{N}_n$:*

(i) $\frac{1}{n}\sum_{i=1}^{n} \|\widehat{r}_m[X_i, \alpha, \alpha_o] - r_m[X_i, \alpha, \alpha_o]\|_E^2 = o_p(\delta_n^2)$;
(ii) $\widehat{r}_m[X, \alpha, \alpha_o] - \widetilde{r}_m[X, \alpha, \alpha_o] = o_p(\delta_n^3)$ *uniformly over* $X \in \mathcal{X}$;
(iii) $\frac{1}{n}\sum_{i=1}^{n} \|\widehat{r}_m[X_i, \alpha, \alpha_o] - \widehat{r}_m[X_i, \Pi_n\alpha^*, \alpha_o]\|_E^2 = O_p(\varepsilon_n^2 \xi_{6n}^2 \delta_n^2)$;
(iv) $\frac{1}{n}\sum_{i=1}^{n} \|r_m[X, \alpha, \alpha_o] - r_m[X, \Pi_n\alpha^*, \alpha_o]\|_E^2 = O_p(\varepsilon_n^2 \xi_{6n}^2 \delta_n^2)$;
(v) $\frac{1}{n}\sum_{i=1}^{n} \|r_m[X_i, \alpha, \alpha_o]\|_E^2 = o_p(\delta_n^2)$.

Proof. Part (i) follows from Corollaries A.1(i) and A.5(ii). Part (ii) follows from applying part A of Lemma A.1 to each element of $r_\rho[Z, \alpha, \alpha_o]$ with $\delta_{1n} = \delta_n^3$, $c_{1n} = \xi_{5n}\delta_n^2$ by Assumption 4.5(i), $c_{2n} = \xi_{2n}\delta_n^2$ by Assumptions 3.6 and 4.4(i). Part (iii) follows from Assumptions 3.1-3.2 and 4.5(ii). Part (iv) follows from Assumptions 3.1 and 4.5(ii). And (v) follows from Corollaries A.2(ii) and A.5(iii). ■

Lemma A.2: *Under Assumptions 3.1-3.4, 3.6-3.7, 3.10, 4.2-4.5, we obtain uniformly over $\alpha \in \mathcal{N}_n$:*

$$\widehat{R}[\alpha, \alpha_o] - \widehat{R}[\Pi_n\alpha^*, \alpha_o] = R[\alpha, \alpha_o] - R[\Pi_n\alpha^*, \alpha_o] + o_p(\varepsilon_n^2).$$

Proof. We write:

$$\begin{aligned}
& \widehat{R}[\alpha,\alpha_o]-\widehat{R}[\Pi_n\alpha^*,\alpha_o]-(R[\alpha,\alpha_o]-R[\Pi_n\alpha^*,\alpha_o]) \\
= & -\frac{1}{2n}\sum_{i=1}^{n}\left(A_i+B_i+C_i+D_i\right),
\end{aligned}$$

where $A_i \equiv A1_i - A2_i$, with

$$\begin{aligned}
A1_i &= \widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*] \\
&\quad +2\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]; \\
A2_i &= m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*] \\
&\quad +2m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*];
\end{aligned}$$

$$B_i = 2\widehat{m}(X_i,\alpha_o)'[\widehat{\Sigma}(X_i)]^{-1}\left(\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\right);$$

$C_i \equiv C1_i - C2_i$, with

$$\begin{aligned}
C1_i &= 2\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{r}_m[X_i,\alpha,\alpha_o] \\
&\quad +2\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\left(\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\right); \\
C2_i &= 2m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\Sigma(X_i)]^{-1}r_m[X_i,\alpha,\alpha_o] \\
&\quad +2m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\Sigma(X_i)]^{-1}\left(r_m[X_i,\alpha,\alpha_o]-r_m[X_i,\Pi_n\alpha^*,\alpha_o]\right);
\end{aligned}$$

and $D_i \equiv D1_i - D2_i$, with

$$\begin{aligned}
D1_i &= \left(\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\right)'\left[\widehat{\Sigma}(X_i)\right]^{-1}\left(\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\right) \\
&\quad +2\left(\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\right)'\left[\widehat{\Sigma}(X_i)\right]^{-1}\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]; \\
D2_i &= \left(r_m[X_i,\alpha,\alpha_o]-r_m[X_i,\Pi_n\alpha^*,\alpha_o]\right)'\left[\Sigma(X_i)\right]^{-1}\left(r_m[X_i,\alpha,\alpha_o]-r_m[X_i,\Pi_n\alpha^*,\alpha_o]\right) \\
&\quad +2\left(r_m[X_i,\alpha,\alpha_o]-r_m[X_i,\Pi_n\alpha^*,\alpha_o]\right)'\left[\Sigma(X_i)\right]^{-1}r_m[X_i,\Pi_n\alpha^*,\alpha_o].
\end{aligned}$$

First, consider the term $\frac{1}{n}\sum_{i=1}^{n}A_i$. Note that

$$\begin{aligned}
& \frac{1}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*] \\
& -m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]\} \\
= & \frac{\varepsilon_n^2}{n}\sum_{i=1}^{n}\left(\begin{array}{c}\widehat{m}_{\alpha_o}[X_i,\Pi_n u^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}_{\alpha_o}[X_i,\Pi_n u^*]- \\ m_{\alpha_o}[X_i,\Pi_n u^*]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]\end{array}\right)=o_p(\varepsilon_n^2),
\end{aligned}$$

where the last equality follows from applying Holder inequality, Corollary A.5(i) and Assumption 3.4. Moreover, applying Corollary A.5(i) and Assumption 3.4,

we obtain:

$$
\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]\\
&-m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]\}=\\
&-\frac{\varepsilon_n}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\Sigma(X_i)]^{-1}\{\widehat{m}_{\alpha_o}[X_i,\Pi_n u^*]-m_{\alpha_o}[X_i,\Pi_n u^*]\}\\
&-\frac{\varepsilon_n}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'\{[\widehat{\Sigma}(X_i)]^{-1}-[\Sigma(X_i)]^{-1}\}m_{\alpha_o}[X_i,\Pi_n u^*]\\
&-\frac{\varepsilon_n}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]-m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]\}'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]\\
&+o_p(\varepsilon_n^2).
\end{aligned}
$$

The first two terms on the right-hand side are $o_p(\varepsilon_n^2)$ by Corollary A.5(i)-(ii) and Assumption 3.4. The last term is decomposed to

$$
\begin{aligned}
&-\frac{\varepsilon_n}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]-\widetilde{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]\}'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]\\
&-\frac{\varepsilon_n}{n}\sum_{i=1}^{n}\{\widetilde{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]-m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]\}'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]\\
&+o_p(\varepsilon_n^2).
\end{aligned}
$$

Corollary A.5(iii) implies

$$
\frac{\varepsilon_n}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]-\widetilde{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]\}'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]=o_p(\varepsilon_n^2).
$$

Notice that

$$
\frac{1}{n}\sum_{i=1}^{n}\{\widetilde{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]-m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]\}'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]
$$

is the product of the respective approximation errors of $[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]$ and $m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]$ by the linear sieve $p^{k_n}(X)$. These approximation errors are in the order of $o_p(\delta_n)$. Therefore, $\frac{1}{n}\sum_{i=1}^{n}A_i=o_p(\varepsilon_n^2)$.

Next, $\frac{1}{n}\sum_{i=1}^{n}B_i=o_p(\varepsilon_n^2)$ follows by applying Corollaries A.1(ii) and A.6(iii) and Assumption 4.5(ii).

Now, consider the term $\frac{1}{n}\sum_{i=1}^{n}C_i$. Applying Corollaries A.5(i) and A.6(i) and Assumption 3.4, we obtain

$$
\frac{1}{n}\sum_{i=1}^{n}\left\{\begin{array}{c}\widehat{m}_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\widehat{\Sigma}(X_i)]^{-1}\widehat{r}_m[X_i,\alpha,\alpha_o]-\\ m_{\alpha_o}[X_i,\alpha-\Pi_n\alpha^*]'[\Sigma(X_i)]^{-1}r_m[X_i,\alpha,\alpha_o]\end{array}\right\}
$$

$$
\begin{aligned}
= \quad & \frac{\varepsilon_n}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,-\Pi_n u^*]-m_{\alpha_o}[X_i,-\Pi_n u^*]\}'[\Sigma(X_i)]^{-1}r_m[X_i,\alpha,\alpha_o] \\
& +\frac{\varepsilon_n}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,-\Pi_n u^*]'\{[\widehat{\Sigma}(X_i)]^{-1}-[\Sigma(X_i)]^{-1}\}r_m[X_i,\alpha,\alpha_o] \\
& +\frac{\varepsilon_n}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,-\Pi_n u^*]'[\Sigma(X_i)]^{-1}\{\widehat{r}_m[X_i,\alpha,\alpha_o]-\widetilde{r}_m[X_i,\alpha,\alpha_o]\} \\
& +\frac{\varepsilon_n}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,-\Pi_n u^*]'[\Sigma(X_i)]^{-1}\{\widetilde{r}_m[X_i,\alpha,\alpha_o]-r_m[X_i,\alpha,\alpha_o]\}+o_p(\varepsilon_n^2).
\end{aligned}
$$

The first term is $o_p(\varepsilon_n^2)$ by Corollaries A.5(i) and A.6(v). The second term is $o_p(\varepsilon_n^2)$ by Assumption 3.4 and Corollary A.6(v). The third term is $o_p(\varepsilon_n^2)$ by Corollary A.6(ii) and Assumption 4.5(ii). The fourth term is the product of the approximation errors of $r_m[X_i,\alpha,\alpha_o]$ and $m_{\alpha_o}[X_i,-\Pi_n u^*]'[\Sigma(X_i)]^{-1}$ respectively by the linear sieve and hence is $o_p(\varepsilon_n^2)$. Also, note that

$$
\begin{aligned}
& \frac{1}{n}\sum_{i=1}^{n}\left\{\begin{array}{c}\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\left(\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\right)- \\ m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\Sigma(X_i)]^{-1}\left(r_m[X_i,\alpha,\alpha_o]-r_m[X_i,\Pi_n\alpha^*,\alpha_o]\right)\end{array}\right\}= \\
& \frac{1}{n}\sum_{i=1}^{n}\{\widehat{m}_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]-m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]\}'[\widehat{\Sigma}(X_i)]^{-1}\left(\begin{array}{c}\widehat{r}_m[X_i,\alpha,\alpha_o]- \\ \widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]\end{array}\right) \\
+ \quad & \frac{1}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'[\widehat{\Sigma}(X_i)]^{-1}\left(\begin{array}{c}\widehat{r}_m[X_i,\alpha,\alpha_o]-\widehat{r}_m[X_i,\Pi_n\alpha^*,\alpha_o]- \\ r_m[X_i,\alpha,\alpha_o]+r_m[X_i,\Pi_n\alpha^*,\alpha_o]\end{array}\right) \\
& +\frac{1}{n}\sum_{i=1}^{n}m_{\alpha_o}[X_i,\Pi_n\alpha^*-\alpha_o]'\{[\widehat{\Sigma}(X_i)]^{-1}-[\Sigma(X_i)]^{-1}\}\left(\begin{array}{c}r_m[X_i,\alpha,\alpha_o]- \\ r_m[X_i,\Pi_n\alpha^*,\alpha_o]\end{array}\right).
\end{aligned}
$$

The first term is $o_p(\varepsilon_n^2)$ by Corollaries A.5(i) and A.6(iii). The second term $o_p(\varepsilon_n^2)$ by Corollaries A.5(iii) and A.6(iii)(iv). The fourth term is $o_p(\varepsilon_n^2)$ by Assumption 3.4 and Corollary A.6(iv).

Finally, $\frac{1}{n}\sum_{i=1}^{n}D_i=o_p(\varepsilon_n^2)$ by applying Corollary A.6(iv) and (v). ■

Corollary A.7. *Under Assumptions 3.1, 4.3-4.4, we have uniformly over $\alpha\in\mathcal{N}_n$,*

$$
\mu_n(m_{\alpha_o}[X_i,\alpha-\alpha_o]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*])=o_p(\varepsilon_n).
$$

Proof. Applying part A of Lemma A.1 to $m_{\alpha_o}[X_i,\alpha-\alpha_o]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*]$ with $c_{1n}=\xi_{4n}\delta_n$, $c_{2n}=\xi_{2n}$, $\delta_{1n}=\xi_{4n}\delta_n^2$, and $p^{k_n}(X)=1$, we obtain

$$
\mu_n(m_{\alpha_o}[X_i,\alpha-\alpha_o]'[\Sigma(X_i)]^{-1}m_{\alpha_o}[X_i,\Pi_n u^*])=o_p(\xi_{4n}\delta_n^2).
$$

The result now follows from $\xi_{4n}\delta_n^2=\varepsilon_n$. ■

Lemma A.3: *Under Assumptions 3.1-3.3, 3.5-3.7, 3.10 and 4.2-4.5, we have uniformly over* $\alpha \in \mathcal{N}_n$,

$$R[\alpha, \alpha_o] - R[\Pi_n \alpha^*, \alpha_o] - E\left\{R[\alpha, \alpha_o] - R[\Pi_n \alpha^*, \alpha_o]\right\} = o_p(\varepsilon_n^2).$$

Proof. Using the same notation as that in the proof of Lemma A.2, we have:

$$R[\alpha, \alpha_o] - R[\Pi_n \alpha^*, \alpha_o] = -\frac{1}{2n}\sum_{i=1}^{n}(A2_i + C2_i + D2_i)$$

From the results (for $A2_i, C2_i, D2_i$) in the proof of Lemma A.2, we have:

$$\begin{aligned} & -\left(R[\alpha, \alpha_o] - R[\Pi_n \alpha^*, \alpha_o] - E\left\{R[\alpha, \alpha_o] - R[\Pi_n \alpha^*, \alpha_o]\right\}\right) \\ = \; & \varepsilon_n \times \mu_n(m_{\alpha_o}[X_i, \Pi_n \alpha^* - \alpha_o]'[\Sigma(X_i)]^{-1} m_{\alpha_o}[X_i, \Pi_n u^*]) + o_p(\varepsilon_n^2) \\ = \; & \varepsilon_n \times \mu_n(m_{\alpha_o}[X_i, \alpha - \alpha_o]'[\Sigma(X_i)]^{-1} m_{\alpha_o}[X_i, \Pi_n u^*]) + o_p(\varepsilon_n^2). \end{aligned}$$

The result now follows by applying Corollary A.7. ■

Lemma A.4: *Under Assumptions 4.3-4.5, we have uniformly over* $\alpha \in \mathcal{N}_n$:

$$E\left[L_n(\alpha) - L_n(\Pi_n \alpha^*)\right] = \frac{1}{2}\left[\|\Pi_n \alpha^* - \alpha_o\|^2 - \|\alpha - \alpha_o\|^2\right] + o_p(\varepsilon_n^2).$$

Proof. With $m(X, \alpha_o) = 0$, write

$$\begin{aligned} -E[L_n(\alpha)] \;=\; & \frac{1}{2}\|\alpha - \alpha_o\|^2 + E\left\{m_{\alpha_o}[X, \alpha - \alpha_o]'[\Sigma(X)]^{-1} r_m[X, \alpha, \alpha_o]\right\} + \\ & \frac{1}{2}E\left\{r_m[X, \alpha, \alpha_o]'[\Sigma(X)]^{-1} r_m[X, \alpha, \alpha_o]\right\}. \end{aligned}$$

Then

$$E\left[L_n(\alpha) - L_n(\Pi_n \alpha^*)\right] - \frac{1}{2}\left\{\|\Pi_n \alpha^* - \alpha_o\|^2 - \|\alpha - \alpha_o\|^2\right\} = A_n + \frac{1}{2}B_n$$

where after some manipulation

$$\begin{aligned} A_n \;=\; & \varepsilon_n E\left\{m_{\alpha_o}[X, \Pi_n u^*]'\Sigma(X)^{-1}\left\{r_m[X, \Pi_n \alpha^*, \alpha_o] - r_m[X, \alpha, \alpha_o]\right\}\right\} \\ & + E\left\{m_{\alpha_o}[X, \alpha - \alpha_o]'\Sigma(X)^{-1}\left\{r_m[X, \Pi_n \alpha^*, \alpha_o] - r_m[X, \alpha, \alpha_o]\right\}\right\} \\ & + \varepsilon_n E\left\{m_{\alpha_o}[X, \Pi_n u^*]'\Sigma(X)^{-1} r_m[X, \alpha, \alpha_o]\right\} \end{aligned}$$

and $B_n =$

$$\begin{aligned} & E\left\{\left\{r_m[X, \Pi_n \alpha^*, \alpha_o] - r_m[X, \alpha, \alpha_o]\right\}'\Sigma(X)^{-1}\left\{r_m[X, \Pi_n \alpha^*, \alpha_o] - r_m[X, \alpha, \alpha_o]\right\}\right\} \\ & + 2E\left\{r_m[X, \alpha, \alpha_o]'[\Sigma(X)]^{-1}\left\{r_m[X, \Pi_n \alpha^*, \alpha_o] - r_m[X, \alpha, \alpha_o]\right\}\right\} \end{aligned}$$

Corollaries A.5(iii) and A.6(iv) imply $A_n = o(\varepsilon_n^2)$. Corollary A.6(iv) implies $B_n = o(\varepsilon_n^2)$. ■

Proof. **(Theorem 4.1)** Define $\widehat{\alpha}^* = \widehat{\alpha} + \varepsilon_n u^*$. Simple calculations give:

$$\widehat{L}_n(\widehat{\alpha}) = \widehat{L}_n(\Pi_n\widehat{\alpha}^*) + \widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n\widehat{\alpha}^*] + \widehat{R}[\widehat{\alpha}, \alpha_o] - \widehat{R}[\Pi_n\widehat{\alpha}^*, \alpha_o].$$

By Lemma A.2, we have:

$$\begin{aligned}
\widehat{L}_n(\widehat{\alpha}) &= \widehat{L}_n(\Pi_n\widehat{\alpha}^*) + \widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n\widehat{\alpha}^*] + R[\widehat{\alpha}, \alpha_o] - R[\Pi_n\widehat{\alpha}^*, \alpha_o] + o_p(\frac{1}{n}) \\
&= \widehat{L}_n(\Pi_n\widehat{\alpha}^*) + \widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n\widehat{\alpha}^*] + E\left[L_n(\widehat{\alpha}) - L_n(\Pi_n\widehat{\alpha}^*)\right] + \\
&\quad + R[\widehat{\alpha}, \alpha_o] - R[\Pi_n\widehat{\alpha}^*, \alpha_o] - E\left\{R[\widehat{\alpha}, \alpha_o] - R[\Pi_n\widehat{\alpha}^*, \alpha_o]\right\} + o_p(\frac{1}{n})
\end{aligned}$$

where the last equality follows from rearranging terms. Lemmas A.3 and A.4 imply

$$\begin{aligned}
\widehat{L}_n(\widehat{\alpha}) &= \widehat{L}_n(\Pi_n\widehat{\alpha}^*) - \frac{1}{2}\left[\|\widehat{\alpha} - \alpha_o\|^2 - \|\Pi_n\widehat{\alpha}^* - \alpha_o\|^2\right] \\
&\quad + \widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n\widehat{\alpha}^*] + o_p(\frac{1}{n}).
\end{aligned}$$

Write

$$\begin{aligned}
\|\Pi_n\widehat{\alpha}^* - \alpha_o\|^2 &= \|\Pi_n\widehat{\alpha}^* - \widehat{\alpha}^*\|^2 + \|\widehat{\alpha} - \alpha_o\|^2 + \varepsilon_n^2 \|u^*\|^2 \\
&\quad + 2\left\langle \Pi_n\widehat{\alpha}^* - \widehat{\alpha}^*, \widehat{\alpha} - \alpha_o\right\rangle + 2\varepsilon_n \left\langle \Pi_n\widehat{\alpha}^* - \widehat{\alpha}^*, u^*\right\rangle + 2\varepsilon_n \left\langle \widehat{\alpha} - \alpha_o, u^*\right\rangle.
\end{aligned}$$

Note that

$$|\langle \Pi_n\widehat{\alpha}^* - \widehat{\alpha}^*, \widehat{\alpha} - \alpha_o\rangle| = \varepsilon_n |\langle \Pi_n u^* - u^*, \widehat{\alpha} - \alpha_o\rangle| \leq \varepsilon_n \|\Pi_n u^* - u^*\| \times \|\widehat{\alpha} - \alpha_o\| = o_p(\varepsilon_n^2)$$

and

$$\varepsilon_n |\langle \Pi_n\widehat{\alpha}^* - \widehat{\alpha}^*, u^*\rangle| = \varepsilon_n^2 |\langle \Pi_n u^* - u^*, u^*\rangle| = o_p(\varepsilon_n^2).$$

It follows that

$$\|\Pi_n\widehat{\alpha}^* - \alpha_o\|^2 = o_p(\varepsilon_n^2) + \|\widehat{\alpha} - \alpha_o\|^2 + 2\varepsilon_n \left\langle (\widehat{\alpha} - \alpha_o), u^*\right\rangle$$

and

$$\widehat{L}_n(\widehat{\alpha}) - \widehat{L}_n(\Pi_n\widehat{\alpha}^*) - \widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n\widehat{\alpha}^*] = \varepsilon_n \left\langle \widehat{\alpha} - \alpha_o, u^*\right\rangle + o_p(\varepsilon_n^2).$$

By definition of $\widehat{\alpha}$, $\widehat{L}_n(\widehat{\alpha}) - \widehat{L}_n(\Pi_n\widehat{\alpha}^*) \geq 0$. And by Theorem 3.1, $\|\widehat{\alpha} - \alpha_o\|^2 = o_p(\delta_n)$. It follows that

$$0 \leq \widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n\widehat{\alpha}^*] + \varepsilon_n \left\langle \widehat{\alpha} - \alpha_o, u^*\right\rangle + o_p(\varepsilon_n^2).$$

Note that $\widehat{\alpha} - \Pi_n\widehat{\alpha}^* = -\varepsilon_n\Pi_n u^*$. We have

$$\widehat{L}_{n\alpha_o}[\widehat{\alpha} - \Pi_n \widehat{\alpha}^*] = -\varepsilon_n \widehat{L}_{n\alpha_o}[\Pi_n u^*]$$

and

$$0 \leq -\varepsilon_n \widehat{L}_{n\alpha_o}[\Pi_n u^*] + \varepsilon_n \langle \widehat{\alpha} - \alpha_o, u^* \rangle + o_p(\varepsilon_n^2).$$

Therefore,

$$0 \geq \widehat{L}_{n\alpha_o}[\Pi_n u^*] - \langle \widehat{\alpha} - \alpha_o, u^* \rangle + o_p(\varepsilon_n).$$

Since this holds for $u^* = \pm v^*$, we obtain:

$$\left| \widehat{L}_{n\alpha_o}[\Pi_n v^*] - \langle \widehat{\alpha} - \alpha_o, v^* \rangle \right| = o_p(\varepsilon_n).$$

This proves that

$$\widehat{L}_{n\alpha_o}[\Pi_n v^*] = \langle \widehat{\alpha} - \alpha_o, v^* \rangle + o_p(\frac{1}{\sqrt{n}}).$$

Applying Corollary A.4, we have:

$$\widehat{L}_{n\alpha_o}[\Pi_n v^*] = -\frac{1}{n} \sum_{i=1}^{n} m_{\alpha_o}[X_i, v^*]' [\Sigma(X_i)]^{-1} \rho(Z_i, \alpha_o) + o_p(\frac{1}{\sqrt{n}})$$

Hence for any fixed non-zero $\lambda \in \mathcal{R}^b$,

$$\lambda'(\widehat{\theta}_n - \theta_o) = -\frac{1}{n} \sum_{i=1}^{n} m_{\alpha_o}[X_i, v^*]' \Sigma(X_i)^{-1} \rho(Z_i, \alpha_o) + o_p(\frac{1}{\sqrt{n}}).$$

Substituting for v^*, we obtain

$$\begin{aligned} &\sqrt{n}(\widehat{\theta}_n - \theta_o) = \\ &- \left[E \left\{ D_{w^*}(X)' (\Sigma(X))^{-1} D_{w^*}(X) \right\} \right]^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} D_{w^*}(X_i)' (\Sigma(X_i))^{-1} \rho(Z_i, \alpha_o) + o_p(1). \end{aligned}$$

The theorem now follows from applying a standard CLT for i.i.d. data. ■

Appendix D: Asymptotic Variance and Efficiency

In this appendix, we first prove the consistency of $\widehat{V}$, then show the efficiency of the optimally weighted minimum distance estimator, and finally establish the asymptotic properties of the three-step estimator.

Proof. (Lemma 5.1): Write

$$\begin{aligned} & \rho(Z, \alpha)\rho(Z, \alpha)' - \rho(Z, \alpha_o)\rho(Z, \alpha_o)' \\ = \; & (\rho(Z, \alpha) - \rho(Z, \alpha_o))(\rho(Z, \alpha) - \rho(Z, \alpha_o))' \\ & + 2(\rho(Z, \alpha) - \rho(Z, \alpha_o))\rho(Z, \alpha_o)' \end{aligned}$$

By Assumption 4.4 and 4.5, we show that $\rho(Z,\alpha)\rho(Z,\alpha)'-\rho(Z,\alpha_o)\rho(Z,\alpha_o)'$ satisfies the envelope condition with coefficient $c_{1n}=\xi_{3n}$. Similarly, we can show that $\rho(Z,\alpha)\rho(Z,\alpha)'-\rho(Z,\alpha_o)\rho(Z,\alpha_o)'$ satisfies the Lipschitz condition with coefficient $c_{2n}=\xi_{2n}$. Applying part A of Lemma A.1 to each element of $\rho(Z,\alpha)\rho(Z,\alpha)'-\rho(Z,\alpha_o)\rho(Z,\alpha_o)'$ with $c_{1n}=\xi_{3n}$, $c_{2n}=\xi_{2n}$ and $\delta_{1n}=\delta_n$, we obtain uniformly over $X\in\mathcal{X}$ and $\alpha\in\mathcal{N}_n$:

$$\widehat{\Sigma}_o(X,\alpha)-\widehat{\Sigma}_o(X,\alpha_o)=E\{\widehat{\Sigma}_o(X,\alpha)-\widehat{\Sigma}_o(X,\alpha_o)|X_1,...,X_n\}+o_p(\delta_n).$$

Next, under Assumption 5.1, we use a slight modification to the proof of part B of Lemma A.1 to obtain uniformly over $X\in\mathcal{X}$ and $\alpha\in\mathcal{N}_n$:

$$E\{\widehat{\Sigma}_o(X,\alpha)-\widehat{\Sigma}_o(X,\alpha_o)|X_1,...,X_n\}=\Sigma_o(X,\alpha)-\Sigma_o(X,\alpha_o)+o_p(\delta_n).$$

The Lemma now follows. ■

Proof. (Theorem 5.1): Given Assumptions 3.1 and 3.4 and Lemma 5.1, to prove the consistency of the estimated covariance matrix, it remains to show that for all $j=1,...,b$, $\widehat{D}_{\widehat{w^{*j}}}(X)$ converges to $D_{w^{*j}}(X)$ uniformly over $X\in\mathcal{X}$. We note that the criterion function

$$\sum_{i=1}^{n}\widehat{D}_{w^j}(X_i)'[\widehat{\Sigma}(X_i)]^{-1}\widehat{D}_{w^j}(X_i)$$

is globally convex in $w^j(.)$. The solution $\widehat{w^{*j}}(.)$ must be bounded by $\left\|\widehat{w^{*j}}(.)\right\|\leq c$ for some constant c. Thus, we only need to concern with the subset $\{\upsilon\in\overline{\mathcal{W}}_n:\|\upsilon\|\leq c\}$. Define

$$\widetilde{D}_{w^j}(X)\equiv\sum_{i=1}^{n}\left(\frac{d\rho(Z_i,\alpha_o)}{d\theta_j}-\frac{d\rho(Z_i,\alpha_o)}{dh'}[w^j(\cdot)]\right)p^{k_n}(X_i)'(P'P)^{-1}p^{k_n}(X)$$

Assumptions 3.1-3.10 imply that $\widehat{\alpha}_n\in\mathcal{N}_n$ with probability approaching to one. This and Assumption 5.2 imply that uniformly over $X\in\mathcal{X}$, $w^j\in\overline{\mathcal{W}}_n$ and $\|w^j\|\leq c$:

$$\widehat{D}_{w^j}(X)-\widetilde{D}_{w^j}(X)=o_p(1).$$

Apply Lemma A.1, we obtain uniformly over $X\in\mathcal{X}$, $w^j\in\overline{\mathcal{W}}_n$ and $\|w^j\|\leq c$:

$$\widetilde{D}_{w^j}(X)-D_{w^j}(X)=o_p(1).$$

Hence $\widehat{D}_{w^j}(X_i)=D_{w^j}(X_i)+o_p(1)$ uniformly over $X_i\in\mathcal{X}$. Employing the same proof as of Theorem 3.1, we show that $\left\|\widehat{w^{*j}}(.)-w^{*j}(.)\right\|=o_p(1)$. Hence, $\widehat{D}_{\widehat{w^{*j}}}(X_i)=D_{w^{*j}}(X_i)+o_p(1)$ uniformly over $X_i\in\mathcal{X}$. The theorem now follows immediately. ■

Proof. (Theorem 6.1): Since the semiparametric efficiency bound is the lower bound of the variances of all $\sqrt{n}$ regular (consistent) semiparametric estimators

and V_o^{-1} is the asymptotic variance of a semiparametric estimator, it follows that V_o^{-1} must be no smaller than the efficiency bound in positive semi-definite matrix sense. To show that it also is the semiparametric efficiency bound, we only need to show that there is a sequence of parametric submodels whose asymptotic covariance matrices converge to V_o^{-1}. Recall that $w^*(\cdot) \in \overline{\mathcal{W}}$ and $-w^*(\cdot)(\theta-\theta_o) \in \overline{\mathcal{H}} - h_o$. Because $\overline{\mathcal{W}}$ is a completion of $\mathcal{W}$ under the Fisher-like norm, there exists a sequence $\{w_k(\cdot) \in \mathcal{W},\ k = 1, 2, ...\}$ satisfying

$$E\left[D_{w_k}(X)'[\Sigma_o(X)]^{-1}D_{w_k}(X)\right] \to V_o \text{ as } k \to +\infty.$$

Note that $-w_k(\cdot)(\theta - \theta_o) \in \mathcal{H} - h_o$. Hence, $g_k(\cdot, \theta) = h_o - w_k(\cdot)(\theta - \theta_o) \in \mathcal{H}$ is a parametric submodel passing through h_o. Let $f_o(Y|X)$ denote the unknown true conditional density of Y given X. Let $f(Y|X, \alpha)$ denote the solution to the following constrained optimization problem:

$$\max E[\log(f(Y|X))] \text{ s.t. } \int \rho(Z, \alpha) f(Y|X) d\mu(Y) = 0;\ \int f(Y|X) d\mu(Y) = 1$$

where $\mu(.)$ denotes the probability measure. Then $f(Y|X, \alpha_o) = f_o(Y|X)$. Let $\lambda(X, \alpha)$ denote the Lagrangian multiplier of the first set of constraints. By calculus variation, we obtain:

$$f(Y|X, \alpha) = \frac{f_o(Y|X)}{1 + \lambda(X, \alpha)'\rho(Z, \alpha)}.$$

Moreover, $\lambda(X, \alpha)$ is the solution to:

$$\max E[\log(1 + \lambda(X, \alpha)'\rho(Z, \alpha))|X].$$

It can be shown that $\lambda(X, \alpha_o) = 0$. Consider the parametric submodel $f(Y|X, \theta, g_k(\cdot, \theta))$ and the maximum likelihood estimation of this submodel. Simple calculations show that the Cramer-Rao lower bound of this parametric submodel is

$$E\left[D_{w_k}(X)'[\Sigma_o(X)]^{-1}D_{w_k}(X)\right]$$

which obviously converge to V_o by construction. This proves that V_o^{-1} is the semiparametric efficiency bound for model (1.3). ■

Proof. (Theorem 6.2): By Lemma 5.1, we have: uniformly over $X \in \mathcal{X}$ and $\alpha \in \mathcal{N}_n$, $\widehat{\Sigma}_o(X, \alpha) = \Sigma_o(X, \alpha) + o_p(\delta_n)$. Notice that the SMD estimator $\widehat{\alpha}_n$ in Step 1 satisfies $\|\widehat{\alpha}_n - \alpha_o\| = o_p(\delta_n)$. Hence, $\widehat{\Sigma}_o(X) = \widehat{\Sigma}_o(X, \widehat{\alpha}_n) = \Sigma_o(X) + o_p(\delta_n)$ uniformly over $X \in \mathcal{X}$. The result now follows from Theorem 4.1 with $\Sigma(X)$ replaced by $\Sigma_o(X)$. ■

Appendix E: Applications

Proof. (Theorem 7.1): For rate calculation, it is sufficient to consider the case $\Sigma(X) = I$.

(i) follows directly from Theorem 3.1, and (a) Assumption 3.8 holds (i.e., for any $\alpha = (\theta', h')' \in \mathcal{A}$, there are $\Pi_n \alpha = (\theta', \Pi_n h')' \in \mathcal{A}_n$ such that $\|\alpha - \Pi_n \alpha\| = o(\delta_n)$); and (b) for any $\alpha_1 = (\theta_1', h_1')'$, $\alpha_2 = (\theta_2', h_2')' \in \mathcal{A}$, $\|\theta_1 - \theta_2\| \leq c \|\alpha_1 - \alpha_2\|$ and $||h_1 - h_2|| \leq c \|\alpha_1 - \alpha_2\|$.

For (a), by Assumption 7.2(ii), we have:

$$\begin{aligned}
\|\alpha - \Pi_n \alpha\| &= \|h - \Pi_n h\| \\
&= \|h - \Pi_n h\|_{\infty,\omega} \sqrt{E\left(\left\| E\{\frac{d\rho(Z,\alpha_o)}{dh'}[\frac{h - \Pi_n h}{\|h - \Pi_n h\|_{\infty,\omega}}]|X\}\right\|_E^2\right)} \\
&\leq c \|h - \Pi_n h\|_{\infty,\omega} = o(\delta_n) \quad \text{(by Assumption 7.1).}
\end{aligned}$$

For (b), given the definition of $\|\theta_1 - \theta_2\|$ and Assumption 7.2(i), we have:

$$\|\theta_1 - \theta_2\| \asymp c \|\theta_1 - \theta_2\|_E \quad \text{(equivalent).}$$

Notice that we can always rewrite $h_1(\cdot) - h_2(\cdot) = -W(\cdot)(\theta_1 - \theta_2)$. Denote

$$A(X) = E\left[\frac{d\rho(Z,\alpha_o)}{d\theta'} - \frac{d\rho(Z,\alpha_o)}{dh'}[W]|X\right].$$

We then have

$$\begin{aligned}
\|\alpha_1 - \alpha_2\|^2 &= (\theta_1 - \theta_2)' E\{A(X)'\Sigma(X)^{-1}A(X)\}(\theta_1 - \theta_2) \\
&\geq (\theta_1 - \theta_2)' E\{D_{w^*}(X)'\Sigma(X)^{-1}D_{w^*}(X)\}(\theta_1 - \theta_2) \\
&\geq c \|\theta_1 - \theta_2\|_E^2 \quad \text{(by Assumption 4.1).}
\end{aligned}$$

hence $\|\theta_1 - \theta_2\| \leq c \|\alpha_1 - \alpha_2\|$. Finally, note that

$$\alpha_1 - \alpha_2 = (\theta_1' - \theta_2', h_o' - h_o')' + (\theta_o' - \theta_o', h_1' - h_2')'.$$

It follows that

$$\|(\theta_o' - \theta_o', h_1' - h_2')'\| \leq \|\alpha_1 - \alpha_2\| + \|(\theta_1' - \theta_2', h_o' - h_o')'\|.$$

Again by Assumptions 7.2(i) and 4.1, we have:

$$||h_1 - h_2|| \leq \|\alpha_1 - \alpha_2\| + \|\theta_1 - \theta_2\| \leq c \|\alpha_1 - \alpha_2\|.$$

(ii) By definition, we have $\|h - \Pi_n h_o\|_{\infty,\omega} \leq \tau_{\infty n} \|h - \Pi_n h_o\|$ for any $h \in \mathcal{H}_n$. This and Assumption 7.1 together imply:

$$\begin{aligned}
\|h - h_o\|_{\infty,\omega} &\leq \|h - \Pi_n h_o\|_{\infty,\omega} + \|\Pi_n h_o - h_o\|_{\infty,\omega} \\
&\leq \tau_{\infty n} \|h - \Pi_n h_o\| + o(\delta_n)
\end{aligned}$$

Note by Assumptions 7.1 and 7.2(ii) we have:

$$\|h - \Pi_n h_o\| \leq \|h - h_o\| + \|h_o - \Pi_n h_o\| \leq \|h - h_o\| + o(\delta_n).$$

Hence

$$\|h - h_o\|_{\infty,\omega} = \max\{O(\tau_{\infty n} \|h - h_o\|), o(\tau_{\infty n}\delta_n), o(\delta_n)\}.$$

Applying (i), we obtain $\left\|\widehat{h}_n - h_o\right\|_{\infty,\omega} = o_p(\tau_{\infty n}\delta_n)$.

(iii) Similarly, we have under Assumptions 7.1, 7.2(ii) and 7.3,

$$\|h - h_o\|_2 = \max\{O(\tau_{2n} \|h - h_o\|), o(\tau_{2n}\delta_n), o(\delta_n)\}.$$

Hence applying (i), we obtain $\left\|\widehat{h}_n - h_o\right\|_2 = o_p(\tau_{2n}\delta_n)$. ■

Proof. (Proposition 7.1): (1) We obtain the convergence rate by applying Theorem 7.1. Assumptions 3.1', 3.2', 3.4 with $\widehat{\Sigma} = \Sigma = I$ are already assumed. Assumption 3.3 is satisfied given A7.1.1 (see Newey and Powell, 1989). Assumption 3.5 is automatically satisfied with $\rho(Z, \alpha) = Y_1 - X_1'\theta - h(Y_2)$,

$$m(X, \alpha) = E[Y_1 - X_1'\theta - h(Y_2)|X] = X_1'[\theta_o - \theta] + E[h_o(Y_2) - h(Y_2)|X]$$

and $\rho_{\alpha_o}[Z, \alpha - \alpha_o] = -X_1'[\theta - \theta_o] - [h(Y_2) - h_o(Y_2)]$, hence

$$\begin{aligned} \|\alpha - \alpha_o\|^2 &= E\left\{\left(X_1'(\theta - \theta_o) + E[h(Y_2) - h_o(Y_2)|X]\right)^2\right\} \\ &= E\{m(X, \alpha)'m(X, \alpha)\}. \end{aligned}$$

For Assumption 7.1, notice that for any $\alpha = (\theta, h) \in \mathcal{A} = \Theta \times \mathcal{H}$, there exists $\Pi_n\alpha = (\theta, \Pi_n h) \in \mathcal{A}_n = \Theta \times \mathcal{H}_n$ with $\mathcal{H}_n$ given in (7.6), such that for any fixed $a > \gamma > 0$, and $c > 0$,

$$||h - \Pi_n h||_{\infty,\omega} = \sup_{y_2} \left|[h(y_2) - \Pi_n h(y_2)](1 + ||y_2||_E^2)^{-a/2}\right| \leq c(k_{1n})^{-\gamma/d},$$

see Haroske (1995). Hence Assumption 7.1 is satisfied with the weighting function $\omega(y_2) = (1+||y_2||_E^2)^{-a/2}, a > \gamma > 0$ and A7.1.5. Now Assumption 7.2(i) is implied by A7.1.1(iii). Since for any $u \in \mathcal{H}$ with $||u||_{\infty,\omega} = 1$ we have

$$E\left(||E[u(Y_2)|X]||_E^2\right) \leq E\{[u(Y_2)]^2\} \leq E\{\left[\omega(Y_2)^{-1}||u||_{\infty,\omega}\right]^2\},$$

Assumption 7.2(ii) is satisfied with A7.1.4. Clearly Assumption 7.3 is also satisfied with the weighting $\omega(y_2)$ and A7.1.4, hence Assumption 3.8 is satisfied, and also

$$||h - \Pi_n h||_2 = \sqrt{E[h(Y_2) - \Pi_n h(Y_2)]^2} \leq ||h - \Pi_n h||_{\infty,\omega} \leq c(k_{1n})^{-\gamma/d}.$$

Assumption 3.9 is satisfied with $\kappa_n = O\left(\sqrt{\frac{k_{1n}}{n}}\right) = o(\delta_n)$ and A7.1.5, see e.g. Chen and Shen (1998).

For Assumption 3.6's envelope condition, note that for any $\alpha \in \mathcal{A}_n = \Theta \times \mathcal{H}_n$ with $\mathcal{H}_n$ given in (7.6),

$$\begin{aligned}\sup_{\alpha\in\mathcal{A}_n} |\rho(Z,\alpha)| &\leq |Y_1| + \sup_{\theta\in\Theta} |X_1'\theta| + \sup_{h\in\mathcal{H}_n} |h(Y_2)| \\ &\leq |Y_1| + \sup_{\theta\in\Theta} |X_1'\theta| + \sup_{h\in\mathcal{H}_n} ||h||_{\Lambda^\gamma} \leq \xi_{3n} c_1(Z),\end{aligned}$$

with $c_1(Z) = |Y_1| + \sup_{\theta\in\Theta} |X_1'\theta| + const.$ and $\xi_{3n} = \ln(1 + k_{1n})$. Hence, given A7.1.3, Assumption 3.6's envelope condition is satisfied with coefficient $\xi_{3n} = \ln(1 + k_{1n})$. Next for Assumption 3.6's conditional smoothness, we note that for any $\alpha \in \mathcal{A}_n = \Theta \times \mathcal{H}_n$,

$$\begin{aligned}m(X,\alpha) &= E[\rho(Z,\alpha)|X] = E[Y_1|X] - X_1'\theta - E[h(Y_2)|X] \\ &= X_1'(\theta_o - \theta) + E[h_o(Y_2) - h(Y_2)|X].\end{aligned}$$

By A7.1.2, there are orthonormal bases $\{\phi_{0k} : k = 1, ..., \infty\}$ (for $\Lambda^{\gamma_0}(\mathcal{X})$ and $L_2(\mathcal{X}, f_X)$), $\{\phi_{1k} : k = 1, ..., \infty\}$ (for $L_2(\mathcal{R}^d, f_{Y_2})$), and corresponding singular numbers $\{\mu_k\}$ with $\mu_1 = 1 \geq \mu_2 \geq \mu_3 \geq ...$ such that

$$\begin{aligned}\frac{f_{X,Y_2}(x,y_2)}{f_X(x) f_{Y_2}(y_2)} &= \sum_{k=1}^{\infty} \mu_k \phi_{0k}(x) \phi_{1k}(y_2), \\ c_1 k^{-(\frac{\gamma_0}{s}+\frac{1}{2})} [\ln(1+k)]^{-1} &\leq \mu_k \leq c k^{-(\frac{\gamma_0}{s}+\frac{1}{2})},\end{aligned}$$

see e.g. Birman and Solomjak (1980, theorem 3.2, corollaries 3.5 and 3.7).

$$\begin{aligned}E[h_o(Y_2) - h(Y_2)|X] &= \int (h_o(y_2) - h(y_2)) \frac{f_{X,Y_2}(X,y_2)}{f_X(X) f_{Y_2}(y_2)} f_{Y_2}(y_2) dy_2 \\ &= \sum_{k=1}^{\infty} \mu_k \phi_{0k}(X) \int (h_o(y_2) - h(y_2)) \phi_{1k}(y_2) f_{Y_2}(y_2) dy_2.\end{aligned}$$

For any $g \in L_2(\mathcal{R}^d, f_{Y_2})$, we denote $\langle g, \phi_{1k} \rangle \equiv \int g(y_2)\phi_{1k}(y_2) f_{Y_2}(y_2) dy_2$. Since $\{\phi_{1k} : k = 1, ..., \infty\}$ is an orthonormal basis for $L_2(\mathcal{R}^d, f_{Y_2})$, we have $||g||_2^2 = \sum_{k=1}^{\infty} \langle g, \phi_{1k} \rangle^2$. Then

$$E[h_o(Y_2) - h(Y_2)|X] - \sum_{k=1}^{k_n} \mu_k \langle h_o - h, \phi_{1k} \rangle \phi_{0k}(X) = \sum_{k=k_n+1}^{\infty} \mu_k \langle h_o - h, \phi_{1k} \rangle \phi_{0k}(X),$$

and

$$\begin{aligned}&\sup_{X\in\mathcal{X}} \left| E[h_o(Y_2) - h(Y_2)|X] - \sum_{k=1}^{k_n} \mu_k \langle h_o - h, \phi_{1k} \rangle \phi_{0k}(X) \right| \\ \leq\ & \sup_{X\in\mathcal{X}} |\phi_{0k}(X)| \times \sqrt{\sum_{k=k_n+1}^{\infty} \mu_k^2} \times \sqrt{\sum_{k=k_n+1}^{\infty} \langle h_o - h, \phi_{1k} \rangle^2} \\ \leq\ & const. \times ||h_o - h||_2 \times \sqrt{\sum_{k=k_n+1}^{\infty} \mu_k^2}\end{aligned}$$

and

$$\sup_{h\in\mathcal{H}_n}\sup_{X\in\mathcal{X}}\left|E[h_o(Y_2)-h(Y_2)|X]-\sum_{k=1}^{k_n}\mu_k\left\langle h_o-h,\phi_{1k}\right\rangle\phi_{0k}(X)\right|$$

$$\leq \quad const.\sup_{h\in\mathcal{H}_n}||h||_2\times\sqrt{\sum_{k=k_n+1}^{\infty}k^{-2(\frac{\gamma_0}{s}+\frac{1}{2})}}\leq c\ln(1+k_{1n})\times(k_n+1)^{-(\frac{\gamma_0}{s}+\frac{1}{2})}$$

hence Assumption 3.6's conditional smoothness is satisfied with $\xi_{3n}=\ln(1+k_{1n})$ and $k_n^{-a_0}=(k_n+1)^{-(\frac{\gamma_0}{s}+\frac{1}{2})}$. For Assumption 3.6's Lipschitz condition, notice that for any $\alpha_1,\alpha_2\in\mathcal{A}_n=\Theta\times\mathcal{H}_n$ with $\mathcal{H}_n$ given in (7.6), we have

$$|\rho(Z,\alpha_1)-\rho(Z,\alpha_2)|\leq|X_1'(\theta_1-\theta_2)|+|h_1(Y_2)-h_2(Y_2)|$$

Again under A7.1.1(iii) and A7.1.7, (note that A7.1.7 is simply Assumption 4.1), following the proof of Theorem 7.1(i), we have:

$$|X_1'(\theta_1-\theta_2)|\leq||X_1||_E||\theta_1-\theta_2||_E\leq c||X_1||_E\left\|\alpha_1-\alpha_2\right\|$$

and

$$\begin{aligned}|h_1(Y_2)-h_2(Y_2)| &\leq [\omega(Y_2)]^{-1}\left\|h_1-h_2\right\|_{\infty,\omega}\\ &\leq [\omega(Y_2)]^{-1}\tau_{\infty n}\left\|h_1-h_2\right\|\leq c[\omega(Y_2)]^{-1}\tau_{\infty n}\left\|\alpha_1-\alpha_2\right\|\end{aligned}$$

Hence under A7.1.4, we have:

$$|\rho(Z,\alpha_1)-\rho(Z,\alpha_2)|\leq c(Z)\tau_{\infty n}\left\|\alpha_1-\alpha_2\right\|,$$

with $c(Z)=||X_1||_E+[\omega(Y_2)]^{-1}$ satisfying $E[c(Z)^2]<\infty$, and Assumption 3.6's Lipschitz condition is satisfied with $\xi_{2n}=\tau_{\infty n}$. By definition,

$$\begin{aligned}\tau_{\infty n} &= \sup_{h_1,h_2\in\mathcal{H}_n:h_1\neq h_2}\frac{\left\|h_1-h_2\right\|_{\infty,\omega}}{\left\|h_1-h_2\right\|}\\ &\leq \sup_{h_1,h_2\in\mathcal{H}_n:h_1\neq h_2}\frac{\left\|h_1-h_2\right\|_{\infty,\omega}}{\left\|h_1-h_2\right\|_2}\times\sup_{h_1,h_2\in\mathcal{H}_n:h_1\neq h_2}\frac{\left\|h_1-h_2\right\|_2}{\left\|h_1-h_2\right\|}\leq\sqrt{k_{1n}}\times\tau_{2n}.\end{aligned}$$

By Condition A7.1.2, we have for any $h_1,h_2\in\mathcal{H}_n$,

$$E\{[h_1(Y_2)-h_2(Y_2)]|X\}=\sum_{j=1}^{k_{1n}}\mu_j\left\langle h_1-h_2,\phi_{1j}\right\rangle\phi_{0j}(X)$$

$$\begin{aligned}\left\|h_1-h_2\right\|^2 &= \sum_{j=1}^{k_{1n}}\mu_j^2\left\langle h_1-h_2,\phi_{1j}\right\rangle^2\\ &\geq (\mu_{k_{1n}})^2\sum_{j=1}^{k_{1n}}\left\langle h_1-h_2,\phi_{1j}\right\rangle^2=(\mu_{k_{1n}})^2\left\|h_1-h_2\right\|_2^2\end{aligned}$$

Hence by definition and by A7.1.2, we have

$$\tau_{2n} = \sup_{h_1,h_2\in\mathcal{H}_n:h_1\neq h_2} \frac{\|h_1-h_2\|_2}{\|h_1-h_2\|} = \frac{1}{\mu_{k_{1n}}} \leq c(k_{1n})^{\frac{\gamma_0}{s}+\frac{1}{2}} \ln(1+k_{1n}).$$

For Assumption 3.10, notice for any $\alpha \in \mathcal{A}_n$ with $||\alpha-\alpha_o|| \leq \eta_n$ and $\eta_n \in [\delta_n, \delta_n^{3/4}]$, we have:

$$\frac{\rho(Z,\alpha)-\rho(Z,\alpha_o)}{\eta_n} = \frac{X_1'(\theta-\theta_o)+[h(Y_2)-\Pi_n h_o(Y_2)]+[\Pi_n h_o(Y_2)-h_o(Y_2)]}{\eta_n}$$

By what we already checked above,

$$\begin{aligned}\left|\frac{\rho(Z,\alpha)-\rho(Z,\alpha_o)}{\eta_n}\right| &\leq [\omega(Y_2)]^{-1}\frac{\xi_{2n}||\alpha-\Pi_n\alpha_o||+||h_o-\Pi_n h_o||_{\infty,\omega}}{\eta_n} \\ &= [\omega(Y_2)]^{-1}\frac{\xi_{2n}\eta_n+o(\delta_n)}{\eta_n}.\end{aligned}$$

$$\begin{aligned}\left|\frac{m(X,\alpha)}{\eta_n}\right| &= \left|E\left[\frac{\rho(Z,\alpha)-\rho(Z,\alpha_o)}{\eta_n}|X\right]\right| \leq E\left[\left|\frac{\rho(Z,\alpha)-\rho(Z,\alpha_o)}{\eta_n}\right||X\right] \\ &\leq E\left[[\omega(Y_2)]^{-1}\frac{\xi_{2n}\eta_n+o(\delta_n)}{\eta_n}\right]\end{aligned}$$

Hence under A7.1.4, $\frac{m(X,\alpha)}{\eta_n}$ satisfies Assumption 3.10 with envelope coefficient $\xi_{4n} = \xi_{2n} = \tau_{\infty n}$.

Finally, Assumption 3.7 is satisfied with $\xi_{0n} = \xi_{1n} = \sqrt{k_n}$ (see Newey (1997)), $\xi_{2n} = \tau_{\infty n} = \xi_{4n}, \xi_{3n} = \ln(1+k_{1n}), N(\mathcal{A}_n, \frac{\delta_n}{\sqrt{k_n}\xi_{2n}}) = \left(\frac{\delta_n}{\sqrt{k_n}\xi_{2n}}\right)^{-k_{1n}} = \left(n^{1/4}\sqrt{k_n}\xi_{2n}^{3/2}\right)^{k_{1n}}$ and A7.1.6.

Hence all the assumptions of Theorem 7.1 are satisfied, and we obtain the convergence rates: $||\widehat{\alpha}-\alpha_o|| = o_p(\delta_n)$ with $\delta_n = n^{-1/4}(\xi_{4n})^{-1/2}$; $||\widehat{h}-h_o||_2 = o_p(\tau_{2n}\delta_n)$; $||\widehat{h}-h_o||_{\infty,\omega} = o_p(\tau_{\infty n}\delta_n)$.

(2) We obtain the $\sqrt{n}$ consistency and efficiency by applying Theorem 6.2. Assumption 4.1 with $\Sigma(X) = \Sigma_o(X)$ is satisfied by A7.1.7. Assumption 4.2 is satisfied by A7.1.8. Assumption 4.3 is implied by Assumptions 3.1', 3.2', A7.1.2, A7.1.6(ii), A7.1.7 and A7.1.8. Assumption 4.4 is satisfied by the verification of Assumptions 3.6 and 3.10 above. Assumption 4.5 is automatically satisfied since $r_\rho[Z,\alpha,\alpha_o] = 0$. Assumption 5.1 is satisfied by the verification above. Applying Theorem 6.2, we obtain $\sqrt{n}(\widetilde{\theta}_n-\theta_o) \Longrightarrow N(0, V_o^{-1})$ with V_o given by (7.7). ■

Table 1. Finite Sample Performance of the SMD Estimator

Estimator	**MEAN**	**STD**	**MAD**	**MSE**	**25%**	**50%**	**75%**
	R=0.9						
	Power series						
Inefficient SMD Estimator of θ	0.964	0.140	0.108	0.145	0.879	0.962	1.049
Efficient SMD Estimator of θ	0.979	0.104	0.089	0.116	0.903	0.975	1.051
Inefficient SMD Estimator of h(.): IMSE=0.478, IMAD=0.977							
Efficient SMD Estimator of h(.): IMSE=0.069, IMAD=0.360							
	Spline						
Inefficient SMD Estimator of θ	0.965	0.122	0.098	0.127	0.879	0.960	1.046
Efficient SMD Estimator of θ	1.016	0.109	0.084	0.110	0.944	1.014	1.083
Inefficient SMD Estimator of h(.): IMSE=0.081, IMAD=0.469							
Efficient SMD Estimator of h(.): IMSE=0.037, IMAD=0.309							
	R=0.1						
	Power series						
Inefficient SMD Estimator of θ	0.993	0.141	0.108	0.141	0.907	0.991	1.083
Efficient SMD Estimator of θ	0.998	0.110	0.086	0.110	0.926	0.997	1.067
Inefficient SMD Estimator of h(.): IMSE=0.503, IMAD=1.004							
Efficient SMD Estimator of h(.): IMSE=0.096, IMAD=0.423							
	Spline						
Inefficient SMD Estimator of θ	0.999	0.127	0.101	0.127	0.918	0.993	1.081
Efficient SMD Estimator of θ	1.014	0.085	0.067	0.087	0.959	1.016	1.070
Inefficient SMD Estimator of h(.): IMSE=0.157, IMAD=0.722							
Efficient SMD Estimator of h(.): IMSE=0.141, IMAD=0.686							